

ABSOLUTELY UNDERSTAND GUITAR

THE ULTIMATE HOME STUDY PROGRAM ON VHS AND DVD!

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THE 6 MAIN AREAS OF MUSIC

It seems that everyone wants to play a musical instrument at some time during their lives. But why do so few people stick with it? Is it because they didn't have the TALENT? Did they not PRACTICE enough? That may be part of it but it's mostly because they simply failed to get the SATISFACTION that they were expecting for all their hard work and over time they just lost interest. You don't want that happening to you...RIGHT?

SATISFACTION

A lot of it is simply INFORMATION. You have to understand what you're doing in order to enjoy playing your instrument. That appears obvious. But then again it seems that a lot of the people around you don't know much about music or about the instrument they claim to play. So why should you? Well perhaps you would like to play BETTER than them. Or perhaps you would like to become a good player in LESS TIME than it took them. Most people waste DECADES worth of time and many never get to where they wanted to in music. Often times people resist learning about music because they have the impression that it is a difficult or impossible subject to understand. That simply isn't true. Music is a very simple subject to understand. It's just that the way it is often taught, they give you the information in the WRONG ORDER so you never get to see how it all comes together. You won't have that problem here!

THE 1st THING THEY SHOULD HAVE TOLD YOU ABOUT MUSIC IS THIS:

MUSIC IS A LANGUAGE !

We mean that quite literally. Don't you feel like you are being communicated with when you listen to music? Doesn't it bring images and emotions into your mind? Like any language, music has an ALPHABET. We select units out of that alphabet and form them into PATTERNS. These patterns have MEANING and are very much like "words". They have SPELLINGS.

YOUR GUITAR IS A TYPEWRITER !

It's just a DUMB MACHINE! When you move your fingers around on the neck of your guitar, you are "TYPING" these patterns. Many people think that all there is to playing the guitar is learning to move their fingers around in these mysterious shapes. That is such a DEAD END! That would be like taping on a typewriter without knowing what any of the words mean or sound like. You don't play guitar with your fingers. You play it with your mind! You have to understand the LANGUAGE of music before you can absolutely understand guitar. The good news is music is a very simple language and easy to understand if you get the facts explained to you in the RIGHT ORDER.

If you want the SATISFACTION, ya' gotta' become a REAL PLAYER. Most people who claim to play the guitar are only PRETENDING to play it. You will be stunned to find out how little people know about music and the instrument itself.

WHAT 2 THINGS DO YOU GOTTA' DO TO BECOME A REAL PLAYER?

① GET TO THE END OF THE SONG!

Notice how a lot of people only play little bits and pieces of songs. Don't confuse playing RIFFS with playing SONGS. Real artists produce real works of art. What possible good would it ever do you to just play part of a song?

② STOP AND ASK YOURSELF "HOW DID I DO?"

To stay interested you have to PROGRESS. To progress you need some system to analyze your DEFICIENCIES... "Check Yourself Out." Try to find a way you can RECORD yourself when you play. Listen to yourself like you're an impartial listener. Do you like what you hear? Now you know what to fix in order to improve. But how are you supposed to ask yourself how you did if you don't know what you're doing in the 1st place? Ya gotta' know what music is before you can judge how well you're playing it! The good news is there are only 6 things going on in music and if you understand these 6 concepts, you can ask yourself these 6 check list questions. Or in other words...

MUSIC IS A LANGUAGE WITH 6 PARTS OF SPEECH The 6 Main Areas of Music

Pitch	High/Low NOTES Frequency 50 "words"	Did I play the right notes ?
	Patterns SCALES CHORDS MELODY HARMONY INTERVALS MODES PROGRESSIONS ARPEGGIOS RIFFS	
Rhythm	Beat Tempo Timing Pulse	Did I keep with the beat ?
Timbre	Tone SOUND QUALITY INSTRUMENTS E.Q. EFFECTS : Distortion	How did I SOUND ?
Dynamics	Volume Loudness Amplitude Balance Up and down through song Size of room	Did I play at the right volume ?
Technique	Physical part Finger movement How to hold the instrument	Did I cooperate with the machine ?
Notation	Reading written music	How is my Reading coming ?

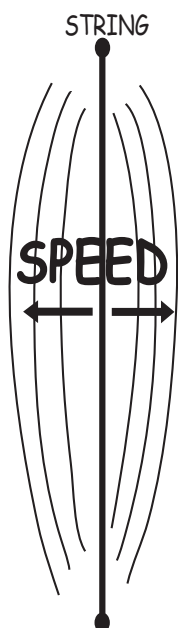
All you gotta' do to be a REAL PLAYER and so satisfy yourself enough to play guitar for the rest of your life is PLAY THE WHOLE SONG and ANSWER YES TO THESE 6 QUESTIONS!

4 WAYS TO KNOW YOUR PITCH PATTERNS

PITCH is the largest area of music. It's all about HIGH and LOW. But high and low what? NOTES...Pitch is all about what notes you play and what PATTERNS you arrange these notes in (like CHORDS and SCALES). The scientific word for pitch is FREQUENCY. You know that everything that is making noise is VIBRATING and sending out SOUND WAVES through the air. The FREQUENCY of the wave is how far apart the waves are SPACED.

Low Frequency 
equals low pitch

High Frequency 
equals high pitch



Your guitar is nothing more than a SOUND WAVE GENERATOR. It can generate waves of very precise and controllable frequencies. But how does it work? A guitar is a stringed instrument. You have pieces of wire stretched between two points. You pluck the string and cause it to vibrate. The SPEED at which the string vibrates back and forth positions the wave peaks where they are and defines the frequency of the sound wave. If the string is vibrating slowly, it sends out waves that are farther apart. If it is vibrating quickly, it sends out waves that are closer together. So you control pitch by controlling the speed that the string vibrates at.

PITCH = FREQUENCY = SPEED OF VIBRATION OF THE STRING

We control the string's vibrating speed using 3 different factors:

1 The MASS of the string

That's why your strings are all different thicknesses. Your low strings are more massive and so vibrate slower.

2 The TENSION of the string

That's what TUNING is. You tighten or loosen your strings to certain predetermined vibrating speeds. This is also how you bend notes.

3 The LENGTH of the string

How do you play different notes on the guitar? You move your fingers to different FRETS, stopping the strings at different lengths.

Like any language, music has an ALPHABET. In order to do anything creative with a language, you first have to learn it's alphabet. All the thousands of tunes you've heard in any musical style are all formed out of this simple set of notes just like all the thousands of words you know in the English language are simply patterns formed out of it's 26 unit alphabet. HOW MANY UNITS ARE THERE IN THE MUSICAL ALPHABET?

The Answer is

How many musical notes are there?

Most people are taught that there are only 7 (A-G) and that the sharps and flats are somehow "different". But that just isn't accurate. You will come to see that the five "black keys" are just as common and just as important as the seven "white keys" on the piano. Your guitar plays these same 12 notes. These notes then repeat in higher and lower OCTAVES.

12

You have to learn how to form about 50 patterns out of this 12 note alphabet.

13 INTERVALS

14 SCALES

20 CHORDS

Think of these as "words" in the language and understand that they have "spellings" out of this 12 note "alphabet". When you move your fingers around on the neck, you are "typing" these musical note patterns. These patterns are all on your SLIDERULE so be sure to assemble it right away.

The FIRST thing you must do is memorize **The MUSICAL ALPHABET**



What do the words SHARP and FLAT mean? SHARP (#) simply means HIGHER in pitch. FLAT (b) means LOWER. A# and Bb are two names for the same note. Soon you will learn when to call a note by its sharp or flat name. One of the most useful things that you can do as a musician is learn to visualize a piano keyboard in your mind. Notice that B and C and E and F have no black key between them and so there are no sharp or flat notes there.

Now...This may be the **MOST IMPORTANT FACT** you will ever learn about music!

You have to learn these 50 PITCH PATTERNS. But like any word in any language, these patterns exist in 4 dimensions. They have 4 different aspects and you need to learn to recognize them no matter which "face" they show you. So there are ...

4 WAYS TO KNOW EACH PITCH PATTERN

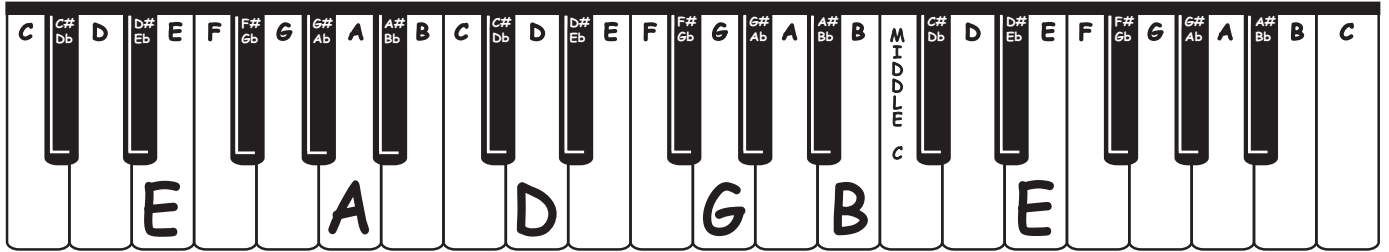
Very few people are ever shown this. This realization will set the stage for all your future musical development!

You have to THINK in the language	In a musical context, this means you know your patterns as CONCEPTS This is the world of MUSIC THEORY	Theory is the part of music that you have to treat like school. You have some stuff to MEMORIZE that is very much like spelling, grammar and vocabulary. Despite what you may have heard, theory is easy and fascinating if presented correctly.
You have to READ in the language	In a musical context, this means you know your patterns as NOTATION All REAL musicians can read music	There are 3 styles of notation for the guitar...STANDARD NOTATION, TABLATURE and the familiar CHORD/SCALE FINGER BOXES. If you ever hope to play more complex styles like jazz or classical, you must learn standard notation.
You have to HEAR in the language	In a musical context, this means you know your patterns as REAL SOUND This is the world of EAR TRAINING	Everyone wants to play "BY EAR". Most people just stumble around. All REAL players go through an EAR TRAINING program. It is the 2nd required course (after theory) at music school. You work out with a coach who trains you to recognize and SING all 50 patterns. You can't really compose or improvise without it.
You have to SPEAK in the language	In a musical context, this means you know your patterns as FINGER PATTERNS To make your guitar "speak" for you	Too many people think that learning to move their fingers is all that the guitar is about. But that is just a DEAD END. Technique is very important and you must memorize your various scale and chord finger charts. But remember, you don't play guitar with your fingers. You play it with your MIND!

THE "DUMB MACHINE"

LESSON 3

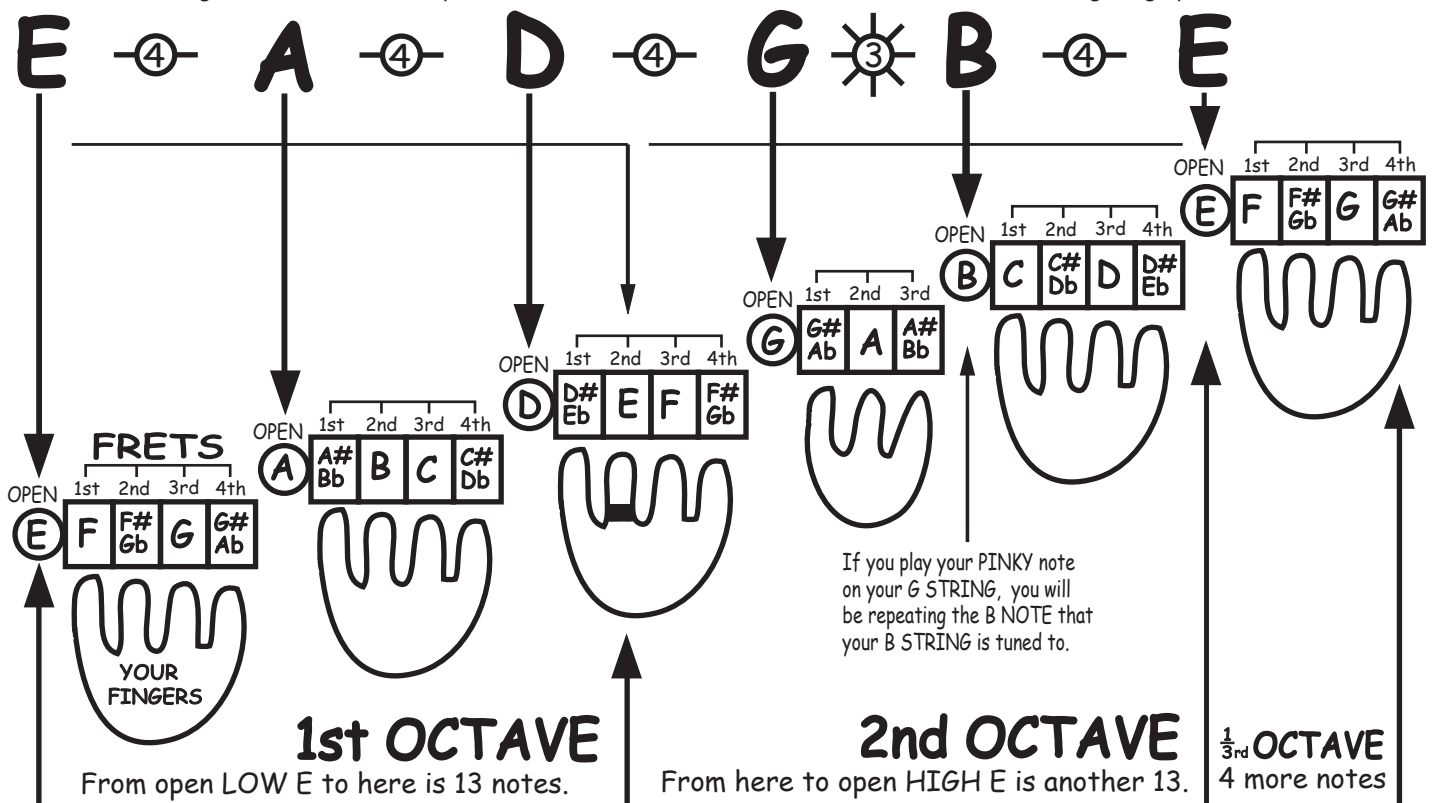
So now you know the 12 note musical alphabet. How do you "type" it on the guitar? How does your instrument actually work? Most people who claim to play the guitar actually don't know! People often ask "Does the guitar have the same LINEARITY as the piano when it comes to addressing these 12 notes?" The answer is YES but it's not quite so evident as on the keyboard. First, you must understand how the guitar is TUNED and where these notes are on a keyboard. The 6 strings of the guitar are tuned from LOW to HIGH **E - A - D - G - B - E** But why? What do these letters actually mean?



Do you see where these notes are on the piano? You know the keyboard is in alphabetical order. Notice that the notes that your guitar strings are tuned to always have 4 keys in between them (except G and B which only have 3). Why did they decide to tune the guitar so that there are mostly 4 NOTES IN BETWEEN EACH OPEN STRING ?

BECAUSE YOU HAVE 4 FINGERS !

To play all 12 notes in a sequence (a pattern called a CHROMATIC SCALE) on the guitar, start with the open LOW E string then play up the first 4 frets with your 4 fingers. After you get to your pinky it is time to move on to the next OPEN A string and so on. Your G and B strings are tuned 3 notes apart because it was discovered that it made more chord fingerings possible.



So from your open LOW E string to your open HIGH E string is actually 2 OCTAVES.

The TOTAL range from open LOW E to the highest fret on your HIGH E string is about 4 OCTAVES.

TUNING YOUR GUITAR

My advice is buy yourself an electronic CHROMATIC TUNER. Beginners often find it impossible and frustrating to tune by ear. If you want, you can develop that skill later. You will not enjoy playing if your instrument is out of tune. Electronic tuners are quick and easy to use. Don't waste your money on a GUITAR TUNER. It won't do FLAT or ALTERED tunings which are quite common.

So the 1st Rule of the GUITAR is

4 FINGERS = 4 FRETS

So to be in control of the guitar is simply to be in control of those 4 frets.

BUT ARE YOU IN CONTROL OR NOT? BEST WAY TO TELL FOR SURE IS...

BE ABLE TO TOUCH ALL 4 FRETS AT ONCE

The main physical skill you must develop to be a guitarist is to be able to control those 4 frets with your 4 consecutive fingers and coordinate this with alternate picking with your pick hand. To practice these **CHROMATIC SCALES**, make sure you are sitting and holding the guitar in the most practical manner...(check VIDEO LESSON 3)...**RIGHT KNEE...HEAD UP...etc.** You will never see the end of this exercise. There is no such thing as being too good at it. You can always get faster...cleaner... **BETTER.**

BUT IT ACTUALLY REQUIRES 5 FRETS TO CONTROL ALL 12 NOTES

At the bottom of the neck, you have the luxury of using the open string as that 5th fret.

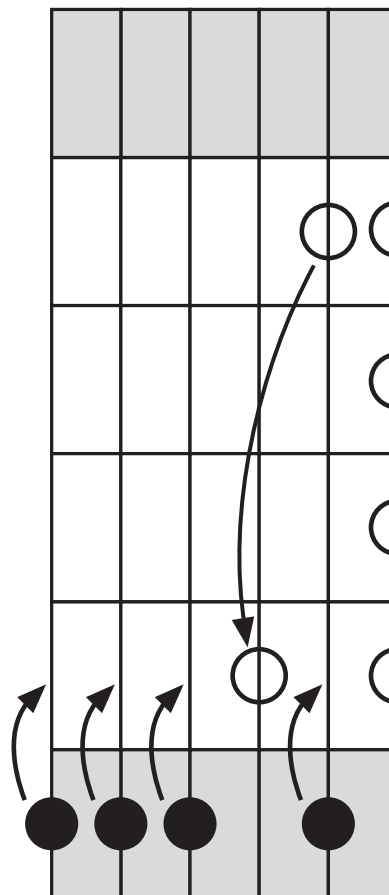
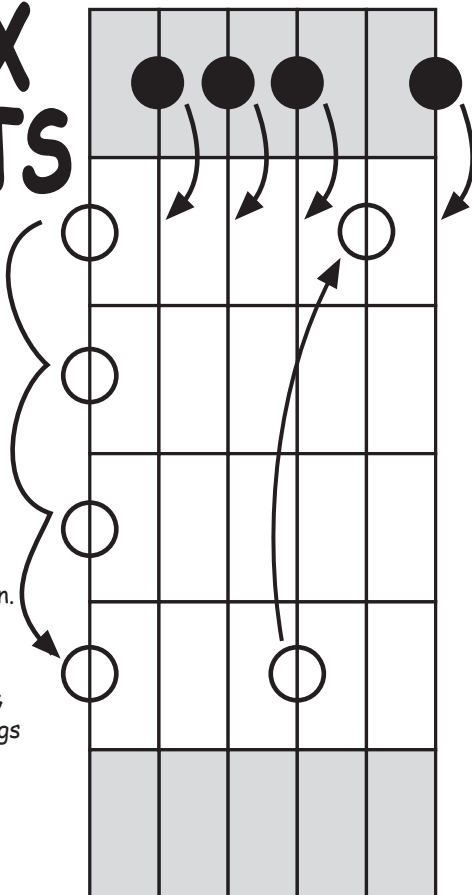
FURTHER UP THE NECK YOU MUST LEARN TO DO FINGER SHIFTS

YOU LEARN TO CONTROL 2 CONSECUTIVE FRETS WITH 1 FINGER !

For now, shift with your **INDEX FINGER** when going up the scale and shift with your **PINKY** coming down the scale. Use **LOGICAL FINGERS**. Try to "**STAY IN THE BOX**". Don't shift **AWAY** from your **HAND**.

INDEX SHIFTS

When going up the scale, start here. Walk up until you get to your pinky. Then reach out an extra fret with your index finger on your next string and then "scoot" it across the fret back onto the fret it would normally have been on and begin walking up that string and so on. No shift required between your pinky on your **G STRING** and your **B STRING** because those strings are tuned one note closer together.



When going down the scale, start with your pinky on your high E string. Walk down until you get to your index finger. Then reach out an extra fret with your pinky on your next lower string and then "scoot" it across the fret back onto the fret it would normally have been on and begin walking down that string. No shift required between your **B STRING** and your **G STRING**. Those strings are tuned one note closer together.

PINKY SHIFTS

If you can play a **CHROMATIC SCALE** on your guitar, you can do anything you want in music just like if you can type the alphabet on the typewriter, you can do what you want in the English language. You need to memorize where the notes are on your neck so make sure you put your **FRETBOARD CHART** together.

LESSON 4

Basic stuff about

CHORDS

CHORDS are MULTIPLE NOTES PLAYED SIMULTANEOUSLY. There are about 50 types of chords in Western Music. These chords all express different "MOODS". The more types of chords you can play, the more types of EMOTION you can express in your music. Common types are

MAJOR-MINOR-6-7-9-11-13-AUG-DIM-SUS-

Like all PITCH PATTERNS, chords have "spellings" and when you place your fingers on the neck you are "typing" that chord. The spelling for all your chords is on your MUSICAL SLIDERULE.

THERE ARE 12 OF EACH TYPE OF CHORD

There are 12 KEYS to play in because there are 12 NOTES. Any chord can be played in any key.

But owing to the physical limitations of the instrument, there are not unique fingerings for each chord.

THERE ARE 5 BASIC POSSIBLE FINGER FORMS (E A D G C forms)

All 12 chords can be played using these 5 forms as movable forms or BAR CHORDS. The open strings are part of the chord spelling just like the notes you have your fingers on. The bar finger replaces the open strings as you move up the neck. You can play any chord in any of the 5 forms. So not only can you play all 12 chords ...

THERE ARE 5 WAYS TO PLAY ANY CHORD

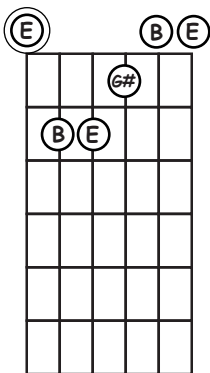
You will find that this is a bit of an over simplification that we will undo later.

The 5 ways to play the chord will sound SIMILAR because they contain the same alphabet letter notes. But they will sound somewhat DIFFERENT because the notes are in a different ORDER or OCTAVE.

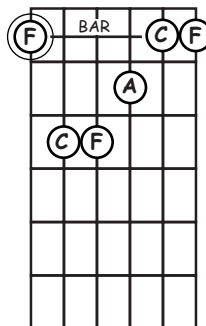
ALL CHORDS HAVE A ROOT NOTE

The ROOT is the LOWEST note in the chord that has the SAME NAME as the chord itself. You must always know where the root note of any chord is. It is the MOST IMPORTANT note in the chord...it's "LAUNCHING PAD".

On your CHORD FINGER CHART, root notes are indicated by ® if they are an OPEN STRING and ● if they are a note played by a finger. Chords that do not have their root in the bass are called SLASH CHORDS and will be discussed later. For now, don't play notes that are lower than the root. X over a string means don't play it (either by missing or muting it).

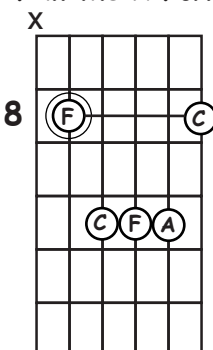


This is the fingering for an E Major chord in the E Form spelled E, G# and B. These letters can be in any order or any octave and repeated or not as you need or choose. Notice that 3 open strings can be used in this spelling. An F Major chord would be the next higher major chord. Each note in it is one note higher than the notes in the E Major chord and so it is spelled F, A and C. How do you play it? There is no F Form. One way might be to raise all the notes in the E Major chord up by one fret.

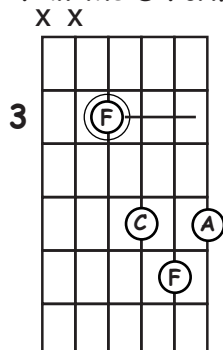


So this is an F Major chord played also in the E Form. Notice how each note has been raised by one fret from the E Major chord including the open strings. These are now being played by the bar finger. If we raised them all up another fret, the chord would become an F# Major chord (still in the E Form) and so on up the neck. We would eventually play all 12 major chords using this one finger shape. But you can also do this with the other 4 finger forms. So there would be 5 ways to play that F Major chord in different areas of the neck as you can see below. Notice where the root note is and note the fret number next to it. Strings lower than the root are X'd off and so not played.

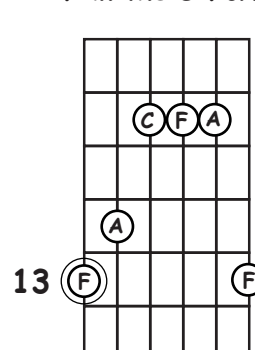
F in the A Form



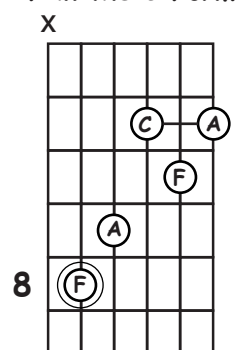
F in the D Form



F in the G Form



F in the C Form



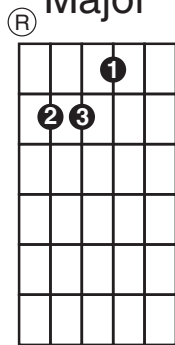


Numbers indicate most common fingerings though others are used. Some chords have variations indicated by adding notes marked ○. (x) means mute string if alternate note is not used.

E Form

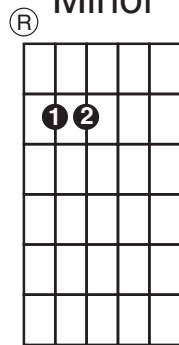
(happy)

Major



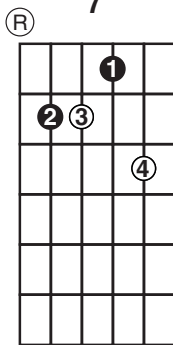
(sad)

Minor



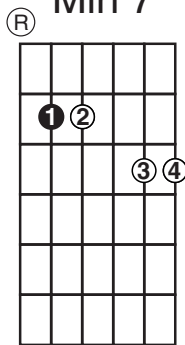
(bluesey)

7



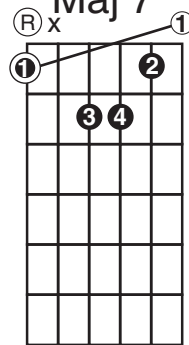
(sad bluesey)

Min 7



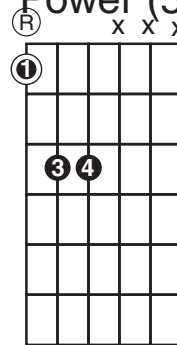
(mellow)

Maj 7

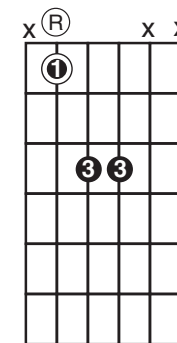
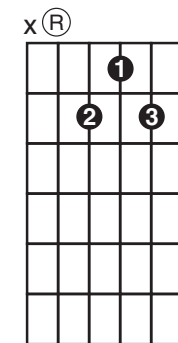
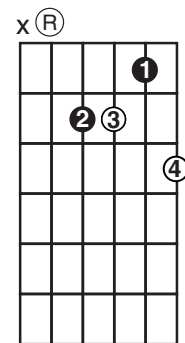
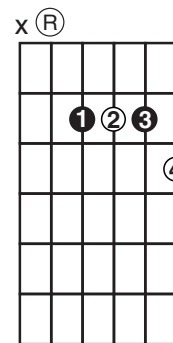
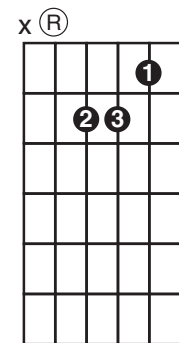
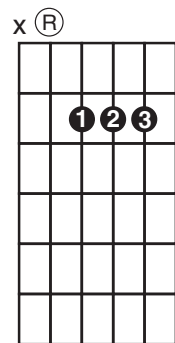


(severe)

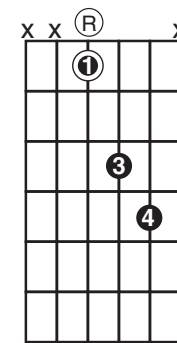
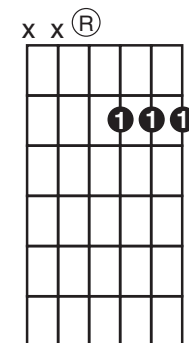
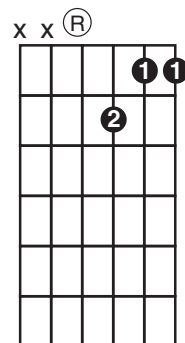
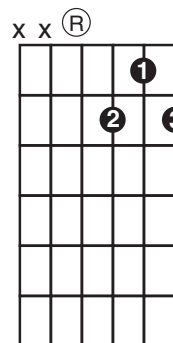
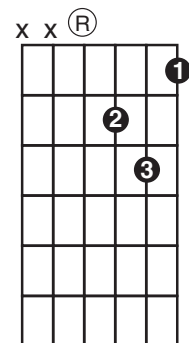
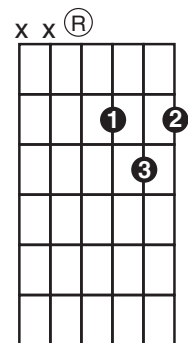
Power (5)



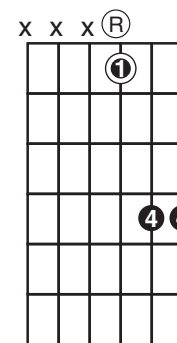
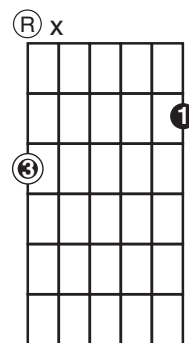
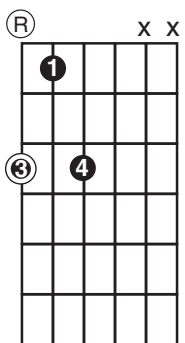
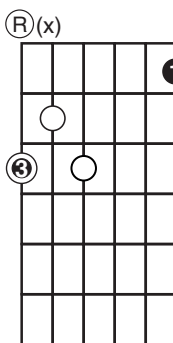
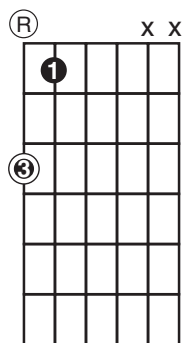
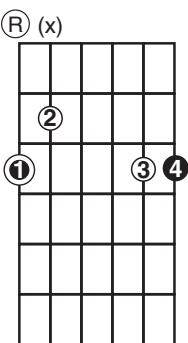
A Form



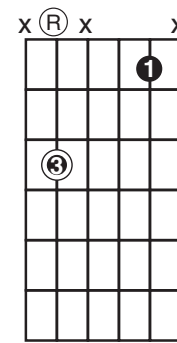
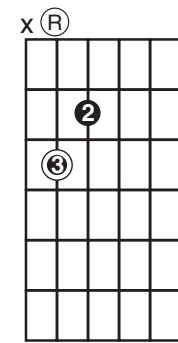
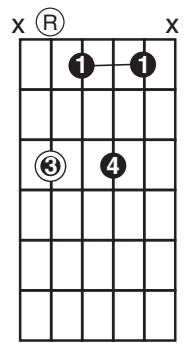
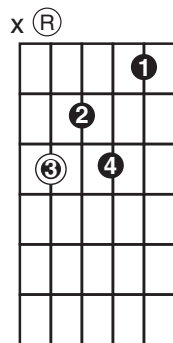
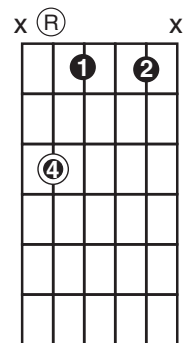
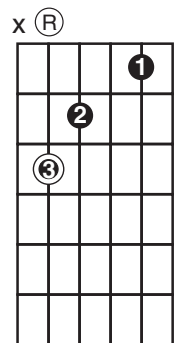
D Form



G Form



C Form



MORE ABOUT CHORDS

LESSON 5

CHORD NAMES Many people know how to PLAY their chords but don't have a clue what they're called. What good is that? You must always know the name of every chord you play. Complex chords can have very long names containing many words and numbers which we will learn about later. **CHORD NAMES HAVE 3 PARTS** (but 3rd part you must figure out yourself).

ROOT NOTE

Bb

TYPE of CHORD

Maj 7

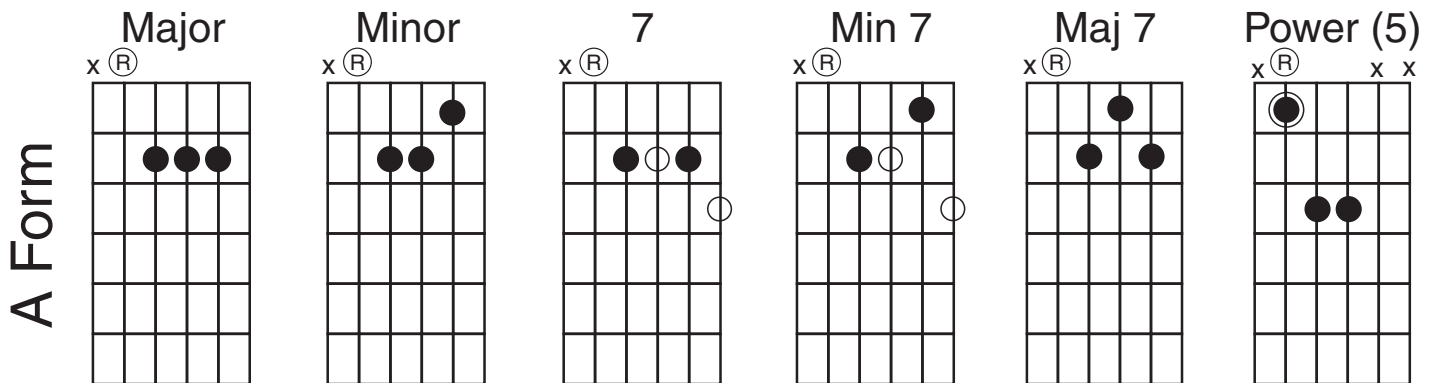
FINGER FORM to play it in

A FORM

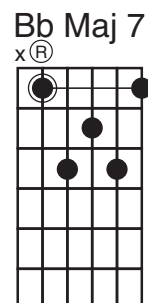
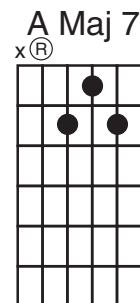
The best way to figure out how to play a chord from it's name is to **WORK BACKWARDS** through the name. Start with the **FORM** (if given) then look at the **TYPE** of chord. Finally you must turn it into a **BAR CHORD** and move it up the neck until you reach the fret (on the correct string) where the actual **ROOT NOTE** is.

SO HOW DO YOU PLAY **Bb MAJ7** in the **A FORM**?

WORK BACKWARDS. FIRST YOU NOTICE IT'S IN THE A FORM SO IT MUST BE IN THIS ROW.



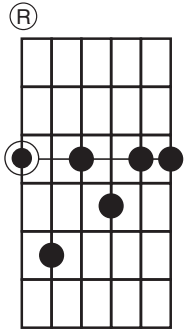
Then you notice it's a MAJ7 so it has to be in this vertical column. So this is going to be your basic finger shape. But since you're in A Form and your Root Note is on the A string, this would be A MAJ7 in the A Form. To play Bb MAJ7 you would have to turn it into a Bar Chord and move it all up one fret so the root becomes Bb. □ □



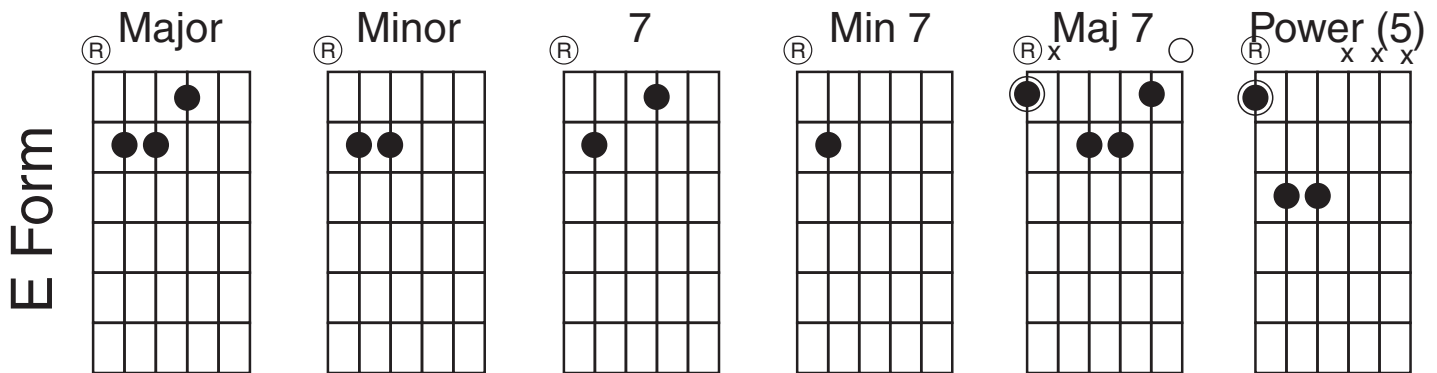
...AND SO HERE IT IS
Bb MAJ7 in the A FORM

So now we have learned how to play a chord by looking at it's name. There is also a REVERSE skill you need to acquire. You will also need to be able to

FIGURE OUT A CHORDS NAME BY LOOKING AT IT'S FINGER SHAPE

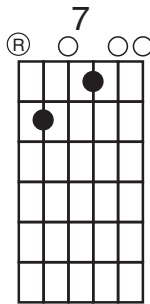


How do you figure out the name of this chord? WORK BACKWARDS. What FORM is it in? Notice where it's ROOT NOTE is. It's on the low E String. Which forms have their root on the low E String? E FORM and G FORM. Since the root is on the same fret with the bar, it must have started out as an open string as we see in E FORM. So it must be in this row



Now, what TYPE of chord is it? Picture the bar as the open strings and you will see that it has the shape of a 7 CHORD in the E FORM. But here the root is the open low E String so this chord would be

E7 in the E FORM



But our chord in question has been turned into a BAR CHORD and raised up 3 frets so it's Root Note has moved to the 3rd fret on the low E String which is a G note. So the name of our chord would be

G7 in the E FORM

DEVELOP THIS SKILL AND YOU WILL BE ABLE TO LEARN SONGS BY WATCHING OTHER PEOPLE PLAY EITHER LIVE OR ON VIDEO ! YOU'LL BE ABLE TO TELL WHAT CHORDS THEY'RE PLAYING JUST BY WATCHING WHERE THEY PLACE THEIR FINGERS ON THE NECK !

So now you UNDERSTAND the basics about chords. We have much more interesting stuff to learn later on. But now you have to learn to actually

PLAY THESE CHORDS !

PLAY THEM CLEANLY...PLAY THEM QUICKLY...WITH THE BEAT !

LESSON 6

PICKING, STRUMMING and PRACTICING

Chords are PITCH PATTERNS and like any "words" must be known 4 ways...THINK, READ, HEAR, SPEAK. You now KNOW something about chords. Later on we will be talking about READING and EAR TRAINING them. Now we must master actually PLAYING them so the guitar can SPEAK. This requires PRACTICE.

HERE IS A VERY USEFUL METHOD FOR PRACTICING CHORDS THAT COULD ALLOW YOU TO PROGRESS AT TWICE THE RATE OF OTHER FOLKS !

The SAD TRUTH about chords is this...

PLAYING CHORDS IS THE EASY PART !
SWITCHING FROM CHORD TO CHORD
WITH THE BEAT IS THE PROBLEM ! So...

PRACTICE CHORDS in PAIRS It's the SWITCHING ya' gotta' work on.

BACK and FORTH every 4 BEATS 80% of Music is counted in even numbers like 2 and 4.

NON-STOP for 5 MINUTES How long is a song? You develop strength and speed from CONSTANT REPETITION.

WITH DRUMS !!!!!!!!!!!!!!! ALL MUSIC HAS RHYTHM. If you're practicing without your DRUM MACHINE on, you're basically WASTING YOUR TIME !

SLOW as you NEED Set yourself a REALISTIC time limit to get your fingers in place. Then CHALLENGE yourself by gradually increasing the TEMPO.

THINK AHEAD It's too late to think about the chord you're actually playing. You'll fall behind the beat. Force your mind ahead to the next chord coming up.

LISTEN to YOURSELF Make sure your chords are coming out CLEAN without muted or buzzing notes. The chord has to SOUND good !

GET A MECHANICAL TIME KEEPER !

IT'S POINTLESS TO PLAY WITHOUT ONE !

The DRUM MACHINES in these KEYBOARDS are better and more fun than a METRONOME. Get one with GOOD SOUND QUALITY and 100 RHYTHMS (about \$100 brand new)

STRUMMING CHORDS: We've talked alot about what your FRETTING hand does when we play chords. But how do you ARTICULATE these chords RHYTHMICALLY with

your STRUMMING hand? We employ the same regular UP/DOWN thinking that we use when plucking individual notes as in our CHROMATIC SCALE exercise. The only way your muscles can really keep a steady rhythm is by moving back and forth regularly like the PENDULUM of a CLOCK. The TIME DURATION separating these strokes can best be expressed using these common and familiar RHYTHM NOTATION symbols. WHEN COUNTING IN 4...

◦ **WHOLE NOTE** lasts for 4 BEATS. Strum DOWN on BEAT 1. Let chord ring for one whole measure (count of 4)

♪ **HALF NOTE** lasts for 2 BEATS Strum DOWN on BEAT 1 and DOWN again on BEAT 3. Each strum lasts 2 beats.

♪ **QUARTER NOTE** lasts for 1 BEAT Strum 4 DOWN strokes...1 per beat.

♪ **EIGHTH NOTE** lasts for 1/2 BEAT Begin hitting UP and DOWN strokes. Count 1 + 2 + 3 + 4 +

♪ **SIXTEENTH NOTE** lasts for 1/4 BEAT DOUBLE TIME UPS and DOWNS. Count 1 e + u 2 e + u 3 e + u 4 e + u

YOU CAN USE THESE DIFFERENT STRUMMING RATES TO HELP YOU LEARN TO SWITCH YOUR CHORDS FASTER AND FASTER !

○ = STRUM * = "BAIL" = TIME YOU HAVE TO FORM 2nd CHORD		
BEAT	STRUM TYPE PER MEASURE	MEASURE 1 STRUM and BAIL on 1st CHORD ↓ MEASURE 2 2nd CHORD MUST BEGIN HERE
4	1 DOWN	①*.....2.....3.....4.....①*.....2.....3.....4.....
2	2 DOWN	① 2 ③*.....4.....① 2 ③*.....4.....
1	4 DOWN	① ② ③ ④*.....① ② ③ ④*.....
$\frac{1}{2}$	DOWN/UP	①+ ②+ ③+ ④+* ①+ ②+ ③+ ④+*
$\frac{1}{4}$	DOUBLE TIME	①+U ②+U ③+U ④+U ①+U ②+U ③+U ④+U

YOU KEEP CHALLENGING YOURSELF TO SWITCH CHORDS FASTER AND FASTER BY STRUMMING MORE AND MORE OFTEN, CUTTING THE TIME YOU HAVE TO SWITCH IN HALF EACH TIME.

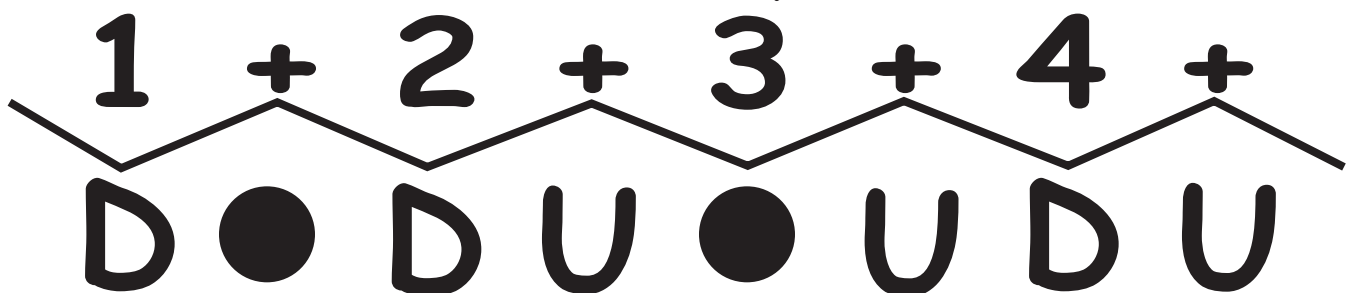
STRUMMING PATTERNS

But songs would sound monotonous if our strumming was so regular. We like to hear articulations of different durations to add interest to the strumming...mixing say Quarter Notes and Eighth Notes in the same measure. How can you do that if your hand is going back and forth so regularly at one pace? The answer is that even though your hand is going back and forth, you don't always hit the strings

on every stroke. A missed stroke allows the string to ring longer creating a longer duration articulation. Every song has it's own strumming pattern and part of learning the song is figuring out how it's particular pattern works. There are probably 20 or so patterns that you hear all the time in different songs. Here is a very common example...

First, you must determine the overall speed that your hand must move at in order to accommodate the quickest duration articulation that you will need in the pattern. Is it the HALF beat (EIGHTH NOTES) or maybe the QUARTER beat (SIXTEENTH NOTES)? In our example, your hand is going up and down at the rate of the HALF BEAT....and so we count 1 + 2 + 3 + 4 + (the + being the half way point between the full beats).

D=Contact Down Stroke U=Contact Up Stroke ●=Missed Stroke



This is just one example. You may need help from friends or other teachers to develop a feel for other strumming patterns. This may drive you crazy at first but stick with it. Later on you add subtillies like individually articulated Bass notes, accents and "pick clicks". There is also FINGER STYLE playing where each of your fingers acts as a separate pick to do fancy "finger picking" style songs.□

MUSIC BY THE NUMBERS

LESSON 7

Remember I said that you had 50 PITCH PATTERNS to learn to master Popular Music? What I meant was there are about 50 basic patterns at work. But each of these patterns can be played in any KEY. What does it mean for something to be in a "KEY"? This is a complex issue with many facets which we will be uncovering gradually as we go through this entire program. For now, think of it this way. No song has just ONE note in it. A song is a swirling mass of notes. But somewhere in that mass is ONE NOTE that is acting as the "anchor" for the whole song. I call it the NUCLEUS NOTE. Once you have identified which of the 12 notes is acting as that nucleus note, you know which key you are playing in. Since there are 12 notes, there are 12 keys. (Later you will learn that there are also MINOR and MODAL keys as well but this is the way to look at it now) If all 50 patterns can be played in all 12 keys, then the actual number of patterns you have to learn is 600 ! But don't panic! It's not so bad. Look at your MUSICAL SLIDERULE. What you see is that each one of these 50 patterns is basically structured the same way no matter which key you play it in. You will come to understand that these patterns are SPATIAL RELATIONSHIPS between notes that can be translated (or TRANSPOSED) into any of the 12 keys. So wouldn't it be nice if we had an ALTERNATIVE SYSTEM for naming the notes that treated music as if there was only ONE UNIVERSAL KEY to play in? Then we could simply look at the structure of each pattern without having to initially memorize it's 12 alphabetic variations. We have such a system. It's called THE MUSICAL NUMBER SYSTEM. But how does it work? Many people waste decades worth of time trying to understand music without getting this system sorted out once and for all. So here's my question...

YOU'VE HEARD OF C7 CHORDS....1-4-5 PROGRESSIONS...MAJOR 3rds...PERFECT 5ths
YOU know there is a NUMBER SYSTEM to NAME THE NOTES making it 11 times easier to memorize PITCH PATTERN SPELLINGS (because it treats music like there is 1 KEY instead of 12). But how does this system work? It's not quite how you might think !

TO WHAT DO THESE NUMBERS CORRESPOND?

THEY NUMBER THE NOTES IN A MAJOR DIATONIC SCALE

WHAT IS THAT? IT'S THAT FIRST THING THEY TRIED TO TEACH US BACK IN GRADE SCHOOL..



A MAJOR SCALE (like any scale) IS A SEQUENCE OF NOTES SEPARATED BY SPECIFIC INTERVALS.

INTERVALS: Our first look at one of the HUGE concepts in MUSIC THEORY. As we said, music is really a set of SPATIAL relationships. It isn't so much what NOTES you play as it is HOW FAR APART those notes are spaced from one another in PITCH.

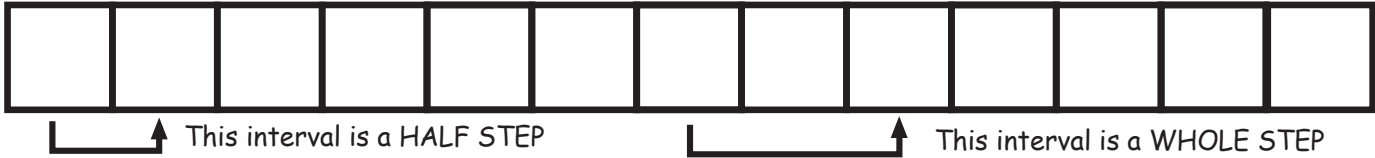
INTERVALS ARE THE PITCH "DISTANCES" BETWEEN THE NOTES.

SCALES: What is a SCALE? Most of you would say that a scale is a SEQUENCE OF NOTES and that is true. But if you're not including the concept of INTERVALS as part of your definition of what a scale is, you're not seeing the whole picture. Scale notes are mostly separated by HALF STEPS or WHOLE.

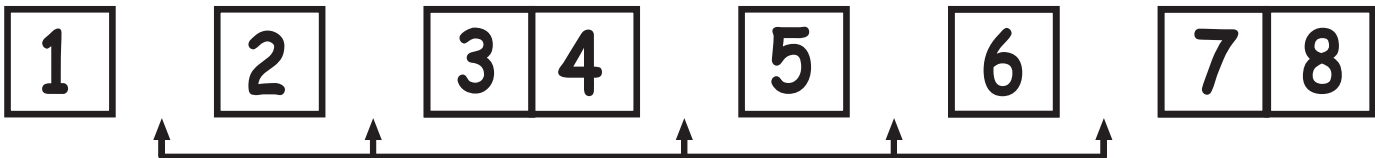
A SCALE IS A SEQUENCE OF NOTES SEPARATED BY SPECIFIC INTERVALS.

For example:

A CHROMATIC SCALE IS A SEQUENCE OF 13 NOTES ALL SEPARATED BY HALF STEPS.



BUT A MAJOR SCALE GOES WHOLE-WHOLE-HALF-WHOLE-WHOLE-WHOLE-HALF
or in other words **MOSTLY WHOLE STEPS EXCEPT BETWEEN 3 & 4 and 7 & 8**.

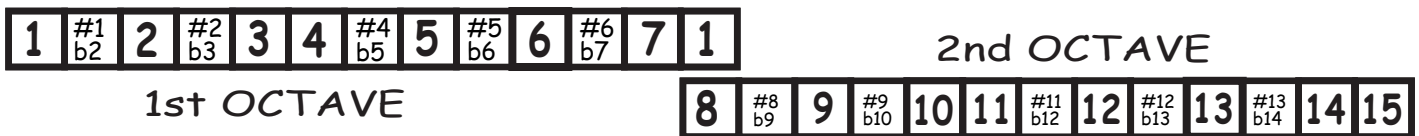


But what about these in between notes? They are important notes too. So it would be a pretty stupid number system if there weren't some way to refer to these notes by number. What we do is **SHARP AND FLAT THE NUMBERS** the same way we do the letters in the **ALPHABET SYSTEM** !



SOMETIMES YOU WILL SEE NUMBERS LIKE 9-10-11-13.

Think of these for now as notes that are up in a **SECOND OCTAVE**. The numbers continue to follow the **MAJOR SCALE INTERVAL PATTERN**. So 8 is the same as 19 is the same as 2 and it is a whole step above 810 and 11 are a half step apart because they are the same as 3 and 4etc.



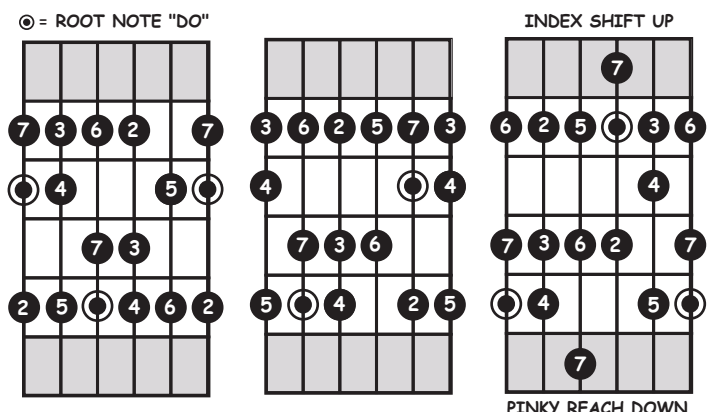
HERE ARE SOME OF YOUR MOST IMPORTANT PITCH PATTERNS EXPRESSED AS NUMBERS.

MAJOR CHORD...1-3-5
MINOR CHORD...1-b3-5
7 CHORD...1-3-5-b7
"POWER CHORD"...1-5-8

NATURAL MINOR SCALE...1-2-b3-4-5-b6-b7-8
PENTATONIC SCALE...1-2-3-5-6-8
MINOR PENTATONIC...1-b3-4-5-b7-8
MOST COMMON CHORD PROGRESSION...1-4-5

THE MAJOR DIATONIC SCALE IS THE CENTRAL PITCH PATTERN IN WESTERN MUSIC

and so it is the pattern they chose to measure all the other ones by. It is also the next **FINGER PATTERN** that you need to begin to practice and ultimately master. There are many different fingerings for this scale but here are the first 3 you would want to know (they are the easiest.... 2 have no **FINGER SHIFTS** and the 3rd has only 1). Note that your finger pattern directly reflects the **MAJOR SCALE INTERVAL PATTERN**. These fingerings can be played anywhere on the neck to play a Major Scale in the 12 different **KEYS**. Begin to practice these every day as you have been practicing your **CHROMATIC SCALES**.



CAVEMAN MUSIC THEORY

LESSON 8

Most people don't really know much about music or our 12 note pitch system. We all "know what we like" but we kind of take the rest for granted. But if you're going to be a **PLAYER**, you need to know music **INTIMATELY**....even things you never even thought to ask yourself. How about this question...

WHERE DID THIS WHOLE 12 NOTE SYSTEM COME FROM ANYWAY?

PITCH is a **CONTINUUM**. There are an infinite number of musical pitches all infinitely close together. Why did we humans decide to "chop it up" into 12 note octaves?

HARMONICS

We discovered **HARMONICS**. Cavemen probably began to notice these spots along a string that produce very loud and clear overtones. They occur at very precise **MATHEMATICAL** divisions of the string. Many guitar players think of them as some funny trick that the guitar will do but they are actually quite profound. Our whole system of pitch is based on them.

THE LOUDEST HARMONICS OCCUR DIRECTLY OVER THE 12th, 7th and 5th FRET.

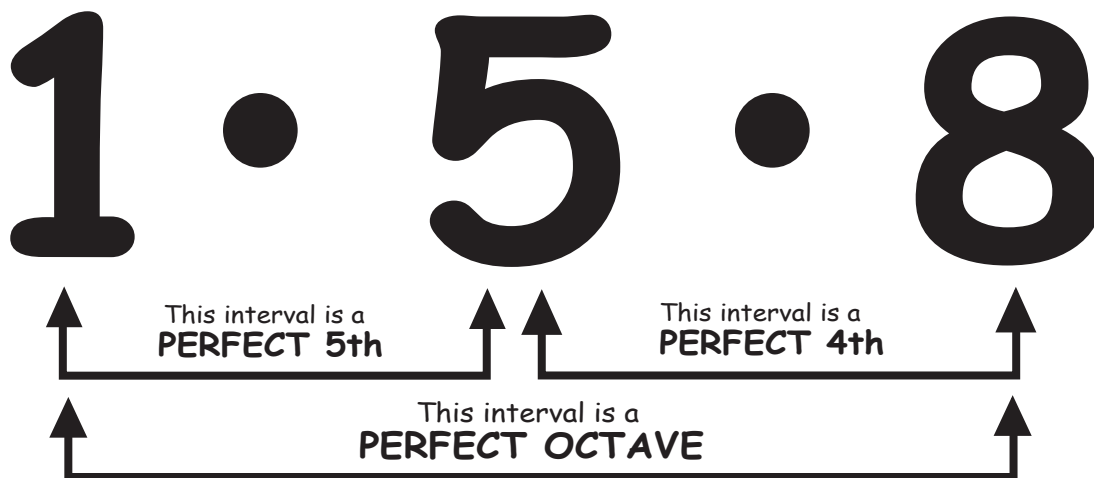
WHAT 3 NUMBER/NOTES CAN BE PRODUCED HARMONICALLY ON ANY STRING ?

Now that we have discovered **HARMONICS**, what pitch relationships do they illuminate to our ears?

12th FRET HARMONIC

7th FRET HARMONIC

5th FRET HARMONIC



WHAT INTERVALS SEPARATE THESE NOTES?

This is the entire history of music right here. These are the 3 notes that kind of hold the whole language together. They are present in almost every moment of music you have ever heard. These notes and intervals are the first ones we discovered ...they led us to the creation of the whole 12 note system ...and they are still the most important pitch relationships in music today! The **OCTAVE** is a huge concept. The 5 is so huge it's called the **DOMINANT**. Think of how often you've heard the 5th mentioned ...**CIRCLE OF 5ths** ...**1-4-5 PROGRESSION** ...**5 CHORDS** (Power Chords are spelled 1-5-8) ...most chords contain 1-5-8 ...most scales ...

1-5-8 IS A PITCH PATTERN AND MUST BE KNOWN 4 WAYS

THINK

You now know some facts about this pitch pattern. There is a lot more to know as well and we will be returning to this concept time and time again as we go through this entire program.

READ

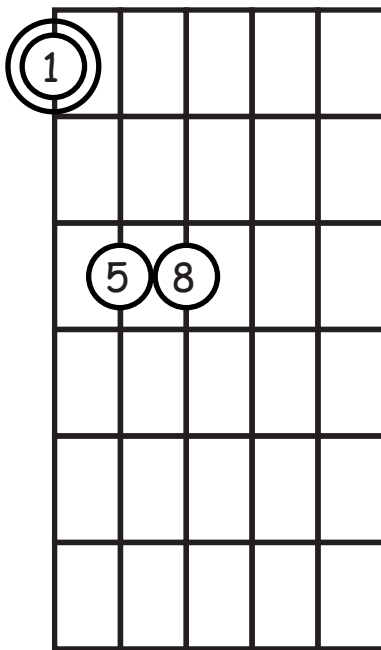
Our very next lessons are about NOTATION and we will be learning how to recognize all of our pitch patterns when we see them on paper including 1-5-8.

HEAR

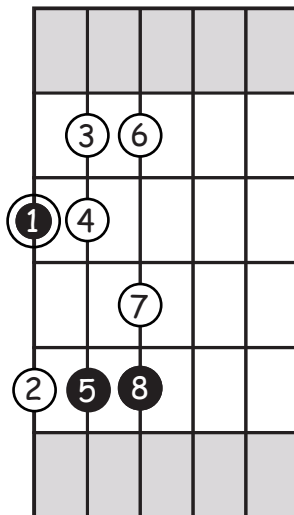
A way to begin EAR TRAINING a particular pattern is to think of some popular piece of music that features the pattern in question. For 1-5-8 think of *ALSO SPRACH ZARATHUSTRA* by R. STRAUSS (Theme to movie 2001)

SPEAK

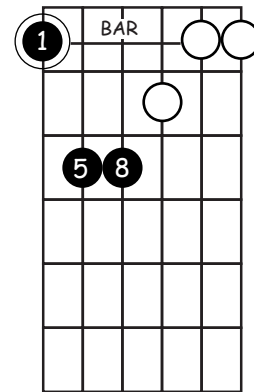
Here's the FINGER PATTERN for 1-5-8. Notice that it's the same shape as a "POWER CHORD". Notice how that same combination of notes winds up in so many of our other common finger patterns.



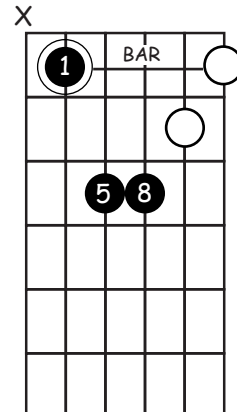
MAJOR SCALE



E FORM MAJOR CHORD



A FORM MINOR CHORD



LESSON 9

But in the end it's not that bad. Your friends had trouble understanding it because it's another one of those things that they don't explain very well. And often we are forced into it when we're too young or don't understand the LANGUAGE well enough in general. Remember, you learned to THINK and SPEAK in the English language long before you learned to READ it. You first have to learn what the various SYMBOLS mean and then practice using them by starting out with very simple music and gradually working your way up to more and more complex music over a period of several exercise book levels and possibly several years of daily practice. Deciphering the multiple notes in CHORDS makes reading for POLYPHONIC INSTRUMENTS (like GUITAR or KEYBOARD) even more challenging.

7 OCTAVES EQUALS
26 LINES AND SPACES

LEAVE THESE OUT

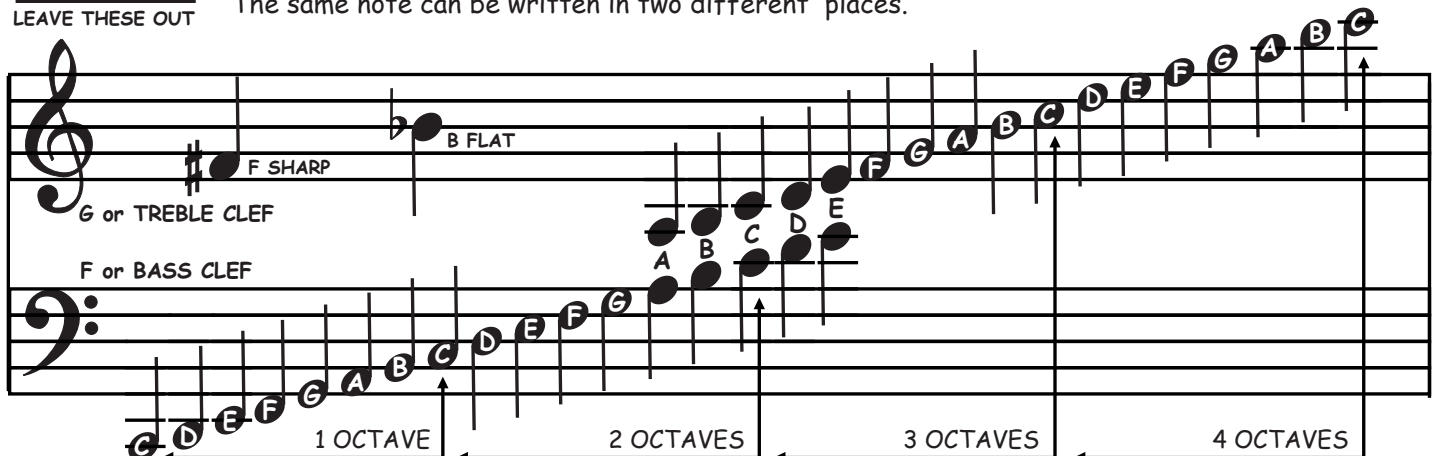
REDUCED DOWN TO
11 LINES AND SPACES

MIDDLE

A B C D

LEAVE THESE OUT

The GRAND STAFF comfortably notates 4 OCTAVES starting 2 LEDGER LINES below the BASS CLEF up to 2 lines above the G CLEF. You might recall that the TOTAL PITCH RANGE of the GUITAR is about 4 OCTAVES. Notice how the LEDGER LINES between the clefs work. The same note can be written in two different places.



This is how the **ACTUAL PITCHES** of your **OPEN STRINGS** would be notated on the **GRAND STAFF**.

GUITAR NOTATION is written 1 **OCTAVE HIGHER** THAN IT **SOUNDS** to better center it on the **G CLEF**.

So this is how the **OPEN STRINGS** actually appear in **GUITAR NOTATION** on the **G CLEF**.

Notes stacked on top of each other are played **SIMULTANEOUSLY** as in a **CHORD**. Here is how some common **CHORDS** would appear.

Chord diagrams and musical notation for G, C, D, Em, and Am chords.

There is more to reading music than just understanding how the **SYMBOLS** work. You have to practice recognizing them until you don't even need to think about it anymore at all. When you read the English language, you don't have to stop on every single word and letter to understand what a sentence means. That is because you are so familiar with the **SHAPES** of certain words and phrases that you recognize them at a glance. It may take you quite some time and disciplined practice to learn to read music as well.

TABLATURE

TABLATURE is a very popular style of notation specifically for the guitar. Many people perceive it as easier to read than **STANDARD NOTATION** but that is mainly because it conveys so much less information. **TAB** only tells you what to do with your fingers. It doesn't tell you what notes you're playing or anything about the **RHYTHM** of the song. It is virtually impossible to learn a song you have never heard before just from the **TAB**. None the less, it still has it's place.

TABLATURE has 6 **LINES** ...one for each string on the guitar. We see **NUMBERS** on these lines indicating which **FRETS** to play on which strings. Numbers stacked on top of each other are played simultaneously as in a **CHORD**. There are various other **SYMBOLS** indicating certain types of **ARTICULATIONS** such as **HAMMER-ONS**, **PULL-OFFS**, **BENDS** and **SLIDES** (These **ARTICULATIONS** are explained in **LESSONS 24** and **25**). A more complete listing of the many **TABLATURE SYMBOLS** can generally be found in any issue of the popular guitar magazines (**GUITAR PLAYER**, **GUITAR WORLD**, **GUITAR ONE**, etc.).

STRINGS									
E									
B									
G									
D									
A									
E									

FRET NUMBERS for a **C MAJOR SCALE**

TAB for a **C MAJOR CHORD**

HAMMER-ON

PULL-OFF

SLIDE UP
DOWN

WHOLE STEP
BEND
AND RELEASE

HALF STEP
PRE-BEND
RELEASE

MUTED
NOTE

PALM
MUTE

MORE ABOUT NOTATION

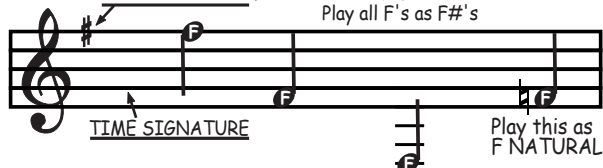
KEY SIGNATURES

LESSON 10

music, ONE of the 12 notes is acting as the "NUCLEUS NOTE". Figure out which note that is and you know what KEY you're in. Each KEY has what is known as it's KEY SIGNATURE. Since the MAJOR DIATONIC SCALE is the CENTRAL PITCH PATTERN in Western Music, we base our ideas of "KEYNESS" (or TONALITY) on that scale. The KEY SIGNATURE tells you which SHARPS or FLATS you must play in order to execute a MAJOR DIATONIC SCALE starting on a given note. In certain KEYS we call the notes by their SHARP names and in others we call them FLATS. This is so we can avoid ALPHABETIC REDUNDANCY in the KEY. Grab your SLIDERULE. Take a look at a G MAJOR SCALE.



KEY SIGNATURE for KEY OF G MAJOR = 1# = F#
Play all F's as F#s



KEY SIGNATURE for KEY OF D $\flat$ MAJOR = 5 b's
Play all B's, E's, A's, D's and G's as b's



There are 12 MAJOR KEYS that you play in in Western Music because there are 12 NOTES (Later we will learn about MINOR KEYS and such).

What does it mean to PLAY IN A KEY? In every piece of conventional music, ONE of the 12 notes is acting as the "NUCLEUS NOTE". Figure out which note that is and you know what KEY you're in. Each KEY has what is known as it's KEY SIGNATURE. Since the MAJOR DIATONIC SCALE is the CENTRAL PITCH PATTERN in Western Music, we base our ideas of "KEYNESS" (or TONALITY) on that scale. The KEY SIGNATURE tells you which SHARPS or FLATS you must play in order to execute a MAJOR DIATONIC SCALE starting on a given note. In certain KEYS we call the notes by their SHARP names and in others we call them FLATS. This is so we can avoid ALPHABETIC REDUNDANCY in the KEY. Grab your SLIDERULE. Take a look at a G MAJOR SCALE.

Notice you play only one SHARPED or FLATTED note. In this KEY that note would be called F# because to call it G $\flat$ would cause the scale to have both a G and a G $\flat$ and no F of any kind and that would be weird! So the KEY SIGNATURE for the KEY of G contains 1# and that is F#. So a single SHARP SIGN is placed on the top F line of the G CLEF and you have to remember to play F# every time you see an F note anywhere on the STAFF unless it has a NATURAL SIGN (n) in front of it.

There are some other more specific rules you will learn about later but this is the basic gist of it and all 12 KEY SIGNATURES work this same way. Some ENHARMONIC KEYS can be spelled 2 different ways. Notice that some KEYS contain 6 or 7 SHARPS or FLATS even though there are only 5 black keys on a piano. That's because, to avoid redundancy, they contain note names like B# (C) or C $\flat$ (B) and E# (F) or F $\flat$ (E).

KEY SIGNATURES help us avoid clutter on the page particularly in KEYS with large numbers of SHARPS or FLATS (like the KEY OF D $\flat$ with it's 5 FLATS). Notes outside of the KEY SIGNATURE that are sharped or flatted are called ACCIDENTALS. There are also DOUBLE SHARPS and DOUBLE FLATS. F DOUBLE SHARP is actually a G for example.

X = DOUBLE SHARP

bb = DOUBLE FLAT

Every MAJOR KEY has it's RELATIVE MINOR KEY rooted on it's 6th DEGREE.

We'll learn all about this in LESSON 22. CLOSELY RELATED KEYS share many notes. You'll learn all about this when we look at a CYCLE OF 5ths in LESSON 31.

HERE ARE THE KEY SIGNATURES FOR ALL 12 MAJOR KEYS, THEIR RELATIVE MINOR KEYS AND ENHARMONIC SPELLINGS.

SHARP KEYS		FLAT KEYS	
G or E$\flat$ = 1 SHARP G A B C D E F# G	C or A$\flat$ = 0 SHARPS or FLATS C D E F G A B C	Ab or F$\flat$ = 4 FLATS A $\flat$ B $\flat$ C D $\flat$ E $\flat$ F G A $\flat$	
D or B$\flat$ = 2 SHARPS D E F# G A B C# D	B or G# = 5 SHARPS B C# D# E F# G# A# B	C$\flat$ or A$\flat$ = 7 FLATS C $\flat$ D $\flat$ E $\flat$ F $\flat$ G $\flat$ A $\flat$ B $\flat$ C $\flat$	E$\flat$ or C$\flat$ = 3 FLATS E $\flat$ F G A $\flat$ B $\flat$ C D E $\flat$
A or F# = 3 SHARPS A B C# D E F# G# A	F# or D# = 6 SHARPS F# G# A# B C# D# E# F#	G$\flat$ or E$\flat$ = 6 FLATS G $\flat$ A $\flat$ B $\flat$ C $\flat$ D $\flat$ E $\flat$ F G $\flat$	B$\flat$ or G$\flat$ = 2 FLATS B $\flat$ C D E $\flat$ F G A B $\flat$
E or C# = 4 SHARPS E F# G# A B C# D# E	C# or A# = 7 SHARPS C# D# E# F# G# A# B# C#	D$\flat$ or B$\flat$ = 5 FLATS D $\flat$ E $\flat$ F G $\flat$ A $\flat$ B $\flat$ C D $\flat$	F or D$\flat$ = 1 FLAT F G A B $\flat$ C D E F

RHYTHM NOTATION

The horizontal aspect of the musical notation graph represents TIME. As we proceed from left to right across the page we are told WHEN to play a note and HOW LONG to hold it before we go on to the next note. We are also told when to be silent. This,

of course, is what RHYTHM is all about ..how the BEAT or PULSE moves through time. There is a set of SYMBOLS that you must understand in order to decipher the rhythmic content of the song. Many of these symbols have been familiar to us since grade school but we still don't really understand what they mean. Actually, these symbols mean different things depending on the TIME SIGNATURE so we begin there.

TIME SIGNATURES

Look at the very beginning of any printed piece of music and you always see the same things. First there is a CLEF SIGN (a G CLEF in guitar music) followed by the KEY SIGNATURE (a collection of SHARP or FLAT SYMBOLS telling you which of the 12 KEYS you

are playing in as we just discussed). Then there is the TIME SIGNATURE. These two numbers stacked on top of each other are the key to understanding the RHYTHMIC STRUCTURE of the song. The TOP number tells you how many BEATS PER MEASURE ...or how many basic beats there are in each repetition. A MEASURE is the distance between each of the vertical BAR LINES in the printed music. The BOTTOM number tells you the TYPE OF NOTE SYMBOL COUNTED AS ONE BEAT. 4 means a QUARTER NOTE ...2 means a HALF NOTE ...8 means an EIGHTH NOTE etc. Add up the beat value of all the notes and rests in a given MEASURE and it will equal the number of beats in the TIME SIGNATURE.

KEY SIGNATURE

= 4 BEATS PER MEASURE

= A QUARTER NOTE LASTS 1 BEAT

TIME SIGNATURE

BAR LINES

◀ ONE MEASURE ▶

4 QUARTER NOTES = 4 BEATS

HERE ARE SOME OF YOUR MOST COMMON TIME SIGNATURES.

= COMMON TIME =

= CUT TIME =

A HALF NOTE LASTS 1 BEAT

= WALTZ TIME

AN EIGHTH NOTE LASTS 1 BEAT

TIME SIGNATURES WHERE THE TOP NUMBER IS 5-7-9-11 etc. ARE RARE.

NOTE SYMBOL AND NAME		BEAT VALUE IN	BEAT VALUE IN	BEAT VALUE IN	RESTS
	WHOLE NOTE	4	2	8	
	HALF NOTE	2	①	4	
	QUARTER NOTE	①	1/2	2	
	EIGHTH NOTE	1/2	1/4	①	
	SIXTNTN. NOTE	1/4	1/8	1/2	

There are 32nd NOTES as well but they are so quick that you don't see them too often. The bottom number of the TIME SIGNATURE could therefore also be 16 or 32.

A DOT IN FRONT ADDS HALF AGAIN THE VALUE OF THE NOTE OR REST TO ITSELF.

3 BEATS 1 and 1/2 BEATS 3/4 of a BEAT

TWO NOTES OF THE SAME PITCH TIED TOGETHER EQUALS ONE NOTE WITH THEIR COMBINED BEAT VALUE.

PLAYED AS ONE NOTE LASTING 2 and 1/2 BEATS

SILENCE and THE NOTE

LESSON 11

CONGRATULATIONS! You are finished with PART 1 of the program! You know all the basic things about the LANGUAGE of MUSIC. Before we go on, take a moment and try to answer the questions in this little quiz. If you can't, you should probably go back and do some review. You won't understand the stuff in PART 2 if you don't get these basic concepts.

GOOD LUCK !

Why do so few people stick with their instrument through their whole lives?

What 2 things do ya' gotta' do to be a REAL PLAYER?

What are the 6 MAIN AREAS of MUSIC?

What is PITCH?

How is TIMBRE different than PITCH?

How many NOTES are there in our system of PITCH?

How many TYPES of PITCH PATTERNS can you name?

What are the 4 WAYS TO KNOW EACH PITCH PATTERN?

What is EAR TRAINING and why is it so IMPORTANT?

What are the NAMES of your STRINGS and why is your guitar TUNED that way?

What is a CHROMATIC SCALE and how do you play them ANYWHERE on the neck?

How many FRETS on each string does it actually require to play CHROMATICALLY?

How do INDEX SHIFTS and PINKY SHIFTS work?

What is a CHORD and how many TYPES of chords are there?

How many MINOR CHORDS are there and how many ways are there to play one?

What does the bar in a BAR CHORD do?

How important is it to own a DRUM MACHINE?

How do you use STRUMMING PATTERNS to learn to SWITCH chords more quickly?

Why is the NUMBER SYSTEM easier than the letters when "spelling" pitch patterns?

What are the NUMBER and INTERVAL patterns for a CHROMATIC SCALE?

What are the NUMBER and INTERVAL patterns for a MAJOR DIATONIC SCALE?

How are MAJOR and MINOR CHORDS spelled and how do they DIFFER?

How many ways can you play a MAJOR SCALE on your guitar?

What are the 3 MOST IMPORTANT NOTES in music and how far apart are they?

In NOTATION, what areas of music do the 2 axes of the musical GRAPH represent?

How do you notate SHARPS and FLATS and how do KEY SIGNATURES work?

How does GUITAR NOTATION differ from standard GRAND STAFF notation?

How do TIME SIGNATURES work and how do they effect note DURATIONS?

What is the most COMMON TIME SIGNATURE?

PART 2

The 6 LEVELS OF PITCH THEORY

Now that you know all the basic stuff about the language of music, it's time to begin acquiring a VOCABULARY. It's time to begin soaking up these 50 PITCH PATTERNS we keep talking about and ultimately learn them all 4 ways:

THINK - READ - HEAR - SPEAK

We are going to do this in ORDER OF COMPLEXITY in 6 different increments starting with the simplest patterns and working our way up to the most complex.

What are the simplest patterns in music? In my way of looking at it, the simplest patterns are so simple that

THERE ISN'T ANYTHING THERE YET AT ALL !

and so the 1st level of complexity to examine is

① SILENCE

What a concept! You didn't come here thinking you were going to learn about SILENCE, but if you think about it ...

WHERE WOULD MUSIC BE WITHOUT SILENCE?

There's a real YIN-YANG thing going on here. The universe can't exist without MATTER and ANTIMATTER. The battery has it's POSITIVE and NEGATIVE pole ...it's like DAY and NIGHT ...BLACK and White ... It's the "canvas" you "paint" the music on ...it's the RESTS ...the pause between PHRASES ...

YA' JUST CAN'T HAVE MUSIC WITHOUT SILENCE !

That's all there is to say about that. So what's the next level of complexity in music?

UP OUT OF THE SILENCE ERUPTS ONE NOTE. THIS IS THE LEVEL I CALL ..

② THE NOTE

SURPRISE AGAIN! There aren't any songs that just have one note in them, but this is a good time to talk about another CENTRAL CONCEPT in music

Remember, a song is a swirling mass of notes. But somewhere in that mass is **ONE NOTE** that is acting as the **NUCLEUS** of the song or pattern and in many ways, all the other notes are whirling around that note like the electrons orbit the nucleus of the atom or the planets orbit the sun. Since this is such a central concept in music, a lot of different TERMS have arisen in different contexts to describe this NUCLEUS NOTE. Often times people don't see that these are all just different ways of saying the same thing.

KEY Once you know which of the 12 notes is acting as the nucleus note, you know what KEY you're in.

TONIC This is perhaps the official academic term. You will see it in text books and magazines.

ROOT Every chord has a nucleus note. Every scale has one too. Indeed the whole song has one.

"DO" SOLFEGGIO is yet a 3rd way of naming the 12 notes using sung syllables. Here we call the root DO.

1⁽⁸⁾

In the musical number system, we call the root note #1. Since the OCTAVE note is the same letter, you can also call the root note #8.

THE FUNDAMENTAL and the concept of RESOLUTION are related subjects as well.

So, after SILENCE and THE NOTE, what are the next most complex PITCH PATTERNS?
Patterns with 2 notes are called DIADS and the PITCH DISTANCES between the notes are called

LESSON 12

③ INTERVALS

Intervals might be the most important concept in MUSIC THEORY.

As we saw before, it isn't really the NOTES you play that defines the structure of these patterns, it's HOW FAR APART they are spaced from one another. You won't really understand your more complex patterns like SCALES and CHORDS if you don't get your INTERVALS down first. You won't ever really understand music well enough to WRITE SONGS or IMPROVISE if you don't have a grasp of these most basic building blocks. So in many ways, this is actually where the work really begins. This is your first big page of pitch pattern that you actually have to sit down and MEMORIZE.

AND AS USUAL, YOU HAVE TO KNOW EACH ONE 4 WAYS !

THINK

You have to UNDERSTAND the basic facts about INTERVALS. There are several TYPES and all different SIZES. You have to MEMORIZE their SPELLINGS. One interval may be known by several different NAMES and you have to be able to associate them with their different FUNCTIONS.

READ

You have to recognize your intervals when you see them written on paper as NOTATION. When you look at 2 notes on a staff, the number of lines and spaces involved between them tells you at least the NUMBER name of the interval that separates the 2 notes.

HEAR

Most importantly, you have to understand that each of these intervals SOUNDS like something and it sounds different from all the other ones. You must eventually EAR TRAIN them and MEMORIZE what they sound like. This is so important when you are learning to IMPROVISE. Initially you try to associate an interval with a familiar MELODY that features the interval.

SPEAK

Finally, you have to know the FINGERINGS for each interval. There are generally more than one. Since intervals get their names from the MUSICAL NUMBER SYSTEM, it is very useful to look at the fingering for a MAJOR DIATONIC SCALE in order to learn the fingerings for your intervals.

There are 2 overall size categories for intervals. SIMPLE INTERVALS are ONE OCTAVE AND SMALLER. COMPOUND INTERVALS are larger than an octave and are generally thought of as simple intervals that have been EXPANDED by X number of octaves. It is beyond the scope of this program to go into compound intervals.

EXAMPLES	FORMULAS		FINGERING
	HALF STEPS	WHOLE STEPS	
PERFECT OCTAVE	12	6	
MAJOR 7th	11	$5\frac{1}{2}$	
MINOR 7th AUGMENTED 6th	10	5	
MAJOR 6th DIMINISHED 7th	9	$4\frac{1}{2}$	
MINOR 6th AUGMENTED 5th	8	4	
PERFECT 5th	7 HALF STEP ABOVE MIDDLE	$3\frac{1}{2}$	
DIMINISHED 5th AUGMENTED 4th	6 MIDDLE OF OCTAVE	3	
PERFECT 4th	5 HALF STEP BELOW MIDDLE	$2\frac{1}{2}$	
MAJOR 3rd DIMINISHED 4th	4	2	
MINOR 3rd AUGMENTED 2nd	3	$1\frac{1}{2}$	
MAJOR 2nd WHOLE STEP TONE	2	1	
MINOR 2nd HALF STEP SEMITONE	1	$\frac{1}{2}$	
MICRO TONES	1/4 STEP		Microtonal Music
PERFECT UNISON	0	0	

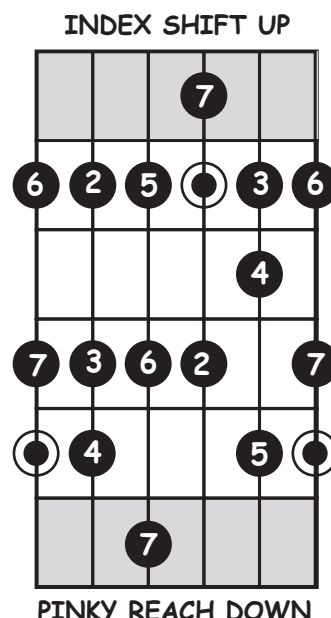
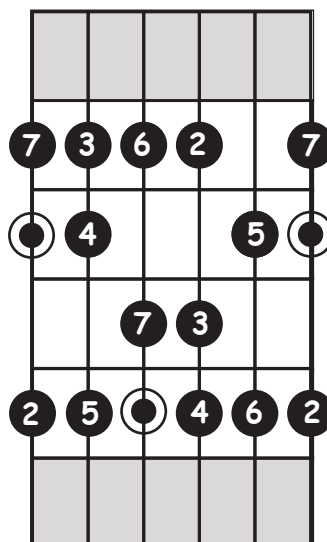
MORE ABOUT INTERVALS

LESSON 13

INTERVAL NAMES COME FROM THE MUSICAL NUMBER SYSTEM BASED ON THE MAJOR SCALE.

1	#1 b2	2	#2 b3	3	4	#4 b5	5	#5 b6	6	#6 b7	7	8
---	----------	---	----------	---	---	----------	---	----------	---	----------	---	---

SO ULTIMATELY, ALL YOUR INTERVAL FINGERINGS CAN BE DERIVED FROM YOUR MAJOR DIATONIC SCALE FINGERINGS. A PERFECT 4th IS THE DISTANCE BETWEEN THE 1st AND 4th NOTE OF A MAJOR SCALE SO THAT'S HOW IT'S FINGERED. SINCE THERE IS MORE THAN ONE WAY TO PLAY A MAJOR SCALE, THERE IS MORE THAN ONE WAY TO PLAY ANY GIVEN INTERVAL.



THERE ARE VARIOUS TYPES OF INTERVALS.

SIMPLE INTERVAL	ALL INTERVALS 1 OCTAVE OR SMALLER
COMPOUND INTERVAL	INTERVALS LARGER THAN 1 OCTAVE
MELODIC INTERVAL	2 NOTES PLAYED IN A SEQUENCE
HARMONIC INTERVAL	2 NOTES PLAYED TOGETHER
MAJOR INTERVAL	GO UP A MAJOR INTERVAL FROM NOTE 1 AND YOU LAND ON A NOTE THAT IS IN A MAJOR SCALE
MINOR INTERVAL	GO UP A MINOR INTERVAL AND YOU LAND ON A NOTE THAT IS NOT IN A MAJOR SCALE
AUGMENTED INTERVAL	EXPAND A PERFECT OR MAJOR INTERVAL BY A HALF STEP AND IT BECOMES AUGMENTED
DIMINISHED INTERVAL	SHRINK A PERFECT OR MINOR INTERVAL BY A HALF STEP AND IT BECOMES DIMINISHED
PERFECT INTERVAL	COME DIRECTLY FROM 1 - 5 - 8

Every interval has it's "MIRROR IMAGE". Every interval can be turned "INSIDE OUT". These are called

INVERTED INTERVALS

1	2	3	4	5	6	7	8
8	7	6	5	4	3	2	1

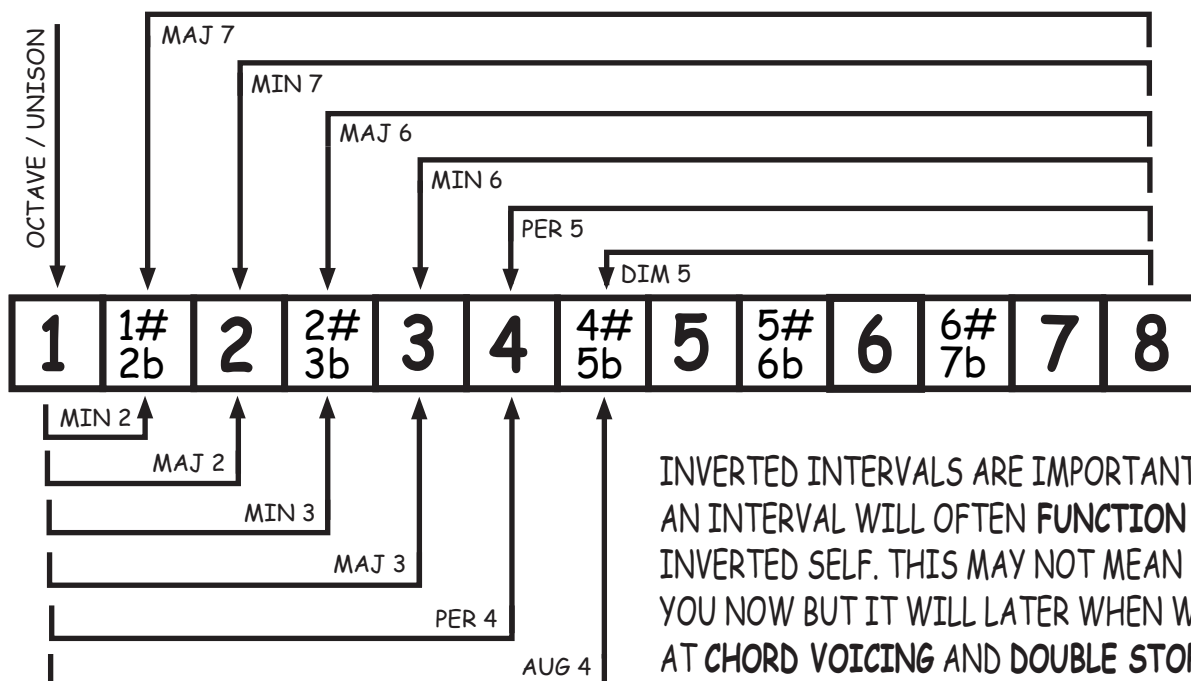
LEARN TO ASSOCIATE THESE NUMBERS. NOTE THAT WHEN ADDED TOGETHER THEY ALWAYS EQUAL 9. THEN YOU HAVE TO "FLIP- FLOP" THE MAJOR TO MINOR AND THE AUGMENTED TO DIMINISHED.

So an inverted **MAJOR 2nd** is a **MINOR 7th**.
an inverted **MINOR 3rd** is a **MAJOR 6th**.
an inverted **AUGMENTED 4th** is a **DIMINISHED 5th**.
and so on ...

PERFECT INTERVALS STAY PERFECT WHEN INVERTED

So an inverted **PERFECT 4th** is a **PERFECT 5th**.
an inverted **PERFECT OCTAVE** is a **PERFECT UNISON**.

AN INTERVAL ADDED TO IT'S INVERSION ALWAYS EQUALS 1 OCTAVE. OR, AS THE CHART BELOW SHOWS, SUBTRACT AN INTERVAL FROM 1 OCTAVE AND WHAT IS LEFT IS IT'S INVERSION.



INVERTED INTERVALS ARE IMPORTANT BECAUSE AN INTERVAL WILL OFTEN **FUNCTION** LIKE IT'S INVERTED SELF. THIS MAY NOT MEAN MUCH TO YOU NOW BUT IT WILL LATER WHEN WE LOOK AT **CHORD VOICING** AND **DOUBLE STOPS**.

④ THE MUST KNOW SCALES

A SCALE IS A SEQUENCE OF NOTES SEPARATED BY SPECIFIC INTERVALS.

The intervals that generally separate the notes in scales are **2nds**.

MINOR 2nds=HALF STEPS

MAJOR 2nds=WHOLE STEPS

AUGMENTED 2nds= 1 and 1/2 STEPS

SCALES are considered to be some of the most important PITCH PATTERNS in music particularly when considering the area of **MELODY**. What is a melody? The melody of the song is that familiar, LINEAR string of INDIVIDUAL NOTES most often SUNG BY THE SINGER. It's that part of the song that you find yourself humming or whistling. It is often considered the HORIZONTAL aspect of music. (HARMONY: how multiple notes sound when played SIMULTANEOUSLY, as in CHORDS, is considered the VERTICAL aspect.) In INSTRUMENTAL MUSIC, of course the melody will be played on an instrument. SOLOS are IMPROVISED MELODIES! Since melodies are thought to be based on scales, those of you interested in JAMMING and IMPROVISING will want to take particular care in learning your scales. There are dozens of scales in the Western World and indeed all over the planet! We will at first focus on what I call the MUST KNOW SCALES, the ones we use everyday in Western popular music. AND AS ALWAYS ...

YOU NEED TO KNOW YOUR SCALES ALL 4 WAYS ...

THINK

There is a lot to KNOW about scales. You have to MEMORIZE their SPELLINGS and their INTERVAL PATTERNS. You have to understand how they are used to form MELODIES and how they interact with CHORDS in songs.

READ

You need to be able to recognize them when you see them written on paper in both STANDARD NOTATION and TABLATURE. You should know how MODES relate to KEY SIGNATURES.

HEAR

Again and most importantly, you need to know what these scales SOUND LIKE. This is the whole thing when it comes to IMPROVISING. This will take lots of work but the rewards are incredible. You can pretty much do anything you want with music.

SPEAK

Finally, you have to master the myriad FINGERINGS for these scales. SCOTTY'S COLOR CODED FINGER CHARTS are awesome tools to use and the MODE JAM TAPE will make it much more fun. But still you have to commit yourself to some DISCIPLINED "grunt work".

THE MUST KNOW SCALES

CHROMATIC		1	#1 b2	2	#2 b3	3	4	#4 b5	5	#5 b6	6	#6 b7	7	8	
MAJOR DIATONIC IONIAN MODE		1	2	3	4	5	6	7	8	I					
LYDIAN		1	2	3	#4	5	6	7	8	IV					
MIXOLYDIAN		1	2	3	4	5	6	b7	8	V					
MELODIC MINOR		1	2	b3	4	5	6	7	8						
DORIAN		1	2	b3	4	5	6	b7	8	ii					
PHRYGIAN		1	b2	b3	4	5	b6	b7	8	ii					
AEOLIAN MODE NATURAL MINOR RELATIVE MINOR		1	2	b3	4	5	b6	b7	8	V					
HARMONIC MINOR		1	2	b3	4	5	b6	7	8						
MAJOR PENTATONIC		1	2	3	5	6	8	I							
MINOR PENTATONIC		1	b3	4	5	b7	8	vi							
SYMMETRICAL	WHOLE TONE	1	2	3	#4	#5	#6	8	+						
	DIMINISHED	1	2	b3	4	b5	b6	bb7	7	8	O				
LOCRIAN		1	b2	b3	4	b5	b6	b7	8	Vii°					

MORE ABOUT SCALES

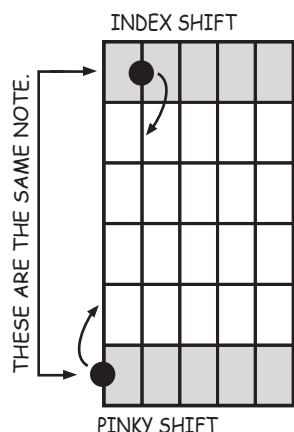
LESSON 15

You probably already know that one big thing about scales is that you have to ultimately know how to PLAY them on the neck of your guitar. All scales have MANY FINGERINGS because you can play any one in any of the 12 keys anywhere on the fret board. You don't need to know all the fingerings, but over your career, you owe it to yourself to soak up as many of them as you can. Your COLOR CODED FINGER CHARTS show 12 ways of playing each scale. These are the possible ways to play the scale within our 6 fret CHROMATIC "BOX" orientation. There are indeed other ways to play these scales as well (many involving SLIDES from one box to another, and you will pick these up over time from various sources). Of course, you could play any given scale on a SINGLE STRING simply by proceeding up the string using the correct pattern of half and whole steps for that scale. But how PRACTICAL a use of your fingers is that? What you're always trying to do is make the BEST and MOST LOGICAL use of all your fingers! That is the only way to attain real speed and control. Many players do not understand these 12 basic fingerings simply because they do not really understand their fret board in the way that YOU DO! You know that you can play all of the 12 notes, within a 5 fret range, for a distance of 2 and 1/3rd octaves anywhere on the neck! You understand how INDEX and PINKY SHIFTS work. What this means is that you should be able to play ANY SCALE... IN ANY KEY... ANYWHERE ON THE NECK! It may be a while before you see how AWESOME it is that you know this stuff and how far ahead of other players this puts you!

Many of these suggested guidelines for scale fingerings may at first seem more difficult or awkward than what you might come up with. But that is often simply because you haven't worked with them enough yet. Your fingers and ligaments haven't stretched enough yet to get the full benefit. You don't have enough EXPERIENCE to see the theoretical significance they hold. A classic example would be all those players who don't use their PINKY because it's such a weak finger. But it's only weak because you DON'T USE IT! You only have 4 fingers to begin with! Don't sacrifice your pinky just because you're too LAZY to get in there and work with it! That's 25% of your hand! It's a lot of work mastering these fingerings, but at the end comes TOTAL FREEDOM of a sort most players never even see is there! Your MODE JAM TAPE will make it more fun to practice your MAJOR DIATONIC MODES and your PENTATONIC SCALES. These are the most common scales and so the first ones you will want to get working on. But you will ultimately have to commit to a FOCUSED and DISCIPLINED practice regimen to master all your scales.

But first we must make sure that we know how to finger scales in general. We have to go back to what we learned about fingering CHROMATIC SCALES.

THE KEY TO UNDERSTANDING SCOTTY'S FAMOUS COLOR CODED FINGER CHARTS



Hopefully you all remember the significance of the 6 FRET CHROMATIC BOX. It takes 5 FRETS to control the 12 notes on the fret board, but that 5th note can be played with either a PINKY SHIFT on one string or an INDEX SHIFT on the next higher string. These can be any 6 frets anywhere on the neck. Remember that your G and B strings are tuned one fret closer together and that is bound to effect your fingering. As a rule, you'll find yourself using index shifts when going up the scale and pinky shifts when coming down but you will see that that isn't always the logical thing to do. Often, using INDEX REACHES when going both up and down help you avoid pinky shifts. When coming back onto the white frets after a FINGER REACH, make sure you use the correct finger that normally would have controlled that fret.

SO WHERE IS THE ROOT NOTE OF THE SCALE?

The root note can be any one of the 12 notes. That's how you tell which key you are playing the scale in. So the root can be under any finger on any fret on any string. It can even be a finger shift note on the shaded frets. That's what accounts for all these different finger patterns for the same scale. Note that as you go from box to box on any one of your charts, the RED DOT which is the ROOT NOTE, moves up one fret each time and so you are playing in the next higher key.

SO HOW DOES THE COLOR CODING SYSTEM WORK?

Remember back in grade school science class you learned the 7 COLORS OF THE SPECTRUM? They go

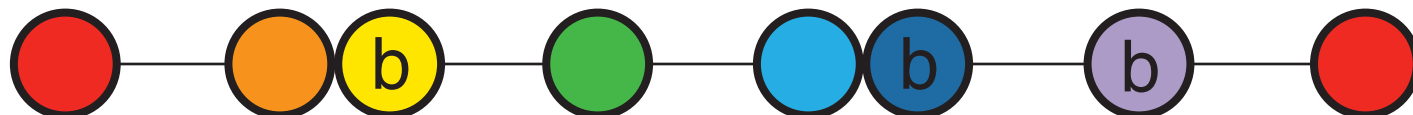
RED - ORANGE - YELLOW - GREEN - BLUE - INDIGO - VIOLET

Notice how most of our most common scales contain 7 NOTES (not including the octave). These are called the DEGREES OF THE SCALE. In our charts, each of the degrees of the scale is represented by each of the corresponding colors in the rainbow. So your ROOT NOTE is RED ...your 2nd is ORANGE ...your 3rd is YELLOW ...your 4th is GREEN ...your 5th is BLUE ...your 6th is INDIGO ...and your 7th is VIOLET. If for example, your 3rd is flatted (as in a minor scale), your YELLOW dot will have a flat sign on it etc. These color coded charts are so awesome because you can see all the degrees of the scale at a glance once you've memorized the color sequence. Compare that to your boring old black and white charts!

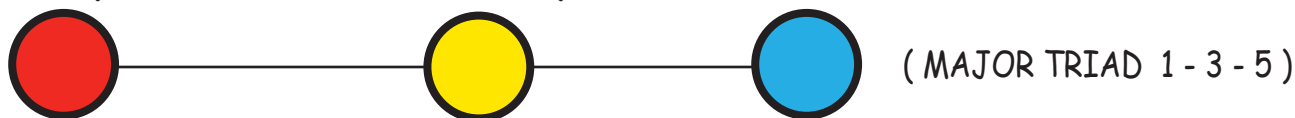
SO A MAJOR DIATONIC SCALE WOULD GO...



AND A NATURAL MINOR SCALE WOULD GO...



AND, SO INTERESTINGLY, THE TRIADS ARE IN PRIMARY COLORS!



STILL MORE ABOUT SCALES

LESSON 16

As we said before, the subject of scales is perhaps the most substantial subject in music theory. As always, we need to know these pitch patterns 4 WAYS ...**THINK - READ - HEAR - SPEAK**. In this lesson, we look deeper into the INTERVALS that separate the notes in scales and begin to listen to them. Eventually you will want to **MEMORIZE** the sound of these scales so you can use them in composition and improvisation.

Many people are shocked to find out how many scales there are ...and all these weird names like MIXOLYDIAN and such. You might have taken quite a lot of music lessons and barely heard of them. Most of us have only heard of the MAJOR SCALE. Did you ever understand a CHROMATIC SCALE before you took this program? You might have heard that PENTATONIC SCALES and BLUES SCALES are big in Rock and Roll. But that's about it. How can Scotty suggest that all these other scales are equally important to know and even equally common in the songs we play? The first thing to know is this ...

THERE IS A SEPARATE SCALE FOR EACH CHORD IN THE SONG.

It is often suggested that an entire song is based on one scale (and certainly you can look at music that way but you might miss some important revelations). In Western Music, that fundamental scale is most often said to be the MAJOR DIATONIC SCALE. That's why it was the first one they tried to teach us back in grade school. A more modern way of teaching theory suggests that it is more useful to consider a separate scale for each chord in the song. How many chords are in the average pop song? Maybe between 3 and 6 ...sometimes more. This means that you are playing 3 to 6 of these scales in every tune you play! We most often see a mixture of MAJOR and MINOR chords in songs. What good is it to view the song as only containing one scale (say the MAJOR SCALE) if you have both major and minor chords? Doesn't it stand to reason that minor scales would work with minor chords? You will find that indeed there are very specific guidelines as to what scale goes with what chord. How do we know which scale works with which chord? This is a complex issue, but first consider this ...

THERE IS A CHORD EMBEDDED INSIDE EVERY SCALE
AND THE SCALE THAT GOES WITH THAT CHORD IS THE SCALE THAT HAS THAT CHORD INSIDE IT!
(Hopefully by this time you remember the number spellings for your major and minor chords)

MAJOR 1-3-5

MINOR 1-b3-5

In the video, we use SILENT NIGHT to demonstrate that even these strange scales like LYDIAN and MIXOLYDIAN show up in the most normal of songs if that is what the chord progression dictates. We will be looking at these relationships more in depth in our next few lessons on MODES and HARMONIZATION.

SO WHAT DO THESE SCALES SOUND LIKE?

CHROMATIC

Hopefully by now you are becoming familiar with the sound of the CHROMATIC SCALE. It is 13 notes separated by 12 half steps. You don't really play the complete scale very often in pop music but a lot of songs do contain brief chromatic passages. You need to know what a half step sounds like.

MAJOR SCALES

We've been listening to the MAJOR DIATONIC SCALE since we were kids. Now we are learning that there is MORE THAN ONE MAJOR SCALE. A major scale has a major chord inside it ...1-3-5 and there are several scale patterns that fit that category. The #4 in LYDIAN may sound strange at first but you will notice it more and more as your ears become accustomed to it. Indeed it often "hides" in the melody to where you didn't even notice it there before. You will come to recognize the b7 in MIXOLYDIAN as being an INCREDIBLY common and important note in many contexts.

MINOR SCALES

There are MINOR CHORDS INSIDE MINOR SCALES ...1-b3-5. Most people would agree that the MINOR 3rd makes things sound SAD. Once again there are several slightly different minor scale patterns ...many again containing that b7. You may find it hard to believe that you will ever be able to pick up on their subtle differences by ear but indeed you can and must. Many pieces of music are written in MINOR KEYS where these scales replace the MAJOR SCALE as the central scale. In songs, minor scales interact with minor chords in very specific ways. Notice that the HARMONIC MINOR SCALE contains an AUGMENTED 2nd (1 and 1/2 step) interval.

PENTATONIC SCALES

PENTATONIC MEANS 5 NOTES. These scales contain two fewer notes per octave than their 7 note DIATONIC cousins. Therefore, these scales contain TWO AUGMENTED 2nd intervals which give them an open and "airy" sound that has attracted a lot of attention by various cultures throughout history. Some of you have heard that these are the ONLY scales you need to know to play Rock and Roll and Blues music but this is a terrible over simplification. Why only try to control 5 notes per octave when there is so much more? The MINOR PENTATONIC SCALE is often thought of as the foundation for the BLUES SCALE. It does contain the MINOR 3rd giving it a sad or "bluesy" feel. The b7 is also a huge note in blues. Try to remember what they sound like.

SYMMETRICAL SCALES

SYMMETRICAL SCALES GO WITH SYMMETRICAL CHORDS like AUGMENTED and DIMINISHED. A WHOLE TONE SCALE is all whole steps. A CHROMATIC SCALE is all half steps. A DIMINISHED SCALE goes whole-half-whole-half etc. Notice that these scales DO NOT CONTAIN NOTE 5 so they are NEITHER MAJOR OR MINOR! There are far fewer of these scales and chords in popular music partially because we are so used to that all important 5th that we don't quite know what to do when it isn't there. It is harder for us to memorize what these patterns sound like. Symmetrical chords are much more common in Jazz and Classical music, often snuck in between major and minor chords. LOCRIAN also has a b5 but notice that it is not symmetrical.

DIATONIC HARMONY A LA MODE

LESSON 17

WHY DO WE SEE THE SAME SETS OF CHORDS OVER AND OVER AGAIN IN DIFFERENT SONGS?

Those of you who have been playing for awhile will agree that there are hundreds of pop tunes that all share the same set of chords. One of the most common combinations in Pop Rock Music certainly is

G MAJOR - C MAJOR - D MAJOR - E MINOR - A MINOR

You might have played dozens of songs that have some or all of these chords in them. Why is that? Well you could just say that perhaps most song writers think that these chords sound good together. But there is a deeper level on which these chords are connected. Consider this statement ...

MOST SONGS CONTAIN ONLY 7 OF THE 12 NOTES. WHICH 7? THE 7 NOTES IN A MAJOR DIATONIC SCALE IN A GIVEN KEY.

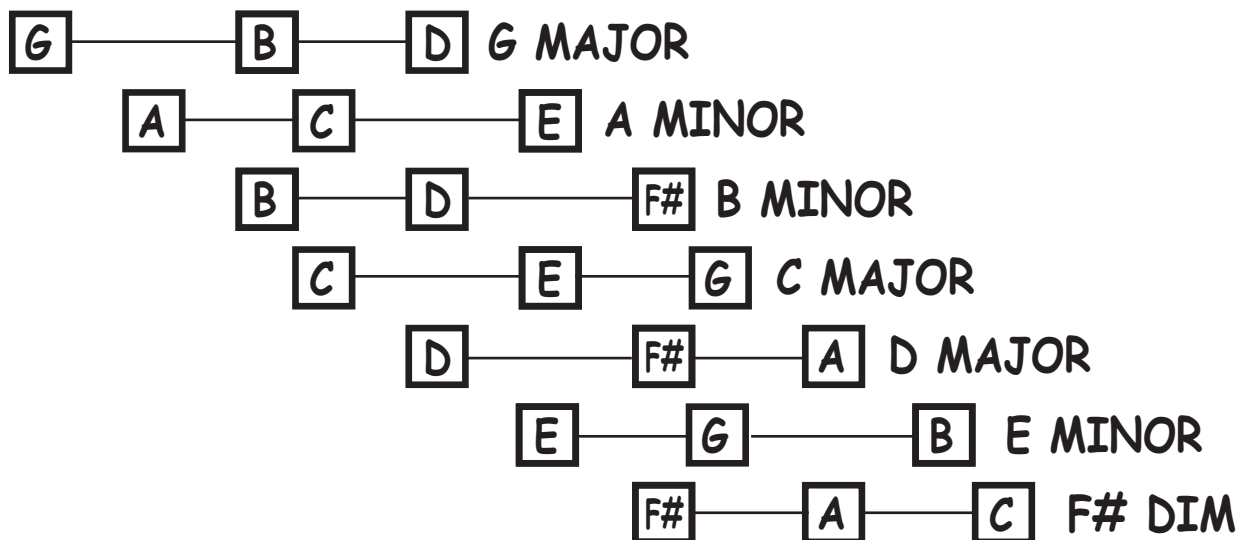
It's our good old friend the MAJOR DIATONIC SCALE again. So much of Western Music can be traced back to it. Now we find that the average song contains only the 7 notes in this scale. Not only is the MELODY of the song constructed out of these notes,

THE SONG CONTAINS ONLY THE CHORDS THAT CAN BE CONSTRUCTED OUT OF THOSE 7 NOTES!

Let's look at an example in the KEY OF G. The 7 notes in a G MAJOR SCALE are



WHAT CHORDS CAN BE MADE OUT OF THESE 7 NOTES?



AS YOU CAN SEE, THIS SET INCLUDES ALL THE CHORDS IN OUR EXAMPLE!

SO WHAT'S THIS GOT TO DO WITH MODES?

This is HUGE. You've learned a lot about PITCH PATTERNS so far in this program. We started with SILENCE and then we learned that all conventional patterns in music have a ROOT NOTE. We then talked about INTERVALS and suggested that the true nature of these patterns involves the SPACING of the notes. Now we're looking at the more complex note patterns like SCALES. You know all this stuff but you still don't see how this all comes together to EXPLAIN the songs we play. Well we are about to look at that now. We're going to do it in a rather unique way involving the concept of MODES. Many people have heard of modes. In fact many music students are taught a great deal about them without ever being shown their significance and relevance to the subject of how SCALES and MELODIES relate to CHORD PROGRESSIONS in conventional songs. The question is ...

HOW DO YOU KNOW WHAT SCALE GOES WITH WHAT CHORD?

Many of us who have done some JAMMING still remember when someone first told us that a single scale (often the MAJOR SCALE or the PENTATONIC SCALE) could be used to create melodies over a given set of chords. Wow! It seemed to make improvising so easy! We sound halfway like ERIC CLAPTON all ready! But if you play this way for long you begin to run out of ideas. You notice that all the notes in the scale sound pretty good but any given note will sound better or worse depending on which chord you play it over. What's going on? What do the guys on the radio know that you don't?

WOW! I CAN PLAY ONE SCALE OVER SEVERAL CHORDS!

NO! WHEN THE CHORD CHANGES, SO MUST THE SCALE CHANGE!

There is a separate scale for each chord.

This makes sense doesn't it? How, for example, could a MAJOR SCALE work over both major and minor chords? Shouldn't we be playing minor scales over the minor chords? But it seems that the majority of people we talk to have never considered this possibility.

They point out that playing one scale sounds pretty good and it would be too hard to switch back and forth from scale to scale every time the chord changes. Who ever heard of such a thing? But there is an answer.

THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES. THE QUESTION IS WHICH NOTE IN THE SET IS "DO"?

THIS IS THE WHOLE THING ABOUT MODES.

MORE ABOUT MODES

LESSON 18

THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES.

This is the whole concept of MODES. But what does this mean and why is it important for you to understand about them? Let's look at 2 octaves of a MAJOR DIATONIC SCALE. We recognize it by it's familiar pattern of intervals ... mostly WHOLE STEPS except between 3 and 4 and 7 and 8 or, in other words ...

WHOLE - WHOLE - HALF - WHOLE - WHOLE - WHOLE - HALF

But if you start on one of the other notes in the set, the interval pattern shifts and it is no longer accurate to view the new sequence of intervals as a Major Diatonic Scale. It is one of the MODES of the Major Diatonic Scale.

WHOLE	WHOLE	HALF	WHOLE	WHOLE	WHOLE	HALF	WHOLE	WHOLE	HALF	WHOLE	WHOLE	WHOLE	HALF
1	2	3 4	5	6	7 8								

This 1st mode is the **IONIAN** ...the MAJOR DIATONIC SCALE itself. Notice that it is a MAJOR SCALE with a MAJOR CHORD (1-3-5) embedded inside it.

1	2 b3	4	5	6 b7	8
---	------	---	---	------	---

Start on the 2nd note and your interval pattern changes. This 2nd mode, **DORIAN** is a MINOR SCALE containing a MINOR CHORD (1-b3-5)

1 b2	b3	4	5 b6	b7	8
------	----	---	------	----	---

Start on the 3rd note and you're playing **PHRYGIAN MODE**. It is a MINOR SCALE. Notice the b2 and b6.

1	2	3	#4 5	6	7 8
---	---	---	------	---	-----

Start on the 4th note and you're playing **LYDIAN MODE**. It is a MAJOR SCALE with a #4.

Start on the 5th note and you are playing the **MIXOLYDIAN**, a very important MAJOR SCALE with a b7.

1	2	3 4	5	6 b7	8
---	---	-----	---	------	---

This 6th mode is the **AEOLIAN**. Notice that it is a NATURAL MINOR SCALE. It is the very important RELATIVE MINOR mode.

1	2 b3	4	5 b6	b7	8
---	------	---	------	----	---

Start on the 7th note and you're playing the **LOCRIAN MODE**. It is the only mode that is neither major or minor. It's b5 and b3 relates it to DIMINISHED.

1 b2	b3	4 b5	b6	b7	8
------	----	------	----	----	---

THERE ARE 7 DIFFERENT SCALES THAT ALL CONTAIN THE SAME NOTES!

Let's look at our example again in the KEY OF G. Set your SLIDERULE up by putting G in the upper left window ...root note of the IONIAN MODE. There are 6 other scales that contain the same notes as a G MAJOR SCALE. They are ...

G	IONIAN (G MAJOR DIATONIC)
A	DORIAN
B	PHRYGIAN
C	LYDIAN
D	MIXOLYDIAN
(E NATURAL MINOR)	E AEOLIAN
F#	LOCRIAN

Not only is this an interesting fact about how scales interrelate with one another, this also tells us everything about why we keep seeing those same **CHORD COMBINATIONS** over and over.

NOTICE THAT EACH OF THESE MODES HAS A CHORD EMBEDDED INSIDE IT.

In our example using the **KEY OF G**, we again see that familiar set of chords. Each chord is rooted on one of the **DEGREES OF THE MAJOR DIATONIC SCALE** and is represented by a **ROMAN NUMERAL**. Some of these chords are **MAJOR** and some are **MINOR** depending on whether the mode it is in has a **MAJOR 3rd** or a **MINOR 3rd** in it. These modes are always in the same order and retain their same major/minor orientation no matter which **KEY** you play them in so you absolutely have to memorize this set of relationships.

THIS ONE SYSTEM OF PITCH ORGANIZATION MAY EXPLAIN THE MELODIES AND CHORD PROGRESSIONS FOR 50-75% OF ALL THE SONGS YOU HAVE EVER HEARD!

I **G** 2 **B** 4 **D** 6 7 8

G MAJOR is the **TONIC CHORD** in the key of **G** and can be found inside **G IONIAN**. But you're not playing this scale over the other chords in the song.

ii **A** 2 **C** 4 **E** 6 b7 8

A MINOR is the **ii CHORD** in the key of **G**. It is embedded inside the **Dorian Mode** and **A DORIAN** is the scale you are playing to it.

iii **B** b2 **D** 4 **F#** b6 b7 8

B MINOR is the **iii CHORD**. You'll be using **B PHRYGIAN** to jam with over this chord

IV **C** 2 **E** #4 **G** 6 7 8

C MAJOR is the **IV CHORD**. The 7 notes are functioning as **C LYDIAN** over this chord.

D MAJOR is the all important **V CHORD** and so **MIXOLYDIAN** is a common and most important scale.

V **D** 2 **F#** 4 **A** 6 b7 8

E MINOR is the **Vi CHORD**. It is the next most common chord after **I-IV-V** so **AEOLIAN**, the **RELATIVE MINOR** mode, is very common.

Vi **E** 2 **G** 4 **B** b6 b7 8

F# DIMINISHED is the **Vii CHORD**. This chord has us playing **LOCRIAN MODE** over it. Note the **b3** and **b5** which make this scale a **HALF DIMINISHED** scale.

Vii **F#** b2 **A** 4 **D** b6 b7 8

IF THESE SCALES ALL CONTAIN THE SAME NOTES AS A G MAJOR SCALE, WHAT'S THE HARM IN THINKING THAT YOU'RE PLAYING G MAJOR OVER ALL THESE CHORDS? WHY DO I NEED TO LEARN ALL THESE NEW SCALES WITH THESE WEIRD GREEK NAMES?

You see it's all about **FUNCTION - FUNCTION - FUNCTION!** Yes, these 7 notes will sound pretty good over these 7 chords but the more you listen, the more you begin to hear that each individual note sounds different depending on which chord you play it over. That is because each note changes **NUMERIC FUNCTION** with each chord change. Sometimes it is an **IMPORTANT** note like the 3 or the 5 and you will want to use it often ...hold it longer ...begin and end **PHRASES** with it. But over another chord, this same note might be functioning as a less important melody note like 4 or 6 and you will need to be more careful with it. Understanding these shifting relationships and being able to control them through **EAR TRAINING** is what separates the great improvisers from the **'LICK BARFERS'** and finger players. Yes it will take more work to learn to jam this way but the rewards are very real. You could learn to solo as well or better than many of your great guitar heroes!

THE MODE JAM TRACKS

LESSON 19

Hopefully you now understand the basic relationship between MELODY, SCALES and CHORD PROGRESSIONS in conventional DIATONIC MUSIC. You will find that there is a lot more to know as well but you've got a lot of it right here. These relationships are all laid out very nicely on your MUSICAL SLIDERULE and you can investigate them in any one of the 12 KEYS. Now it's time to begin mastering these relationships. You have a lot of FINGER PATTERNS to memorize and a lot of EAR TRAINING to do. It is going to take most of you YEARS to actually accomplish this. But don't let that put you off. All real musicians have to go through this and if you are destined to be a REAL PLAYER, you will not find the process too difficult or boring. By all means, take it at your own pace

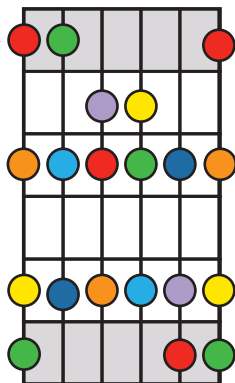
Scotty's famous COLOR CODED FINGER CHARTS are awesome tools to help you learn your scales and modes. They allow you to see all the DEGREES OF THE SCALE at a glance and where all the octave notes are as well. They may seem a bit intimidating at first ...all these pages with all these patterns on each one. But it's not as bad as it looks. Over time you will notice a lot of REDUNDANCY built into these finger patterns. For example ...

THERE ARE 7 SCALES THAT ALL HAVE THE SAME NOTES IN THEM SO THE SAME FINGERING CAN BE USED TO PLAY ALL 7 SCALES!

Look at the charts for your 7 DIATONIC MODES ...IONIAN, DORIAN, PHRYGIAN, LYDIAN, MIXOLYDIAN, AEOLIAN and LOCRIAN. Notice that all the dots are actually in the same place on each page. The finger patterns you use to play these scales are all the same so you don't have 12 new shapes to master on each page. The only thing that changes from chart to chart is the COLOR of the dots indicating that the NUMERIC/HARMONIC FUNCTION of each note changes depending on which mode you are playing. A different note is functioning as the ROOT NOTE in each mode. Notice each of the finger boxes has a number. Lets compare some of them. Lets look at

IONIAN 1

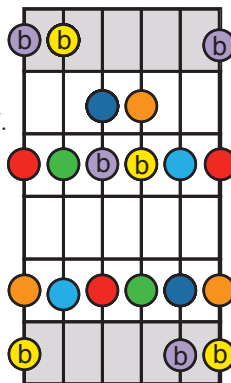
Notice that the dots are all in the same place on each of these charts. RED is the ROOT NOTE and it's octaves. IONIAN is the 1st MODE and in this box it's root note is the lowest note you are playing.



DORIAN 1

Though the dots are all in the same place, the COLORS have all shifted up by 1 SCALE DEGREE.

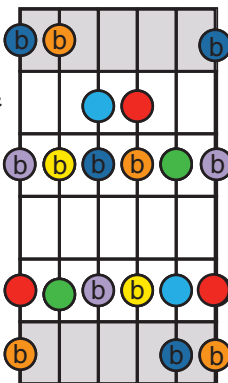
Since DORIAN is the 2nd MODE, the ROOT (red) has moved up to what was the 2nd note of the previous mode, IONIAN.



PHRYGIAN 1

The same fingering can be used to play PHRYGIAN. It is the 3rd MODE. The dot colors have all shifted up by one more scale degree from the DORIAN and 2 up from the IONIAN.

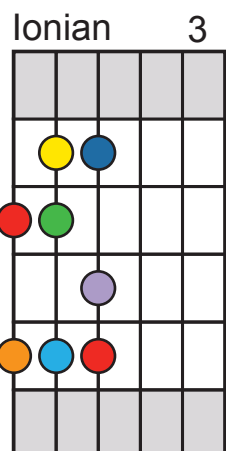
It is rooted on the 3rd DEGREE of IONIAN MODE.



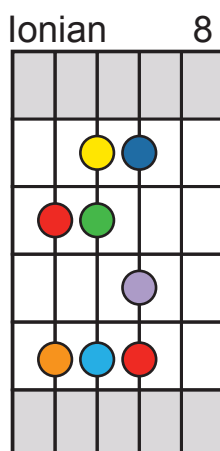
CHECK OUT YOUR OTHER MODE FINGER CHARTS AND YOU WILL SEE THIS PROCESS CONTINUING.

Here's another bit of redundancy to note. You know that all your strings are tuned 4 NOTES APART except G and B. So a scale fingering should remain the same when you move it over by a string as long as the ROOT NOTE IS UNDER THE SAME FINGER. The pattern will only change when you encounter the tuning difference between your G and B string. Notice how in these 3 examples, the root is always under your 2nd finger but on different strings. The finger patterns are then identical until you hit the B string. This is just one example of many ...

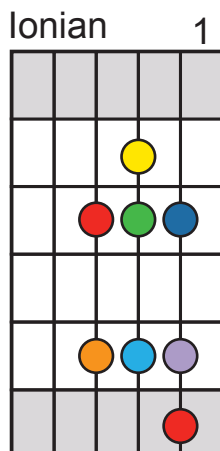
Here is the 1st full octave of a MAJOR SCALE from the box IONIAN 3. It requires no finger shifts. Notice the familiar MAJOR SCALE interval pattern. The ROOT NOTE is under your 2nd finger on your low E string.



In IONIAN 8, notice that the root note is also under your 2nd finger but it has moved over to be on your A string. Since this is also a MAJOR SCALE it retains the same interval pattern and so the same finger shape as IONIAN 3 just moved over one string.



In IONIAN 1 the root is still under your 2nd finger but now it has moved to your D string. We see the same fingering again until we get to the B string. These last 3 notes must be shifted up a fret to adjust for the tuning.



Pity the poor guitar student who must slave away hour after hour with his boring old black and white finger charts and no real sense of what these scale notes sound like MELODICALLY over the chords in a song. Your MODE JAM TRACKS provide you with a RHYTHM SECTION (Bass, Drums and Rhythm Guitar) to IMPROVISE over anytime you want. Start by using it to simply make memorizing your scale fingerings less of a solitary pursuit. You get to hear how each note relates to each chord you play it over. This is EAR TRAINING. If you can stand it, try singing and saying the number of each note as you play it until you can sing the scale independent of hearing it on your guitar. Remember ...you don't solo with your FINGERS ...you solo with your MIND and your EARS! Eventually you will want to find a way to create your own jam tracks by acquiring recording gear of some sort. Put your drum machine on and strum the chords in the solo section of any song you want to learn to jam over or get the guys in the band to play that section over and over while you sit out and record it. Don't waste rehearsal time. It often takes a while to develop good melodic ideas from your scales. Take that recording home and jam your brains out until you get it right!

But for now, these tracks will be a good place to start. They contain several 2 chord progressions utilizing the most common changes in DIATONIC MUSIC. There is also a BLUES PROGRESSION that you can jam to a little later after we learn about BLUES HARMONY. We recorded these progressions all in the KEY OF A but you will want to learn to apply this way of thinking to all 12 keys. You have a lot of patterns to master particularly if you ever want to play JAZZ.

Start by cutting your scale charts into FLASH CARDS. This way you can choose any you want and put them right next to each other for visual comparison. You might look first at the ones with no FINGER SHIFTS (boxes 3 and 10) then 1 shift (box 5) then 2 shifts and so on. This may seem like a lot more work than you were expecting but the more fingerings you know for a scale, the more FREEDOM you have to get your melodic ideas out of your head and on to your fretboard quickly and smoothly.

Different mode types with the same BOX NUMBER will have the same finger pattern. For example, the dots on IONIAN 3 and LYDIAN 3 will be positioned in the same place but their color coding will be different showing that they have a MODAL RELATIONSHIP to one another. So to explore your mode fingerings, you will be choosing LIKE BOX NUMBERS to play over your mode jam progressions. Here is the key to the progressions included in your MODE JAM TRACKS....

TUNING NOTES		If your ELECTRONIC TUNER tells you that these tuning notes are sharp or flat by say 10 cents, tune your guitar accordingly.
I PROGRESSION 1 IV A MAJOR - D MAJOR		A IONIAN TO D LYDIAN A very common progression. Grab an IONIAN and a LYDIAN flash card with the same BOX NUMBER. Locate the place on the neck where the RED DOTS on the IONIAN box will be A's. Eventually try to memorize as many boxes as you can.
I PROGRESSION 2 V A MAJOR - E MAJOR		A IONIAN TO E MIXOLYDIAN Grab an IONIAN and a MIXOLYDIAN card with the same box number. Combine this thinking with the previous progression and you've got your I-IV-V PROGRESSION. Again find where you have to be to play in the key of A. Try all the boxes.
I PROGRESSION 3 Vi A MAJOR - F# MINOR		A IONIAN TO F# AEOLIAN A MAJOR CHORD and a MINOR CHORD in that all important RELATIVE MINOR relationship. The Vi CHORD is probably the next most common after I-IV-V making this a very common and important scale/chord relationship to master.
Vi PROGRESSION 4 ii F# MINOR - B MINOR		F# AEOLIAN TO B DORIAN When we learn about MINOR KEYS, you will find that this is better thought of as a I-IV PROGRESSION in the key of F# MINOR. Since we are working off the AEOLIAN MODE make sure your ROOT NOTE IS F# INSTEAD OF A.
I IV V PROGRESSION 5 A-D-E 12 BAR BLUES		A MINOR PENTATONIC BLUES MUSIC is different than normal DIATONIC MUSIC, We will be talking about it in a few more lessons. At that time, you will find this progression useful to jam over with an A MINOR PENTATONIC SCALE.

You will come to find out that these scale/chord relationships are not carved in stone. The really clever players can use ANY NOTE OVER ANY CHORD IN ANY KEY AT ANY TIME. But this requires very well trained ears and we approach this level of playing by first learning how to integrate this more conventional thinking.

OTHER SCALES, OTHER MODES LESSON 20

You probably know that there are many more scales in the world than those that we have looked at so far. This program is geared towards people mostly interested in playing everyday AMERICAN POP/ROCK MUSIC. Those of you who belong to various ethnic groups might have grown up listening to music that utilizes scales that sound quite different from the ones on our MUST KNOW LIST. I am not an ETHNOMUSICOLOGIST and it is beyond the scope of this program to address the music of other cultures. I have included a list of EXOTIC SCALES that I have gathered over the years from various sources and you may find it interesting to play and listen to them. Some of them do wind up in Western Pop on occasion and you will recognize them. Others can be found in more advanced WESTERN FINE ART MUSIC and the experimental music of the AVANT-GARDE.

Most of you will first want to concentrate on the MUST KNOW LIST. But there is an expanded list of scales that do show up in more conventional Western Music and we will look at them here.

Many people know about the system of MODES that we have been talking about for the last few lessons. But most people don't know that the system of modes based on the MAJOR DIATONIC SCALE is not the only set of modes that there are. Indeed, ANY SCALE can generate a system of modes. Often they aren't worth investigating (as in the case of the SYMMETRICAL SCALES) because no new and useful scale patterns will be discovered. But there are a few scales on our list that will generate meaningful systems of modes.

Consider the MELODIC MINOR SCALE. It has a system of 7 modes, several of which can show up in pop/rock music...not so much in ordinary DIATONIC chord progressions ...but in odd or advanced progressions containing MODULATIONS (where the song actually changes key) or CHORD SUBSTITUTIONS. Two of these scales, LYDIAN b7 and MIXOLYDIAN b13, are actually quite common and after you have gained a fair proficiency with the DIATONIC and PENTATONIC SCALES, you might want to turn your attention to these. There are finger charts for all 7 MELODIC MINOR MODES in your support material. Notice that they only differ from your familiar DIATONIC MODES by one note and are named accordingly.

START WITH 2 OCTAVES OF A MELODIC MINOR SCALE.

□	□ □	□	□	□	□ □	□ □	□	□	□	□ □
1	2 b3	4	5	6	7 8	MELODIC MINOR				

1 b2	b3	4	5	6 b7	8	DORIAN b9
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1	2	3	#4	#5 6	7 8	LYDIAN #5
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1	2	3	#4 5	6 b7	8	LYDIAN b7
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MIXOLYDIAN b13	1	2	3 4	5 b6	b7	8
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LOCRIAN b9	1	2 b3	4 b5	b6	b7	8
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(Notice the b4=3)	ALTERED	1 b2	b3 b4	b5	b6	b7	8
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You may see these scales labeled differently by different publishers. For example, DORIAN b9 may be called DORIAN b2 etc. and since each of these scales is close to 2 DIATONIC MODES, LYDIAN b7 could also be called MIXOLYDIAN #4 etc.

EXOTIC SCALES

1	b2	b3	4	5	b6	7	8			
1	b2	b3	4	5	6	7	8			
1	b2		3	4	b5	6	b7 8			
1	b2		3	4	5	b6	7 8			
1	b2		3	#4	#5	#6	7 8			
1	2	b3		5	b6		8			
1	2	b3		#4	5	b6	7 8			
1		#2	3	#4	5	6	b7 8			
1	b2		4	5	b6		8			
1	b2		4	b5		b7	8			
1	2	3	4	5	b6	b7	8			
1	b2	b3	3	4	b5	b6	b7 7 8			
1	b2	b3		5		b7	8			
1	2	b3		#4	5	b6	b7 8			
1	b2		3	4	5	b6	b7 8			
1	2	3	4	b5	b6	b7	8			
1	2	3	#4	5	b6	b7	8			
1	2	3	#4	5	6	b7	8			
1	2	3	#4	#5	#6	7	8			
1	2	3	4	b5	b6	b7	8			
1	b2	b3		5	b6		8			
1	b2		3	4	5	b6	7 8			
1	2	b3	4	5	b6		7 8			
1	b2	b3	4	5	6	b7	8			
1	b2		3	4	b5	b6	7 8			
1	2	b3		#4	5	b6	7 8			
1	b2		3	4	5	b6	7 8			
1	2	b3	4	5	6		7 8			
1	b2		3	4	5	b6	b7 8			
1	2	3		5	6		8			
1	2	b3	3	4	5	b6	6	b7	7	8
1	b2		3	4	5	b6	b7		8	
1	2		4	5		b7		8		
1	b2		4	5	b6			8		
1	2	3		5	6			8		
1		3	#4	5				7 8		
1	2	3	4	5	b6			7 8		

Neapolitan Minor
Neapolitan Major
Oriental
Double Harmonic
Enigmatic
Hirajoshi
Hungarian Minor
Hungarian Major
Kumoi
Iwato
Hindu
Spanish 8 Tone
Pelog
Hungarian Gypsy
Major Phrygian
Major Locrian
Lydian Minor
Overtone
Leading Whole Tone
Arabian
Balinese
Gypsy
Mohammedan
Javanese
Persian
Algerian
Byzantine
Hawaiian
Jewish
Mongolian
Ethiopian
Spanish
Egyptian
Japanese
Chinese
Chinese 2
Harmonic Major

PENTATONIC MODES

LESSON 21

PENTATONIC SCALES are some of the best known scales in popular music. The MAJOR and MINOR PENTATONIC SCALES are on our MUST KNOW LIST. PENTATONIC means 5 NOTES. These scales are said to have 5 notes. But I will encourage you to include the OCTAVE note and so they actually have 6 notes. Since these scales have fewer notes than the DIATONIC SCALES, their notes are separated by LARGER INTERVALS. Pentatonic Scales contain no half steps (minor 2nds) and each contains two 1 and 1/2 step intervals (augmented 2nds). As a result, these scales have an open and uncluttered sound that gives them many special properties. They can actually function as several different DIATONIC SCALES at the same time. For example, since a MAJOR PENTATONIC SCALE has no 4th or 7th, it could function as a simplified version of the IONIAN, LYDIAN or MIXOLYDIAN MODES since the 4 and 7 are the only notes these scales don't share.

Our MUST KNOW LIST contains only two PENTATONIC SCALES ...MAJOR and MINOR. But many people don't know that PENTATONIC SCALES HAVE MODES. There are in fact many Pentatonic Scales. How do we know this? Remember when we first learned about DIATONIC MODES? We talked about how many people first learn to improvise. We looked at that argument between the three guys ...

WOW! I CAN PLAY ONE SCALE OVER SEVERAL CHORDS!

NO! WHEN THE CHORD CHANGES, SO MUST THE SCALE CHANGE!

THIS IS THE THING ABOUT MODES. THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES. THE QUESTION IS WHICH NOTE IN THE SET IS "DO"?

We first looked at this within the context of DIATONIC HARMONY involving the modes of the MAJOR DIATONIC SCALE. But often, our first experience with scales in improvisation involves the PENTATONIC SCALES. The Pentatonic Scale (or Minor Pentatonic) is often the first scale we are shown as fledgling Rock guitarists. You might have thought you were playing a Pentatonic Scale over the multiple chords in a progression, but in fact, as in the case of the Diatonic Modes, you were actually playing various MODES of the PENTATONIC SCALE over these chords. Again, this is significant because the NUMERIC/HARMONIC FUNCTION of each note will change depending on which chord you are playing it over. In most contexts, we are using these Pentatonic Modes in the same way we used the Diatonic Modes ...over the same types of progressions. It's just that the Pentatonic Modes contain TWO FEWER NOTES. At first, this seems to make these modes easier to control ...less notes to worry about. For example, the 4 and the 7, notes that are left out of a Pentatonic Scale, are notes that you have to be more careful with when constructing melodies. They won't sound as good as the 1st, 3rd and 5th, notes that are in the chord itself. So you may not want to start or end melodic phrases with them. You may not want to dwell on them as long as other notes. Remember, when using a note in a melody you not only have to consider it's PITCH but also it's DURATION. Notes that add spice and interest when used sparingly can actually sound like wrong notes when you hang on them too long. Once your ears are trained, you will be able to hear these relationships more clearly and you will notice that a lot of players play a lot of dumb notes because they aren't really listening.

Some players believe that the solution is to simply leave these notes out and just play Pentatonic Scales. But I'm not so sure. Why would you want only 6 notes in your arsenal when you could have 8? That's 25% of the notes you could use to make your melodies 25% more interesting! Better we should train our ears and learn to control these difficult notes rather than just leaving them out. In fact, what you will come to see is that the really clever players can use ANY OF THE 12 NOTES OVER ANY CHORD IN ANY KEY ANYTIME! So you may not want to be a Pentatonic player ...you may not even want to be a Diatonic player. I think that all players who really understand music are trying to become true CHROMATIC PLAYERS. This doesn't mean that you try to cram every last note into every little space and OVERPLAY (as you will hear some players do). It just means that you have more choices at any given moment and that's a good thing. In fact many of the people telling you that Pentatonic Scales are so significant don't really understand enough about what they're doing to have an intelligent opinion. There are a lot of "meat heads" in Rock guitar. They'll tell you that they use Pentatonic Scales but if you go back and analyze what they just played, you'll find that they actually were using notes outside of the Pentatonic Scale without even knowing it!

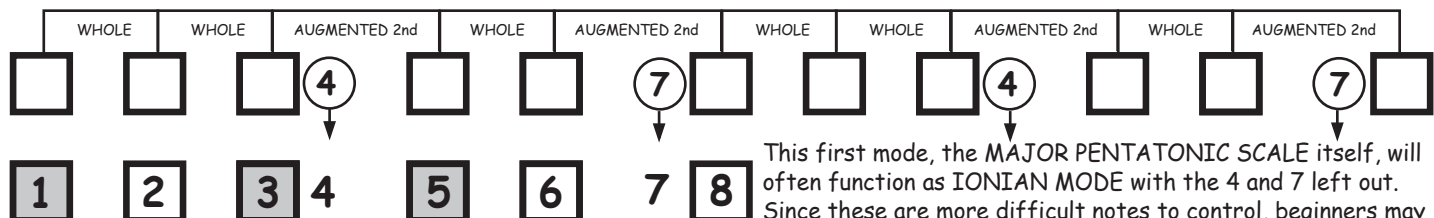
There may have been some justification for leaving out the 4 and 7 over the I chord in the progression, but as you investigate the PENTATONIC MODES over the other chords in the progression, you will see that you are leaving out more useful notes like the 3rd and the 5th. In fact, over the IV chord and the VII chord, you will be leaving out the ROOT NOTE! How dumb is that?

PENTATONIC SCALES HAVE MANY USES BUT YOU WILL MOST OFTEN SEE THEM AS SIMPLIFIED VERSIONS OF DIATONIC MODES

That is they work over normal DIATONIC PROGRESSIONS but each mode has 2 fewer notes. Sometimes these are the more difficult notes to control but not always. As you go through the PENTATONIC MODES you will see yourself missing some of your most useful and powerful notes.

START WITH 2 OCTAVES OF A MAJOR PENTATONIC SCALE.

IT'S LIKE A MAJOR DIATONIC SCALE WITH THE 4 AND 7 LEFT OUT. NOTICE THE TWO AUGMENTED 2nd INTERVALS.



This first mode, the MAJOR PENTATONIC SCALE itself, will often function as IONIAN MODE with the 4 and 7 left out. Since these are more difficult notes to control, beginners may have some justification for skipping them.



This 2nd mode will often function as DORIAN MODE with the b3 and 6 left out so I call it PENTATONIC DORIAN. But why would you want to skip a strong and useful note like b3 over a minor chord?



PENTATONIC PHRYGIAN leaves out the b2 and the 5. Since 5 is your 2nd most powerful note, I wouldn't just skip it. Only amateurs routinely play like that.

This is the 4th mode, PENTATONIC LYDIAN. Played over the IV chord, this is a common mode.



But here you are leaving out your ROOT NOTE ...your most useful and powerful note! Train your ears so you won't miss opportunities.

PENTATONIC MIXOLYDIAN is your 2nd most common mode since it is played over the V chord. Once again we skip important notes



Why leave out the 3 over a MAJOR CHORD? And b7 is also a huge note particularly in BLUES.

Recognize this? The MINOR PENTATONIC SCALE is the 6th mode of the MAJOR PENTATONIC SCALE! Very important to know. It's that RELATIVE MINOR thing.



But here you are skipping the 2 and the b6.

In PENTATONIC LOCRIAN, we once again wind up leaving out the ROOT NOTE. Though DIMINISHED CHORDS are much less common in Pop Music, when they do show up, it would be pretty dumb to leave out b5.



TAKE A LOOK AT YOUR MAJOR AND MINOR PENTATONIC FINGER CHARTS.

Since these two scales are MODES of each other (the Minor Pentatonic is the 6th mode of the Major Pentatonic), the same fingerings can be used to play either scale. Notice that the ROOT of the Minor Pentatonic (red dot) is the 6th (indigo dot) of the corresponding Major Pentatonic box. Many players don't know whether they're using these fingerings as Major or Minor at any given moment. The question is are you playing it over a Major or Minor chord or chord progression?

Minor Pentatonic Scales are very big in BLUES and BLUES ROCK MUSIC and we will be looking at them more in depth in a few lessons when we look at these styles of music.

I don't mean to sound like I am discouraging you from taking Pentatonic Scales seriously. Their uncluttered sound makes them very useful in many contexts. It's just that too many players who don't know their theory and haven't trained their ears miss a lot of opportunities when playing strictly Pentatonically over conventional progressions. I myself am striving to learn to control more of the 12 notes ...not less! Many players play that way because they just don't know any better.

MUSIC IN MINOR KEYS

LESSON 22

THIS IS HUGE! HOW MANY KEYS ARE THERE TO PLAY MUSIC IN?

Up to now we have been saying that there are 12. But those are just the MAJOR KEYS ...based on the MAJOR DIATONIC SCALE and it's MODES and embedded TRIADS. There are also 12 MINOR KEYS to play in. Perhaps half of our songs are in these MINOR KEYS. How do you tell if a song is in a MINOR KEY?

How do you generally tell what key any song is in? As I have said before, this is a complex issue with a number of different facets. You kind of develop evidence towards a particular conclusion and when enough of these indicators point toward one answer, you make the judgment as to what key you are in. What is the FIRST CHORD in the song? What is the LAST CHORD in the song? What KEY SIGNATURE is printed in the sheet music? None of these alone is guaranteed to give you the absolute answer but they are all helpful.

I always ask myself which chord feels like the "HOME CHORD". Which chord begins and ends the most PHRASES? Which chord feels like the point from which everything else flows and then ultimately returns to? If that chord is a MAJOR CHORD, you are probably in a MAJOR KEY. If that chord is a MINOR CHORD, you may well be in a MINOR KEY (although in both cases you might also be in a MODAL KEY which we will discuss in our next lesson). You may have trouble hearing which chord is the HOME CHORD at first and so this becomes one of your earliest EAR TRAINING challenges. Since Minor Chords sound SAD, songs in a MINOR KEY are apt to be the more melancholy songs you hear on the radio.

We spent several hours learning about MAJOR DIATONIC HARMONIZATIONS (Major Key modes and progressions). Since MINOR KEY music is just as big, you might think that it will take hours to learn about them as well. But the good news is ...if you understand MAJOR DIATONIC HARMONY, you already know most of what there is to know about MINOR KEYS. In fact, I'm fond of saying that the whole subject can be summed up in three simple words ...

Vi BECOMES i

SIX BECOMES ONE. The 6th DEGREE OF THE MAJOR DIATONIC SCALE becomes the TONIC for the MINOR KEY. This is the main and true meaning of the concept of RELATIVE MINOR. Every MAJOR KEY is said to have it's RELATIVE MINOR KEY rooted on it's 6th DEGREE. So, for example, the key of A MINOR is the RELATIVE MINOR KEY of the key of C MAJOR because A is the 6th of C. The RELATIVE MINOR of G MAJOR is E MINOR. The RELATIVE MINOR of Bb MAJOR is G MINOR etc. All these relationships are laid out on your MUSICAL SLIDERULE. The MAJOR KEY and it's RELATIVE MINOR KEY share the SAME KEY SIGNATURE since they both have the same notes in them.

MAJOR KEY music involves HARMONIZING a MAJOR DIATONIC SCALE. As we learned in our previous few lessons, all our basic SCALE / CHORD PROGRESSION / MELODY relationships are obtained by looking at the MODES of the MAJOR DIATONIC SCALE and noticing that each of these modes has a chord embedded inside it.

The 6th mode of the Major Scale is the AEOLIAN MODE. In this context it is more often referred to by it's other name, THE NATURAL MINOR SCALE. To understand the SCALE / CHORD PROGRESSION / MELODY relationships in MINOR KEY MUSIC, we are now going to learn to HARMONIZE the NATURAL MINOR SCALE and observe it's MODES and embedded TRIADS. Since the AEOLIAN is one of the modes of the IONIAN, we are not going to encounter any new scales here. We will see our old familiar DIATONIC MODES simply starting this time with AEOLIAN. They are even in the same order as they were in the Major Key. Look at your SLIDERULE. Notice that each of your DIATONIC MODES has a ROMAN NUMERAL in front of it. By now you should know all about these (the DEGREES of a MAJOR DIATONIC HARMONIZATION). Now notice that there is a second Roman Numeral at the END of each mode. These correspond to the DEGREES of the NATURAL MINOR SCALE. Notice that AEOLIAN is now mode #1. All the other modes fall right into line. Locrian is now mode #2. IONIAN (the TONIC MODE in a Major Key) is now the 3rd mode etc. As always, there are chords embedded in these modes.

THIS SHIFT GENERATES A WHOLE NEW TYPE OF CHORD PROGRESSION.

i MINOR - (AEOLIAN) NATURAL MINOR

ii DIMINISHED - (LOCRIAN)

III MAJOR - (IONIAN)

iV MINOR - (DORIAN)

V MINOR - (PHRYGIAN)

VI MAJOR - (LYDIAN)

VII MAJOR - (MIXOLYDIAN)

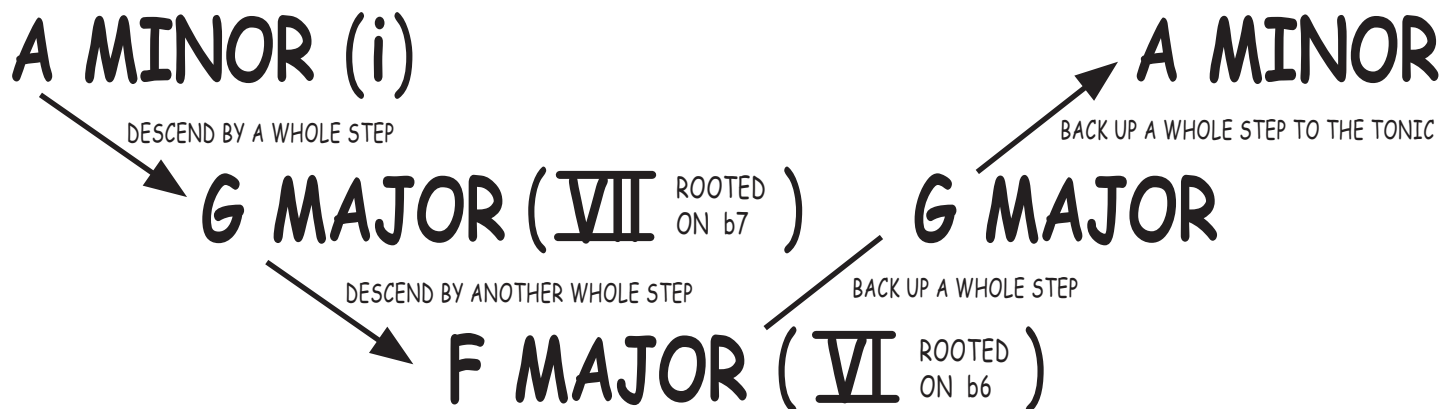
LET'S LOOK AT 2 OCTAVES OF A NATURAL MINOR SCALE (AEOLIAN)

	WHOLE	HALF	WHOLE	WHOLE	HALF	WHOLE	WHOLE	WHOLE	HALF	WHOLE	WHOLE	HALF	WHOLE	WHOLE
i	1	2 b3	4	5 b6	b7	8	The first mode in a MINOR KEY is AEOLIAN, the NATURAL MINOR SCALE itself. So the HOME CHORD is a MINOR CHORD. As a result, MINOR KEY MUSIC sounds SAD.							
ii ^ø	1 b2	b3	4 b5	b6	b7	8	The 2nd mode is LOCRIAN. It is a HALF DIMINISHED SCALE. A FULL DIMINISHED SCALE would contain a bb7 (which is actually a 6).							
III	1	2	3 4	5	6	7 8	IONIAN is the 3rd mode but notice it is rooted on the b3 of the NATURAL MINOR SCALE. It is the TONIC MODE in the RELATIVE MAJOR KEY.							
iV	1	2 b3	4	5	6 b7	8	The 4th mode is the old 2nd mode, DORIAN. So the iV CHORD in a MINOR KEY is a MINOR CHORD.							
V	1 b2	b3	4	5	b7	8	PHRYGIAN is the 5th mode so the V CHORD is also a MINOR CHORD.							
LYDIAN, the 6th mode is rooted on b6 of the NATURAL MINOR SCALE so the VI CHORD is a MAJOR CHORD.	VI	1	2	3	#4 5	6	7 8							
The 7th mode, MIXOLYDIAN is rooted on b7 so the VII CHORD is MAJOR, rooted a WHOLE STEP below the TONIC in MINOR KEY MUSIC.	VII	1	2	3 4	5	6 b7	8							

SO IN MINOR KEYS, **i-iV-V ARE ALL MINOR CHORDS !**

THIS IS BECAUSE, IN A MAJOR KEY, THE SPACING OF I-IV-V IS THE SAME AS THE SPACING OF Vi-ii-iii. So when Vi becomes i, the old ii and iii (Dorian and Phrygian) become the new iV and V in the RELATIVE MINOR KEY.

Another classic MINOR KEY CHORD PROGRESSION involves STARTING ON THE i CHORD (MINOR), DESCENDING by a WHOLE STEP to the VII CHORD (MAJOR) and then DOWN another WHOLE STEP to the VI CHORD (MAJOR) and then back UP AGAIN. How many hundreds of songs have you heard like this? Try this example in the key of A MINOR ...



THERE IS STILL A BIT MORE TO KNOW ABOUT MINOR KEYS AND WE WILL LOOK AT THAT IN OUR NEXT LESSON.

MODAL MUSIC

LESSON 23

Actually, we'll talk about MODAL MUSIC in just a minute.. There is still one more BIG thing to know about MINOR KEY MUSIC.

REMEMBER WE SAID THAT A

i-iv-V PROGRESSION IN A MINOR KEY CONTAINS ALL MINOR CHORDS.

WELL THAT ISN'T ALWAYS TRUE. IN FACT, MORE OFTEN THEN NOT,

**THE V CHORD CAN BE A MAJOR CHORD
THE SAME WAY IT IS IN MAJOR KEY MUSIC !**

That's why we created the **HARMONIC MINOR SCALE.**

There is yet another HARMONIZATION that you should know about. HARMONIC MINOR is not a scale that you PLAY that often. It is more a THEORETICAL CONSTRUCT that justifies a certain type of CHORD PROGRESSION. If we look at the MODES and EMBEDDED TRIADS that occur when we HARMONIZE a HARMONIC MINOR SCALE, we see that the V CHORD becomes a MAJOR CHORD. What is the only difference between NATURAL MINOR and HARMONIC MINOR? Notice that in HARMONIC MINOR, the 7th has been raised by a half step up from b7 to 7. (This also opens up that AUGMENTED 2nd INTERVAL between the b6 and the 7.) In a NATURAL MINOR HARMONIZATION, the b7 would translate down to become the b3 in the 5th MODE causing the V CHORD to be a MINOR CHORD. By shifting the b7 up to 7 in HARMONIC MINOR, we shift the b3 in the 5th MODE up to become a MAJOR 3rd, changing the V CHORD INTO A MAJOR CHORD!

1	2 b3	4	5 b6	b7	8	NATURAL MINOR
			↓	↓		
			V	1 — Minor 3rd Interval — b3		In the 5th mode of NATURAL MINOR, the 5 translates down to become the 1 and the b7 becomes the b3 making the V CHORD a MINOR CHORD.

1	2 b3	4	5 b6	Aug 2nd Interval	7 8	HARMONIC MINOR
			↓		↓	
			V	1 — Major 3rd Interval — 3		In the 5th mode of HARMONIC MINOR, the 5 becomes the 1 again but the 7 now translates down to become the MAJOR 3rd making the V CHORD a MAJOR CHORD.

Of course the 5th is not the only mode effected by this shifted 7th. All 7 modes are altered and two other EMBEDDED TRIADS also change. Again, all these are laid out on your MUSICAL SLIDERULE. Here is a full HARMONIC MINOR HARMONIZATION.

i	1	2 b3	4	5 b6 → 7 8	HARMONIC MINOR
ii°	1 b2	b3	4 b5 → 6 b7	8	DIMINISHED
III+	1	2	3 4 → #5 6	7 8	AUGMENTED
iv	1	2 b3 → #4 5	6 b7	8	MINOR
V	1 b2 → 3 4	5 b6	b7	8	MAJOR
VI	1 → #2 3	#4 5	6	7 8	MAJOR
VII°	→ 1 b2	b3 b4	b5	b6 bb7 → 8	DIMINISHED

The III CHORD changes from MAJOR to AUGMENTED because the shifted note changes from 5 to #5. But most POP SONGS in MINOR KEYS seem to retain the III CHORD as a MAJOR CHORD as in NATURAL MINOR.

It seems that the majority of songs in MINOR KEYS contain a V CHORD that is a MAJOR CHORD because perhaps we simply prefer the way it sounds. There are however many songs where the V CHORD is MINOR.

Here the ROOT is the shifted note, making the VII CHORD DIMINISHED

MODAL MUSIC

SOME SONGS AREN'T IN MAJOR OR MINOR KEYS

EVEN THOUGH THEIR SCALE/CHORD PROGRESSION/MELODY RELATIONSHIPS STILL RELATE TO THE MAJOR SCALE.

Grab your SLIDERULE and consider this simple CHORD PROGRESSION. What KEY is it in?

Em - A

The PROGRESSION begins with Em. Which CHORD would you end with? Probably still Em. Which CHORD feels like the "HOME CHORD"? The answer is again Em. So you're thinking this PROGRESSION is in the KEY of E MINOR. So you start to jam using E AEOLIAN. But you notice there is a really bad wrong note in there. What's up with that? So you grab your SLIDERULE and set it up to be in Em (put E in the RELATIVE MINOR window in the Vi MODE, AEOLIAN and use the second set of ROMAN NUMERALS at the end of each MODE). Em is now the i CHORD and A is the iv CHORD. What's wrong with this picture?

IT CAN'T BE IN E MINOR BECAUSE A IS THE iv CHORD AND SHOULD BE MINOR.

So you think somehow you miscalculated. Maybe it's in the KEY OF A MAJOR. So you put A in the 1st IONIAN window at the upper left.

BUT IT'S NOT IN A MAJOR. E WOULD BE THE V CHORD AND SHOULD BE MAJOR.

So what KEY is it in? The KEY OF D MAJOR is the only KEY that has an Em and an A CHORD in it. So put D in the upper left window. Em is now the ii CHORD (DORIAN-MINOR) and A is the V CHORD (MIXOLYDIAN-MAJOR). So you begin jamming with a D MAJOR SCALE (you're actually playing E DORIAN and A MIXOLYDIAN) and everything sounds good. That wrong note is gone.

BUT IT'S NO GOOD THINKING YOU'RE IN D MAJOR IF THERE IS NO D ROOT IN THE SONG.

So what KEY are we in? Since Em feels like the HOME CHORD and is occupying the DORIAN position in this particular progression,

THIS IS A PIECE OF MODAL MUSIC SAID TO BE IN THE KEY OF E DORIAN.

Now play this very common PROGRESSION. The D sounds like the "HOME CHORD".

D - C - G - D

So you think maybe it's in the KEY OF D MAJOR. But when you begin jamming with a D MAJOR SCALE you hear that wicked wrong note again. What's wrong this time? You grab your trusty SLIDERULE and set it up in D MAJOR. But notice that there is no C at all in the KEY OF D. Then maybe you notice that G-C-D is actually I-IV-V in the KEY OF G MAJOR. So you begin jamming with a G MAJOR SCALE and once again the wrong note goes away.

BUT WE CAN'T BE IN G MAJOR BECAUSE G DOESN'T FEEL LIKE THE "HOME CHORD".

So what KEY are we in? Since D feels like the HOME CHORD and occupies the MIXOLYDIAN position in this particular KEY,

THIS IS A PIECE OF MODAL MUSIC IN THE KEY OF D MIXOLYDIAN.

DORIAN AND MIXOLYDIAN ARE BY FAR THE MOST COMMON MODAL HARMONIZATIONS AND SO THE ONLY ONES WE WILL INCLUDE HERE.

DORIAN HARMONIZATION

1	2 b3	4	5	6 b7	8	i MINOR
PHRYGIAN						
1	b2	b3	4	5 b6	b7	8 ii MINOR
LYDIAN						
1	2	3	#4 5	6	7 8	III MAJOR
MIXOLYDIAN						
1	2	3 4	5	6 b7	8	IV MAJOR
AEOLIAN						
V MINOR	1	2 b3	4	5 b6	b7	8
LOCRIAN						
Vi DIM.	1	b2	b3	4 b5	b6	b7 8
IONIAN						
VII MAJOR	1	2	3 4	5	6	7 8

MIXOLYDIAN HARMONIZATION

1	2	3 4	5	6 b7	8	I MAJOR
AEOLIAN						
1	2 b3	4	5 b6	b7	8	ii MINOR
LOCRIAN						
1	b2	b3	4 b5	b6	b7	8 iii DIM.
IONIAN						
IV MAJOR	1	2	3 4	5	6	7 8
DORIAN						
V MINOR	1	2 b3	4	5	6 b7	8
PHRYGIAN						
Vi MINOR	1	b2	b3	4	5 b6	b7 8
LYDIAN						
VII MAJOR	1	2	3	#4 5	6	7 8

What else is there to know about HARMONY IN WESTERN MUSIC? Some CHORD PROGRESSIONS CHANGE KEY right within the song. This is called MODULATION and we will be talking about that in LESSON 31. Some music is just so weird that it can't be analyzed in any conventional context. This can run the gamut from advanced EXPERIMENTAL MUSIC to FREE IMPROVISATION to know-nothings making a bunch of weird, ugly noise which, sadly, so much ROCK MUSIC has become.

But there is one more style of HARMONIZATION that is very popular and important to understand. That is this stuff we call THE BLUES. Let's look at it now.

PLAYIN' THE BLUES

LESSON 24

FOR SEVERAL LESSONS NOW, WE HAVE BEEN LOOKING AT THE VARIOUS FORMS OF DIATONIC HARMONY. MAJOR KEY MUSIC - MINOR KEY MUSIC - MODAL MUSIC

We saw that many SCALE/MELODY/CHORD PROGRESSION relationships evolve out of the MODES of the MAJOR DIATONIC SCALE. We created a MODE JAM TAPE so that you can begin practicing, playing and listening to these SCALES and MODES. This will help you ANALYZE and ultimately CREATE your own MELODIES either as a COMPOSER/ SONG WRITER or as a SOLOIST in an IMPROVISATORIAL setting. I'm sure you can hear that a huge percentage of WESTERN MUSIC is structured this way.

But a lot of music we hear has a certain different kind of sound ...a kind of edgy, harder sound ...a kind of dirty ...gritty ...sexy kind of feel ...dark and passionate ...raw and visceral. Your DIATONIC MELODIES just don't seem to have it. It all sounds a little too pretty ...too normal. What is this "bad ass" kind of funk feel? Where do I find that kind of soul? You, my friend, have got THE BLUES.

BLUES is a fairly recent but enormously popular style of HARMONIZATION. Evolving out of BLACK FOLK MUSIC of the 19th Century and catching hold in the expanding urban, industrial society, BLUES had just the right balance of pleasure and pain to seem like "real life" to a people just beginning to see how good it could be ...and just how BAD it could get. CLASSICAL MUSIC seemed just a little too clean ...too pretty.

The guitar has been an integral part of the evolution of BLUES MUSIC. From the earliest pioneers, through the first super stars like ROBERT JOHNSON, LEADBELLY, LONNIE JOHNSON, BLIND LEMON JEFFERSON, BLUES began to fracture into a thousand different and very personal forms. ... JOHN LEE HOOKER, ELMORE JAMES, HOWLIN' WOLF, B.B. KING, MUDDY WATERS, T-BONE WALKER, ALBERT KING, BUDDY GUY, ALBERT COLLINS, SON SEALS ...the list goes on and on. DELTA, CHICAGO, MEMPHIS, DETROIT, NEW ORLEANS, TEXAS BLUES. ...expanding in complexity and spawning JAZZ with EDDIE LANG, DJANGO REINHARDT, CHARLIE CHRISTIAN, HERB ELLIS, BARNEY KESSEL, TAL FARLOW, JIM HALL, JOE PASS, KENNY BURRELL, CHARLIE BYRD, WES MONTGOMERY, JOHN McLAUGHLIN, GEORGE BENSON, PAT METHENY ...RHYTHM AND BLUES, COUNTRY BLUES, SOUL, GOSPEL, MOTOWN ...The BLUES goes electric and ROCK AND ROLL is born ... CHET ATKINS, LES PAUL, EDDIE COCHRAN, SCOTTY MOORE, DUANE EDDIE, BUDDY HOLLY, BO DIDDLEY, CHUCK BERRY, PAUL BUTTERFIELD, ELVIN BISHOP, MICHAEL BLOOMFIELD, JOHN MAYALL ...and into the familiar modern BLUES/ ROCK GUITAR HEROES like ROY BUCHANAN, JOHNNY WINTER, KEITH RICHARDS, ERIC CLAPTON, JIMI HENDRIX, FRANK ZAPPA, STEPHEN STILLS, DUANE ALLMAN, STEVIE RAY VAUGHAN, JIMMY PAGE, JEFF BECK, ROBERT CRAY ...the FUNK of SLY AND THE FAMILY STONE, THE JACKSONS, PRINCE and PARLIAMENT ...morphing into the HARD ROCK heyday of the 70's and 80's ...AEROSMITH, LYNRYD SKYNYRD, BRUCE SPRINGSTEIN, MOTLEY CRUE, VAN HALEN, KISS, AC/DC, ZZ TOP, GUNS & ROSES, LENNY KRAVITZ, JOE SATRIANI, STEVE VAI ... and then into HEAVY METAL mayhem ...BLACK SABBATH, DEEP PURPLE, IRON MAIDEN, POISEN, METALLICA, MEGADETH ...and into the 21st Century with such diverse styles as HIP HOP and RAP ...KENNY WAYNE SHEPPARD, RAGE AGAINST THE MACHINE, BEN HARPER ...indeed every popular artist in the modern era has felt the awesome power of THE BLUES.

BLUES is a highly IMPROVISATORY style of music. We see lots of SOLOS where the players get to make up improvised MELODIES on the spot. Ask anyone and they'll tell you that the essence of THE BLUES is hard to define ...and that may indeed be true. People want to talk about "feel" and "soul" and such and that's cool ...but they can't really tell you much about what they're doing in terms of MUSIC THEORY. But that's not because BLUES is unknowable in terms of conventional HARMONY. It's just that the average BLUES player is often not the most knowledgeable MUSICAL THEORETICIAN you might meet. Remember ...BLUES is a form of FOLK MUSIC and a lot of it's tenants have been passed down through the generations by emulation and imitation. BLUES players "jam" and learn the tricks of the trade from one another ...not so much from THEORY books. So they just can't explain what they do very well. But there is a lot of interesting stuff that you can know about THE BLUES now that you know the basics of how HARMONY works. Though they share a lot of common concepts, there are some readily observable differences between BLUES and DIATONIC MUSIC. BLUES players simply choose to play different notes. These notes are collectively known as "BLUE NOTES".

ALL THE MOST CONVENTIONAL FORMS OF BLUES TUNES EMPLOY A I-IV-V CHORD PROGRESSION

Some people think that this progression defines THE BLUES but as you know, this is the most common CHORD PROGRESSION in all styles of music. In a piece of MAJOR KEY DIATONIC MUSIC, these CHORDS would all be MAJOR and we would expect to find ourselves using our now familiar MAJOR DIATONIC MODES to jam with.

BUT THE SCALE WE MOST OFTEN ASSOCIATE WITH BLUES IS THE MINOR PENTATONIC SCALE



How can this be? We have been taught that you always play MAJOR SCALES over MAJOR CHORDS. But things are different in THE BLUES. The distinction between MAJOR and MINOR gets a little blurred. Playing b3rds over MAJOR CHORDS (called a BLUES 3rd) is one of the odd note choices that gives BLUES it's dirty "bad ass" kind of sound. It's the all time classic BLUE NOTE. Many BLUES RIFFS use the 3 and the b3.

Not all BLUES MUSIC uses MAJOR CHORDS. There are MINOR KEY BLUES where the CHORDS are MINOR and the BLUES 3rd is not so apparent. 5 CHORDS (so called POWER CHORDS) are also common in BLUES, HARD ROCK and HEAVY METAL. They don't contain a 3rd of any type.

TAKE A MINOR PENTATONIC SCALE.
ADD THE BLUES 3rd AND A COUPLE
OF OTHER CHROMATIC "PASSING
TONES" AND WHAT YOU HAVE IS A

BLUES SCALE



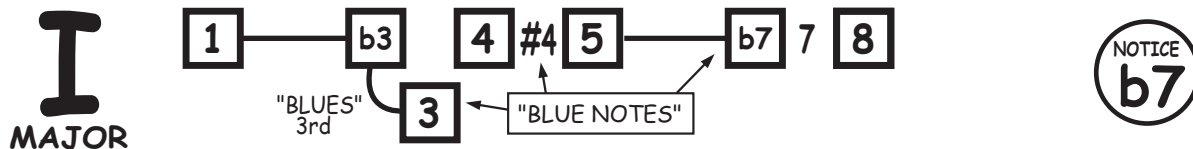
People have different ideas about what actually constitutes a BLUES SCALE and so how to finger it. In your support material you have COLOR CODED FINGER CHARTS for both a MINOR PENTATONIC SCALE and a BLUES SCALE. Note that I left out the MAJOR 3rd in the BLUES SCALE because that is the way it is most often described. But don't forget about the BLUES 3rd and make sure you know how to play it. As with all SCALE/CHORD RELATIONSHIPS, none of this stuff is "carved in stone". These are not the only notes that you hear in THE BLUES. The really clever players can use all 12 notes to jam with but this requires a lot more skill, experience and EAR TRAINING.

SO HOW DO BLUES SCALES RELATE TO THE CHORDS IN A I-IV-V PROGRESSION?

Here we begin to HARMONIZE the BLUES SCALE. We will be looking at it's 4th and 5th MODES as we jam over the IV CHORD and the V CHORD. If we compare this to our normal DIATONIC HARMONIZATION we begin to see where our BLUE NOTES come from.

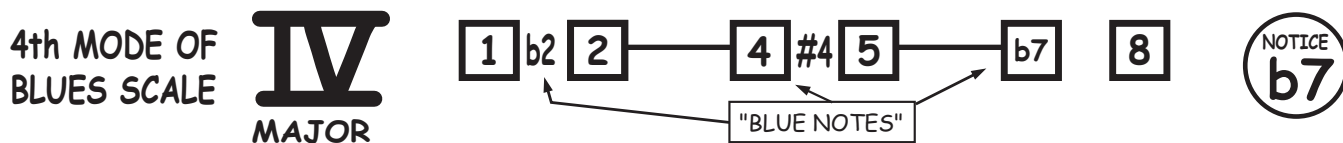
IN A DIATONIC HARMONIZATION WE WOULD EXPECT TO USE THE IONIAN MODE OVER THE I CHORD.

But IONIAN doesn't contain the BLUES 3rd, the #4 or the b7 so these would be considered our first BLUE NOTES.



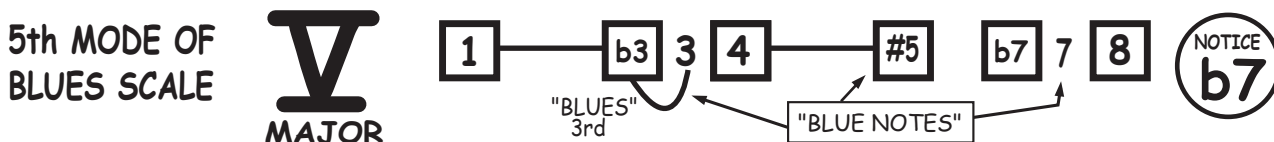
IN A DIATONIC HARMONIZATION WE WOULD EXPECT TO USE THE LYDIAN MODE OVER THE IV CHORD.

But LYDIAN doesn't have the 4, the #2 or the b7 contained in the 4th MODE of the BLUES SCALE ...more BLUE NOTES.



IN A DIATONIC HARMONIZATION WE WOULD USE THE MIXOLYDIAN MODE OVER THE V CHORD.

But MIXOLYDIAN doesn't have the BLUES 3rd, the #5 or the 7 contained in the 5th MODE of the BLUES SCALE.



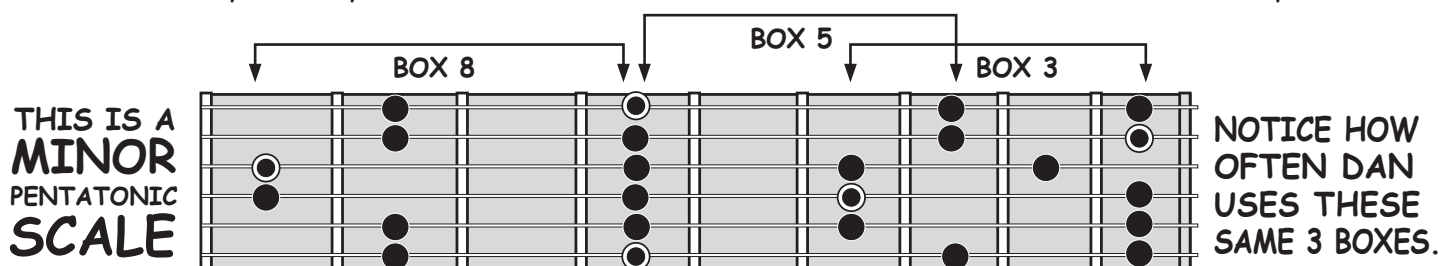
So as you can see, b7 is a huge BLUE NOTE. This makes MIXOLYDIAN another popular and important SCALE in BLUES improvisation. It is a MAJOR SCALE with that all important b7 ...made to go with those "bluesy" 7 CHORDS.

THERE ARE MANY WAYS TO FINGER A BLUES SCALE.

But by far the most common BOX that players use is NUMBER 5.

BOX 8 and BOX 3 are also very common. Notice how these boxes all interconnect up and down the neck.

Lots of times you find yourself SLIDING back and forth between these three BOXES as you solo.



BLUES JAM WITH DAN LAWSON LESSON 25

BEFORE WE GET TO JAMMIN' WITH DAN, LET'S TALK A BIT MORE ABOUT

CHORDS IN BLUES MUSIC

ALL THE MOST CONVENTIONAL FORMS OF BLUES TUNES EMPLOY A **I-IV-V CHORD PROGRESSION**

IN THE
KEY of A
THAT IS

A - D - E

HERE IS THE RHYTHM TAB FOR THE 12 BAR BLUES "SHUFFLE" IN YOUR MODE JAM TRACKS.

12 "BARS" means 12 MEASURES. Then it starts over again. These CHORDS begin as 5 CHORDS (POWER CHORDS) but as you can see, one of the notes keeps moving up and down by a whole step. If you have truly memorized your INTERVAL fingerings, you can easily see that this is the 5th in the CHORD going up to the 6th and back again. This is what is known as a "SHUFFLE" and I bet you've heard this in thousands of BLUES tunes. Later in the jam, I switch to DOMINANT 7 CHORDS (a MAJOR CHORD with a b7 ...written like A7) and that is how I labeled the CHORDS in this PROGRESSION. 7 CHORDS (and EXTENDED 7 CHORDS like 9's and 13ths) are huge in BLUES and b7 is going to emerge as one of our most significant BLUE NOTES. We will be learning more about 7 CHORDS in LESSON 27.

A7 (I)	D7 (IV)	A7 (I)	D7 (IV)
2 24 42 24 4 0 00 00 00 0	2 24 42 24 4 0 00 00 00 0	2 24 42 24 4 0 00 00 00 0	2 24 42 24 4 0 00 00 00 0

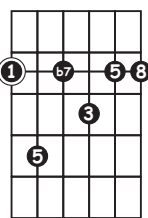
A7 (I)	E7 (V)	D7 (IV)	A7 (I)	E7 (V)
2 24 42 24 4 0 00 00 00 0	2 24 42 24 4 0 00 00 00 0	2 24 42 24 4 0 00 00 00 0	2 24 42 24 4 0 00 00 00 0	2 24 42 24 4 0 00 00 00 0

BLUES MUSIC HAS LOTS OF 7 CHORDS

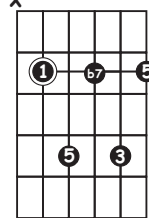
Officially known as DOMINANT 7 CHORDS (not to be confused with DIMINISHED 7 CHORDS), these very common CHORDS all contain the b7 and so are spelled 1-3-5-b7. This means that you

might see MIXOLYDIAN and AEOLIAN MODES showing up in BLUES SOLOING as well. EXTENDED DOMINANT CHORDS like 9's and 13's are really just 7 CHORDS that have been "fancied up" by adding the 2 or 6. We'll be learning more about them in LESSON 27. Here are some common fingerings.

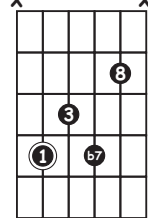
E FORM



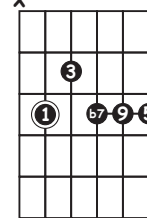
x A FORM



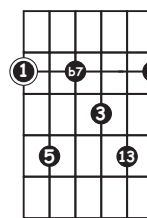
x C FORM x



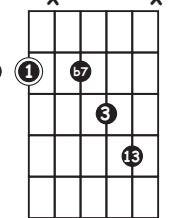
x 9 CHORD



13



x 13 x



We will be learning more about these fascinating CHORDS when we get to LESSON 27. FINGERINGS for these CHORDS can be found in the COLOR CODED CHORD CHARTS that came with your support material package.

BLUES JAM WITH DAN LAWSON

I first met Dan Lawson back in 1986. His band THE KEEP was part of a big benefit project we all did to raise money for THE MARCH of DIMES. Over the years, I've had the privilege of working on several projects with Dan and when I first thought to start looking for the perfect guy to show us all these cool BLUES/ROCK RIFFS, I realized there was really only one man to call.

Come on down to Cape Cod on a Friday or Saturday night and you're bound to catch the fabulous DAN LAWSON BAND rockin' at one or another of their favorite clubs. With an awesome new rhythm section to back him up, Danny is finally beginning to attract the attention his fine playing and singing deserves. As we mentioned in the show, Dan recently took 2nd place (and many there that night thought he should have been #1) in an international JIMI HENDRIX BLUES GUITAR COMPETITION sponsored by the Hendrix family themselves! You can learn more about THE DAN LAWSON BAND, hear cuts from their new album and get club dates and info by going to their web site at www.danlawson@jenrikentertainment.com.

Danny wows us again here in LESSON 25 with his massive vocabulary of classic BLUES CLICHES'. A lot of BLUES guitar involves funny little "finger tricks" that players have created over the years and passed along to other players through live performance and recordings. You can hear RIFFS that MUDDY WATERS created 40 years ago on the new ERIC CLAPTON album and so on. Danny has been studying the masters since he was a kid. His uncle ROY PARKS was an internationally known player who recorded with folks like TEX RITTER, DOTTY WEST, THE BLUE RIDGE MOUNTAIN BOYS and many others. I guess Dan's got THE BLUES in his blood!

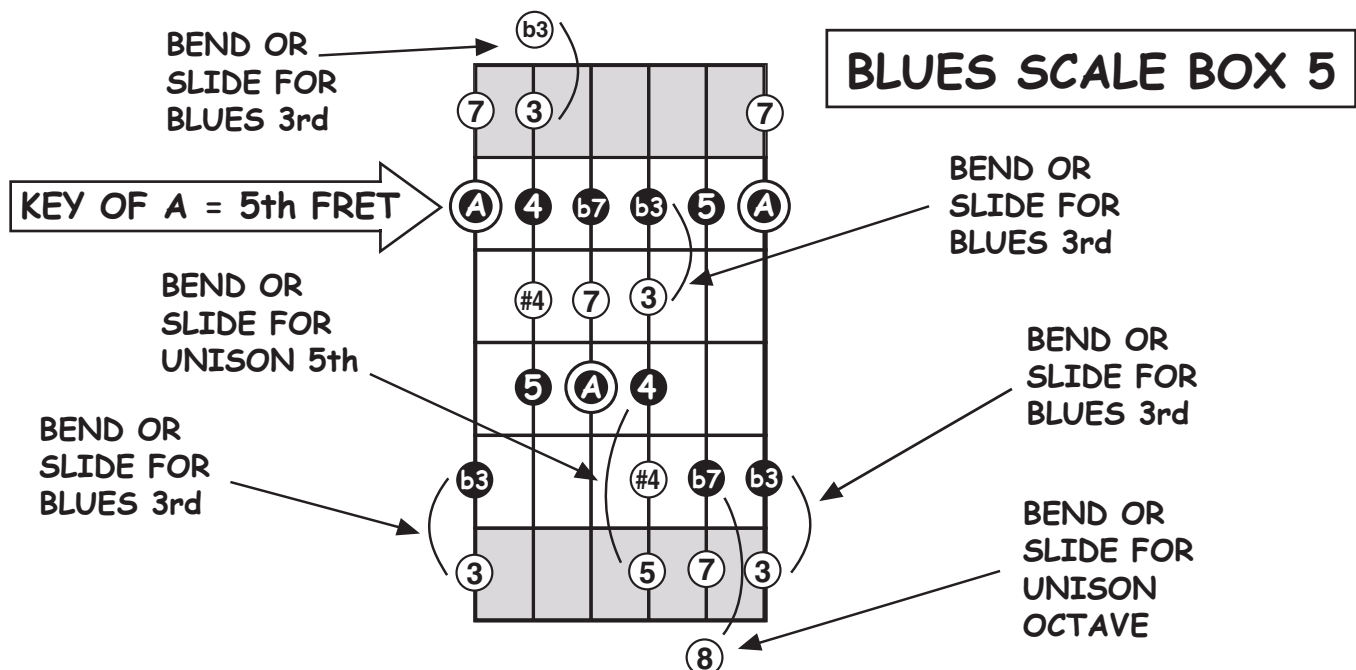
About 8 minutes into our video here, you will see a small TIME CLOCK appear in the lower right corner of your screen. This clock will sync up with the TABLATURE that I have transcribed showing you how to finger the particular riff that Danny is playing at that moment on the tape. It will be tough mastering these LICKS from the TAB. Dan makes it look so easy! You will have to play them dozens of times to get them smooth and up to speed. But keep at it. These BENDS, SLIDES, HAMMER-ONS, PULL-OFFS, TRILLS and PALM MUTES will add a classic BLUES "spark" to your soloing. You'll be feelin' low down and "BAD ASS BLUE" in no time!

THE TAB FOR ALL THESE CLASSIC LICKS IS IN YOUR SUPPORT MATERIAL PACKAGE.

If you need to review READING TABLATURE, refer back to LESSONS 9 and 10. Master these riffs and then try them out yourself using the BLUES JAM PROGRESSION (#5) of your MODE JAM TRACKS dubbed onto the 2nd half of the same disc or tape with BLUES JAM WITH DAN LAWSON (#25). It is essentially the same CHORD PROGRESSION that Dan and I use to do the demonstrations on the show itself. It's all in the KEY OF A. Many of these same RIFFS would also work with PROGRESSIONS 2, 3 and 4 but listen carefully. You may have to CHANGE KEY (for example you are in F# in PROGRESSION 4) and so change frets.

NOTICE THAT DAN IS SO OFTEN PLAYING OUT OF BOX 5.

HERE'S A BREAKDOWN OF WHAT NOTES HE'S PLAYING AND HOW THE BENDS ACTUALLY WORK.



⑤ TRIADS

LESSON 26

We are now going to begin talking about **CHORDS** ...LEVEL 5 of Scotty's 6 LEVELS OF PITCH THEORY. We have already talked about **SILENCE** and **THE NOTE**. We talked a great deal about **INTERVALS**. Then we went on to the huge subject of **SCALES AND MODES**. In the last few lessons, we've begun to see that **SCALES ALL HAVE CHORDS EMBEDDED INSIDE THEM** and the **SCALE** that goes with the **CHORD** is the **SCALE THAT HAS THAT CHORD INSIDE IT**. Look at the **CHORD PROGRESSION** embedded inside a system of **MODES** and you are **HARMONIZING A SCALE**. Scales are thought of as the main ingredient in **MELODIES**, the **HORIZONTAL ASPECT** of **MUSIC**. Now it is time to learn more about the **STRUCTURE** of the **CHORDS** inside these **HARMONIZATIONS** ...the **VERTICAL ASPECT** of music.

We talked a bit about **CHORDS** in **LESSONS 4-6**. We learned that there are about 50 **TYPES** of chords and any chord can be **PLAYED IN ANY KEY**. There are 5 **BASIC FINGER FORMS** so any chord can be played 5 different ways. This is what **BAR CHORDS** are all about. We learned how to practice chords using a **KEYBOARD DRUM MACHINE**.

But we still don't know much about the "**SPELLING**" of these **CHORDS** or their **INTERVAL PATTERNS**. For many of you, playing **CHORDS** is the **BIG THING** when it comes to playing guitar. It's the thing you want to understand most and best, Well in that sense, we've saved the best for last. Learning about **CHORDS** is fun and easy. There are just a few things to know.

CHORDS ARE MULTIPLE NOTES PLAYED SIMULTANEOUSLY.

As in **SCALES**, these notes are separated by very specific **INTERVALS**. You will recall that the notes in **SCALES** are most often separated by 2nds ...Half Steps = Minor 2nds Whole Steps = Major 2nds 1 and 1/2 Steps = Augmented 2nds
One generalization you can make is that the notes are spaced **FARTHER APART** in **CHORDS** than they are in **SCALES**.

IN OUR MOST FUNDAMENTAL CHORDS, THE NOTES ARE MOST OFTEN A MAJOR OR MINOR 3rd APART.

This is known as **TERTIAN HARMONY**. So much of the **HARMONY** in **WESTERN MUSIC** involves taking a note ...adding the note that is a 3rd above that ...and then adding the note that is a 3rd above that ... and another 3rd ... and so on. You will see so many of your common **CHORDS** structured that way. (One of the secrets to **MODERN JAZZ HARMONY** is an increased use of what are called **QUARTEL HARMONIES** ...involving notes separated by 4ths. But **POP/ROCK** is still mostly involved with **HARMONY** based on 3rds.)

TRIADS ARE YOUR MOST BASIC 3 NOTE CHORDS.

THERE ARE FOUR TYPES OF TRIADS DIVIDED INTO TWO CATEGORIES AND ONCE AGAIN IT'S ALL ABOUT THE PATTERN OF INTERVALS.

Some of this you already know because we have covered it in other parts of the program. Examples are written here in **KEY of C** but, of course, you'll see them in all 12 keys.

UNEQUAL INTERVAL TRIADS

Each note has a specific position within the chord
ONE ROOT

C	MAJOR TRIAD
Cm ^{C-}	MINOR TRIAD

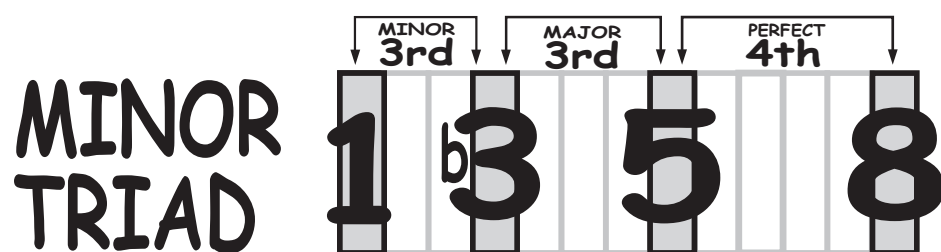
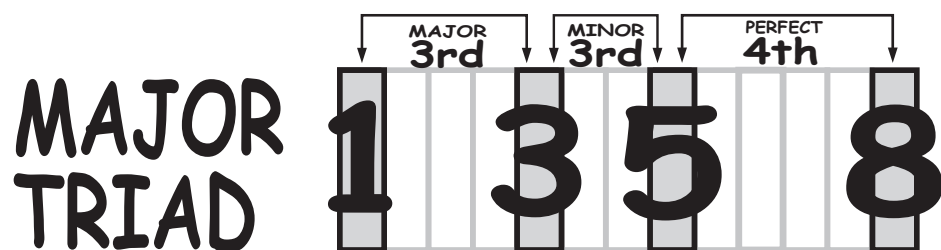
EQUAL INTERVAL TRIADS

Each note can assume any role within the chord.
Any of the notes could be the **ROOT**.

Caug ^{C+}	AUGMENTED TRIAD
Cdim ^{C°}	DIMINISHED TRIAD

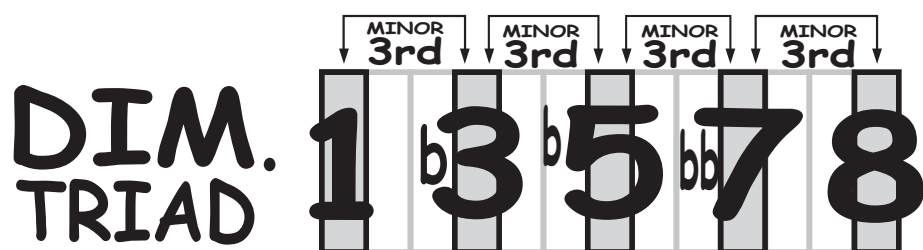
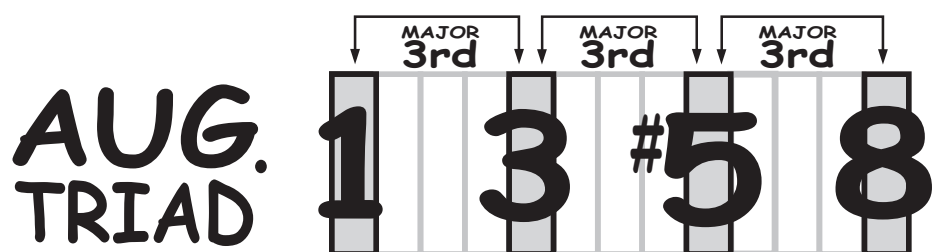
THESE CHORD SPELLINGS ARE ALL NICELY LAID OUT ON YOUR **SLIDERULE**.

THE UNEQUAL INTERVAL TRIADS



MAJOR and MINOR TRIADS are our most popular chords. The HAPPY sounding MAJOR TRIAD starts with a MAJOR 3rd INTERVAL and then a MINOR 3rd is stacked on top of it. The SAD sounding MINOR TRIAD starts with a MINOR 3rd and then stacks a MAJOR 3rd on top of it. The very UNEQUALNESS of the spacing in these TRIADS is the thing that allows each note to occupy it's own PRIVILEGED position within the pattern. Your ear will tell you that there is only one note that will sound like the ROOT. These chords are easiest to COMPOSE with because they fit so nicely into our DIATONIC HARMONIZATIONS.

THE EQUAL INTERVAL TRIADS



These are the SYMMETRICAL TRIADS. The notes in an AUGMENTED TRIAD are all a MAJOR 3rd apart. In EQUAL INTERVAL TRIADS, any one of the notes in the pattern could sound like the ROOT. These TRIADS do not have note 5 in them. Notes in a DIMINISHED TRIAD are all a MINOR 3rd apart. They sound TENSE. Their SYMMETRY makes them ideal as PIVOT CHORDS in songs that MODULATE (change KEY) and as CHORD SUBSTITUTIONS. These chords show up most in more sophisticated Pop/Rock, JAZZ and CLASSICAL. Rock players often don't know how to handle this stuff at all. You use your SYMMETRICAL SCALES to play over these chords

Take a DIMINISHED TRIAD and add the note another MINOR 3rd above it (the bb7...known as the DIMINISHED 7th is actually NOTE 6) and you have a FULL DIMINISHED 7th CHORD. A HALF DIMINISHED CHORD would have the b3 and b5 but not the bb7. It would have the b7. It is also called a MINOR 7 b5 CHORD. The LOCRIAN MODE works better when jamming here.

TRIADS are your most basic chords. When we begin to stack other notes on top of them (most often also a 3rd apart), we get our familiar "fleshed out" NUMBERED CHORDS. CHORD NAMES are a "secret code" that tells you what notes are in the chord itself. You are most often given a TRIAD and then told what NOTES to ADD TO IT.

SO A C6 CHORD IS

1 - 3 - 5 - 6

AND A Cm6 CHORD IS

1 - b3 - 5 - 6

BUT JUST WHEN YOU START TO GET THAT DOWN, ONCE AGAIN YOU FIND OUT IT'S NOT QUITE THAT SIMPLE !

CHORDS

LESSON 27

TAKE NOTES AND ADD THEM TO A TRIAD AND YOU GET YOUR NUMBERED CHORDS. THAT'S HOW IT'S SUPPOSED TO WORK. BUT JUST WHEN YOU GET TO THINKING THAT YOU UNDERSTAND THAT, YOU FIND THERE ARE A NUMBER OF

EXCEPTIONS AND SURPRISES INVOLVING THE NUMBER SYSTEM WHEN NAMING CHORDS

C7.....1-3-5-b7

DOMINANT 7

Cmaj7..1-3-5-7

Cm7....1-b3-5-b7

Cm^{maj7}...1-b3-5-7

The DOMINANT 7 CHORD is the most common numbered chord in all styles of Western Music and yet it could be your first point of confusion when it comes to CHORD STRUCTURE. You might think it should go 1-3-5-7 but it doesn't. The number 7 very often means note b7 when naming chords. The DOMINANT 7 is the b7. Think back to the DEGREES OF THE MAJOR DIATONIC SCALE. The V is called the DOMINANT. It is the domain of the V CHORD and so very important. Notice that the 5th MODE, MIXOLYDIAN, is a MAJOR SCALE with a b7. The V CHORD is so often a DOMINANT 7 CHORD that it is often referred to as the $\text{V}7$ CHORD. The chord that actually goes 1-3-5-7 is called a MAJOR 7 CHORD and the abbreviation maj is spelled out in the chord name. A MINOR 7 CHORD is a MINOR TRIAD with the DOMINANT 7. And then there is the rare and interesting MINOR MAJOR 7...a MINOR TRIAD with the MAJOR 7th.

C^{SUS 4}1-4-5

C^{SUS 2}1-2-5

C7^{SUS 4}1-4-5-b7

SUSPENSION

SUSPENSION isn't something you add to a TRIAD, it's a way of MODIFYING the TRIAD. A SUSPENSION REPLACES THE 3rd IN THE CHORD. This sets up some interesting TENSIONS that are very commonly used. You FEEL SUSPENDED like the music is restless ...UNRESOLVED. This is the classic meaning of CONSONANCE and DISSONANCE in MUSIC THEORY. A SUS4 is kind of like the "AH-" in "AH-MEN" at the end of the CHURCH HYMN. You don't feel complete and at rest again until the SUSPENDED 4th is released and drops back down to the MAJOR 3rd at "-MEN"...it's point of RESOLUTION and CONSONANCE within a MAJOR CHORD.

C9.....1-3-5-b7-9

EXTENDED CHORDS

C^{add 9}.....1-3-5-9

Cm9....1-b3-5-b7-9

Cmaj9..1-3-5-7-9

C11....1-(~~3~~)-5-b7-9-11

Cm11..1-b3-5-b7-(9)-11

C13....1-3-5-b7-(9)-(11)-13

Cm13..1-b3-5-b7-(9)-(11)-13

Add yet another note above 7 and you are talking EXTENDED CHORDS.

The first EXTENSION is the 9th. Most all of these have 7 in them.

If it contains b7, it is called an EXTENDED DOMINANT CHORD. An ADD9 CHORD is simply a TRIAD with the 9 added to it. The MAJ9 CHORD CONTAINS a MAJ7. The 9 CHORD is common in BLUES.

The next EXTENSION is the 11th. It's like a 4 in the next octave. NOTICE THAT IT RETAINS THE PREVIOUS EXTENSION (the 9). The MAJOR 3rd is usually left out of the 11 CHORD because it clashes with the 11. They have a kind of "otherworldly" sound because they are classic POLYCHORDS ...a chord made up of elements of other chords all "smashed" into one and competing for your ears attention.

THE 13 CHORD is very big in JAZZ, ROCKABILLY and to a lesser extent, BLUES. You have a lot of VOICING options with these chords ...often simply coming down to what's POSSIBLE with your fingers! The 13 CHORD is often played as a DOMINANT 7 CHORD with an added 6th.

NOW YOU KNOW THE CODE. YOU SHOULD BE ABLE TO LOOK AT ANY CHORD AND FIGURE OUT WHAT NOTES ARE IN IT. THESE EXAMPLES ARE ALL NOTED AS THEY WOULD APPEAR IN THE KEY OF C.

C.....	MAJOR.....	1 3 5
Cm.....	MINOR.....	1 b3 5
Caug...C+.....	AUGMENTED.....	1 3 #5
Cdim...Co.....	DIMINISHED TRIAD.....	1 b3 b5
Csus2.....	SUSPENDED SECOND.....	1 2 5
Csus4.....	SUSPENDED FOURTH.....	1 4 5
Cb5.....	FLAT FIVE.....	1 3 b5
C6.....	SIX.....	1 3 5 6
Cm6.....	MINOR SIX.....	1 b3 5 6
C6/9.....	SIX ADD NINE.....	1 3 5 6 9
Cm6/9.....	MINOR SIX ADD NINE.....	1 b3 5 6 9
Cdim7.....	DIMINISHED 7th CHORD. (FULL DIM.)	1 b3 b5 bb7 (6)
C7.....	DOMINANT SEVEN.....	1 3 5 b7
Cm7.....	MINOR SEVEN.....	1 b3 5 b7
Cmaj7.....	MAJOR SEVEN.....	1 3 5 7
C7sus4.....	SEVEN SUSPENDED FOUR.....	1 4 5 b7
Cm/maj7.....	MINOR MAJOR SEVEN.....	1 b3 5 7
C7#5.....	SEVEN SHARP FIVE.....	1 3 #5 b7
C7b5.....	SEVEN FLAT FIVE.....	1 3 b5 b7
Cm7b5...Cø.....	MINOR SEVEN FLAT FIVE. (HALF DIM.)	1 b3 b5 b7
C7#5#9.....	SEVEN SHARP FIVE SHARP NINE.....	1 3 #5 b7 #9
C7#5b9.....	SEVEN SHARP FIVE FLAT NINE.....	1 3 #5 b7 b9
C7b5b9.....	SEVEN FLAT FIVE FLAT NINE.....	1 3 b5 b7 b9
C7#9.....	SEVEN SHARP NINE.....	1 3 5 b7 #9
C7b9.....	SEVEN FLAT NINE.....	1 3 5 b7 b9
Cadd9.....	ADD NINE.....	1 3 5 9
C9.....	NINTH.....	1 3 5 b7 9
Cm9.....	MINOR NINE.....	1 b3 5 b7 9
Cmaj9.....	MAJOR NINE.....	1 3 5 7 9
Cm9maj7.....	MINOR NINE MAJOR SEVEN.....	1 b3 5 7 9
C9#5.....	NINE SHARP FIVE.....	1 3 #5 b7 9
C9b5.....	NINE FLAT FIVE.....	1 3 b5 b7 9
C11.....	ELEVENTH.....	1 (3) 5 b7 9 11
Cm11.....	MINOR ELEVENTH.....	1 b3 5 b7 9 11
C13.....	THIRTEENTH.....	1 3 5 b7 (9) (11) 13
Cm13.....	MINOR THIRTEENTH.....	1 b3 5 b7 (9) (11) 13
C13b9.....	THIRTEEN FLAT NINE.....	1 3 5 b7 b9 (11) 13
C13b5b9.....	THIRTEEN FLAT FIVE FLAT NINE.....	1 3 b5 b7 b9 (11) 13

THESE FORMULAS ARE ALL ON YOUR SLIDERULE. YOU CAN CONVERT ALL YOUR NUMBER PATTERNS INTO ALPHABET SPELLINGS USING IT.

CHORD VOICING

LESSON 28

KNOWING THE FORMULAS FOR YOUR CHORDS IS VERY IMPORTANT ... BUT THAT'S ONLY THE BEGINNING !

Remember, those notes simply have to be PRESENT. They don't have to be in any particular ORDER and they don't have to be in any particular OCTAVE. You can REPEAT them as you choose or not. So you can probably see that there are actually a lot of different ways to ARTICULATE any given CHORD FORMULA. This is the fascinating area of CHORD VOICING. You might say that a CHORD has only one FORMULA but it has many VOICINGS. Not only do you need to know what CHORD to play, you also need to know which way to play it. If you're trying to figure out a COVER and you want to play it exactly like the original artist, you must identify not only the CHORD, but also the FORM that they are playing it in. Remember back in Lesson 4 we learned that there are 5 basic POSSIBLE FINGER FORMS that you can play guitar CHORDS in ...E-A-D-G and C. We are about to UNDO that OVERSIMPLIFICATION. There are in fact many more ways to VOICE any CHORD on the guitar (but since they could all be considered SUB VOICINGS, derived from our original 5 FORMS, this 5 FORM concept still retains much validity). You may need to go back and REVIEW these FORMS.

THERE ARE MANY WAYS TO VOICE ANY CHORD.

CLOSED VOICINGS

CLOSED VOICED CHORDS are where the notes are CLOSE TOGETHER ...most often a MAJOR or MINOR 3rd apart in music based on TERTIAN HARMONY. You do see a lot of notes only separated by 2nds in CHORDS as well. Take, for example, the distance between the 5 and the 6 in a C6 chord ...or the 4 and 5 in a SUS4 CHORD. They are only a whole step apart.

OPEN VOICINGS

In OPEN VOICINGS, the notes can be any distance apart that you want. A simple MAJOR TRIAD could span SEVERAL OCTAVES. Here we could see a lot of COMPOUND INTERVALS and INVERTED INTERVALS. In CLOSED VOICINGS, you see a lot of MAJOR and MINOR 3rds. In OPEN VOICINGS, you will see a lot of MAJOR and MINOR 6ths separating the notes. Here, they are FUNCTIONING as INVERTED 3rds (see INVERTED INTERVALS Lesson 13).

INVERSION

By changing the ORDER that you play the notes in, you create various INVERSIONS of the CHORD. If the ROOT note is the lowest note you are playing, the CHORD is said to be in ROOT

POSITION. If you shift the notes all up one so that the 3rd is the lowest note, this is called 1st INVERSION. If the 5th is on the bottom, you are playing in 2nd INVERSION. TRIADS would only have a ROOT POSITION and TWO INVERSIONS but more complex CHORDS have several. Here are the INVERSIONS of a DOM7 CHORD.

1-3-5-b7
ROOT POSITION

3-5-b7-1
1st INVERSION

5-b7-1-3
2nd INVERSION

b7-1-3-5
3rd INVERSION

SLASH CHORDS

CHORDS where the ROOT NOTE is not the lowest PITCH are also known as SLASH CHORDS and are written with a SLASH in the name ..for example, C/G.

This would be a C MAJOR CHORD with G as the BASS NOTE played under it. Changing the BASS NOTE under a CHORD can give it a whole different SOUND and FEEL. Beginners trying to figure out songs BY EAR are often confused by them. It is harder to hear which note is actually the ROOT. Most often, this SLASH BASS will be a note that is ALREADY IN THE CHORD ANYWAY (In our example C/G, the G is the 5th of C and is always in a C MAJOR CHORD. It just isn't usually in the BASS). Here then are the common SLASH CHORDS in the KEY of C ...

C/E

E is the 3rd
in this MAJOR CHORD
1st INVERSION

C/G

G is the 5th
in this MAJOR CHORD
2nd INVERSION

Cm/Eb

Eb is the b3
in this MINOR CHORD
1st INVERSION

Cm/G

G is the 5th
in this MINOR CHORD
2nd INVERSION

C7/Bb

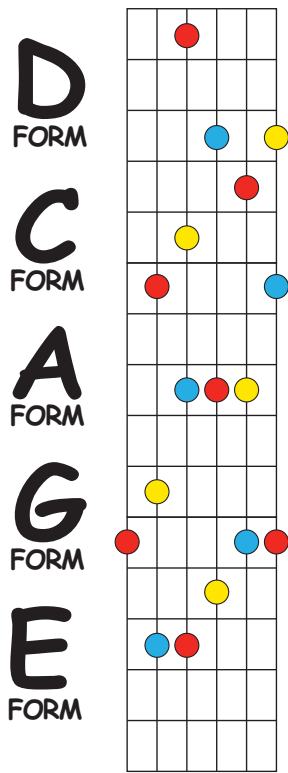
Bb is the b7
in this DOM7 CHORD
3rd INVERSION

Often times, SLASH CHORDS pertain more to what the BASS PLAYER in the band is doing and don't affect the guitar part. You could see a D/F# in the music and you would continue to play a plain old D MAJOR CHORD. But notice that the note the BASS PLAYER is playing has changed to F#. Other ALTERED BASS NOTES can be part of some BASS RIFF or "WALKING" BASS LINE or the BASS NOTE in a POLYCHORD ...where more than one TONAL CENTER competes for your ears attention.

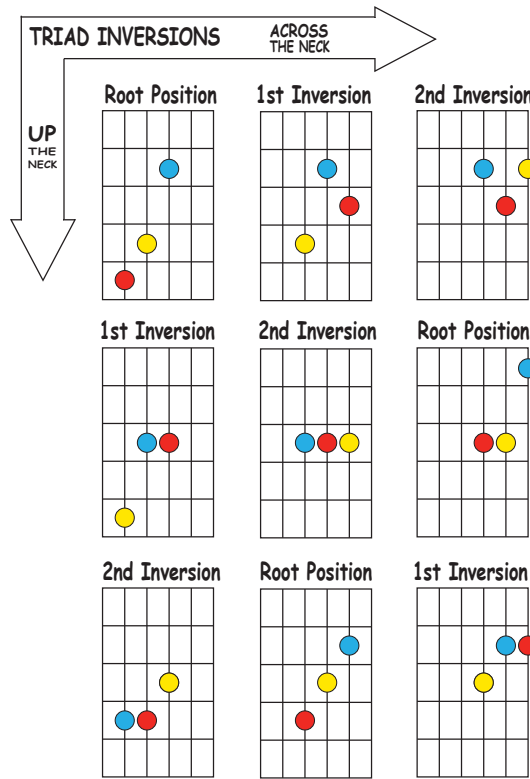
As an example, here are MAJOR and MINOR CHORD and TRIAD VOICINGS of various types.

MAJOR CHORDS

In case you haven't noticed
THE 5 CHORD FORMS
ALL CONNECT UP
AND DOWN THE NECK.

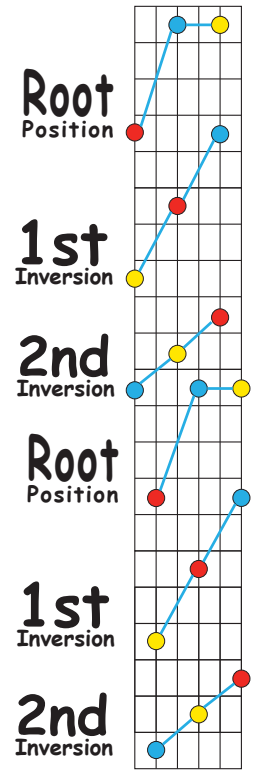


MANY SMALLER TRIAD SHAPES CAN BE
DERIVED FROM THE 5 FORMS AS YOU INVERT
ACROSS AND UP AND DOWN THE NECK.

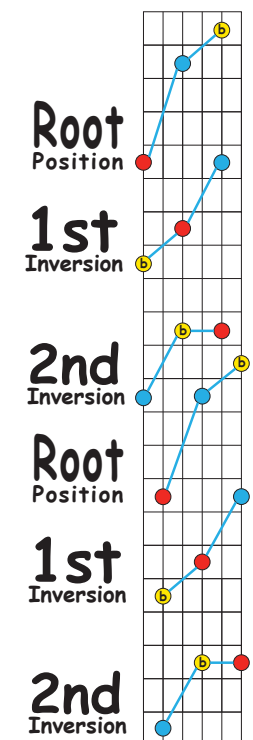
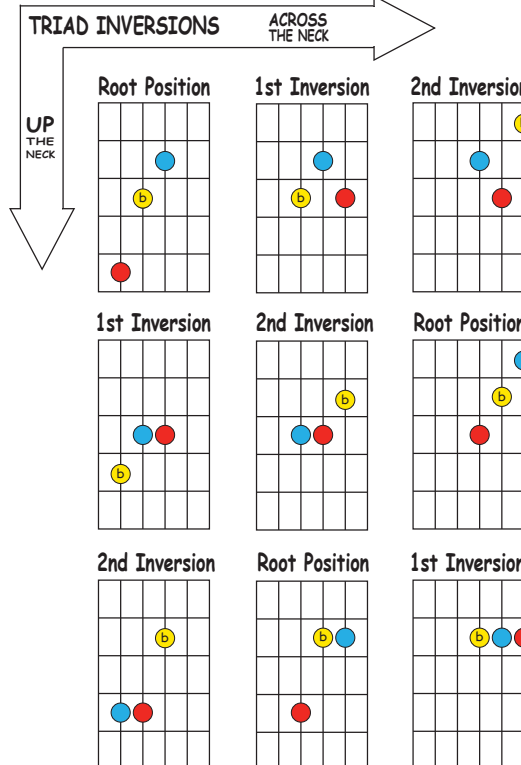
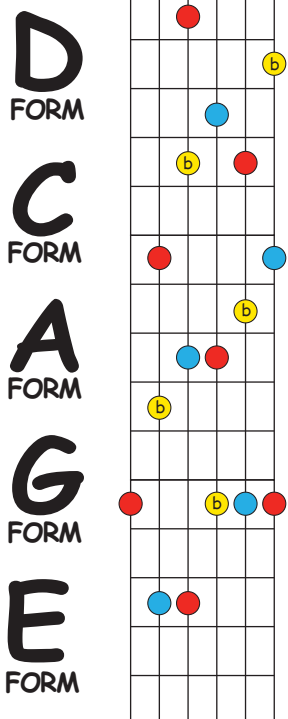


OPEN VOICINGS

Notice the notes here are often a MAJOR or MINOR 6th apart acting as INVERTED 3rds. You must FINGER PICK or MUTE the in between strings when playing these shapes.



MINOR CHORDS



These are just MAJOR and MINOR CHORDS. Of course, a comparable chart could be drawn up for ANY TYPE of CHORD.

MORE ABOUT CHORD VOICING

LESSON 29

SO YOU HAVE TO LEARN THESE CHORDS AND, LIKE ALL PITCH PATTERNS, YOU HAVE TO KNOW THEM ALL 4 WAYS.

THINK * READ * HEAR * SPEAK

By now, you've learned quite a bit about the THEORY and STRUCTURE of CHORDS. Like all PITCH PATTERNS, they are like WORDS in the language of music and they have SPELLINGS created out of our 12 NOTE CHROMATIC MUSICAL ALPHABET. When you arrange your fingers on the neck of your guitar to play a CHORD, you are "TYPING" that "word" on your "MUSICAL TYPEWRITER"/GUITAR. All these SPELLINGS are on your MUSICAL SLIDERULE. We will begin looking further into their FINGERING in just a moment.

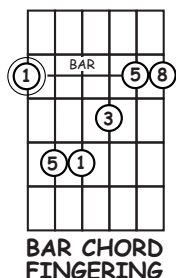
We know that each CHORD has it's own SOUND and "FEELING" and through the process of EAR TRAINING, we can learn to recognize these CHORDS when we hear them in the songs around us (or when we hear them INSIDE OUR OWN HEADS when we are COMPOSING MUSIC ...writing songs). Often, the easiest way to tell what kind of CHORD you're hearing is by noticing how it makes you FEEL or what PICTURES come to your mind. MAJOR sounds HAPPY ...MINOR sounds SAD ...DIM sounds TENSE ...6 sounds PASTORAL ..DOM7 sounds "BLUESEY" ... MAJ7 sounds RELAXED ...POWER CHORDS (5) sound ANGRY and SEVERE ... and the list goes on and on. You will have fun and derive great SATISFACTION from being able to recognize more and more CHORDS BY EAR.

SCOTTY'S COLOR CODED CHORD CHARTS

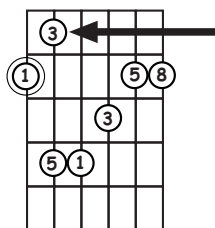
This is not meant to be a complete chord encyclopedia though I think you will find it very useful. There are many fine chord books available and my advice is to check them all out. These are the forms that I find most useful, common and representative of what the chord is trying to SOUND like. Complex chords that do not have the ROOT NOTE in the BASS (or sometimes not present at all) can be confusing to the ear. These chords are generally only present in more complex JAZZ and CLASSICAL playing anyway and so are not included here. The symbol (x) indicates a redundant, optional or unnecessary note and shapes that have several of these may have two or more sub-shapes within them. Try to find them all. Notice that in some complex chords the 5 is SACRIFICED in order to facilitate the playing of other necessary notes. You can get away with this because the 5 is such a loud OVERTONE that your ear will kind of hear it even if you don't actually play it. I did not include any forms where the 3rd is sacrificed though some of these can be useful. Many of these finger shapes involve STRETCHES and MUTES that beginners will find difficult or impossible to play at first. Don't become discouraged! If you continue to practice regularly, your fingers, ligaments and joints will loosen up and you will get to where you can play them. These are all MOVABLE FORMS but note that any of them can be used as OPEN CHORDS in certain keys by moving them down the neck until the LOWEST FINGER DOTS BECOME OPEN STRINGS. There are many interesting and novel CHORD VOICINGS that use OPEN STRINGS in particular keys. These are too many and varied to include here. You will have fun discovering and exploring them on your own as you become more and more familiar with your fretboard. When confronted with a new and novel CHORD VOICING, don't just play it. ANALYZE IT! You can learn so much that will help you when you compose your own music!

Once again, the DEGREES OF THE SCALE are represented by the COLORS OF THE SPECTRUM. Notice that the TRIADS are in the PRIMARY COLORS ...RED-YELLOW-BLUE.

ARPEGGIOS



BAR CHORD FINGERING



ARPEGGIO FINGERING

Take the NOTES and INTERVALS in a CHORD and articulate them in a SEQUENCE and you are playing an ARPEGGIO. They are very common in all styles of music. When you hear someone FINGER PICKING, they are actually playing ARPEGGIOS. IMPROVISED SOLOS can get pretty boring if you constantly use only the smaller INTERVALS (2nds) in SCALES to separate the notes in your MELODY. Learn to "jump around" using the larger INTERVALS (3rds and 4ths) present in ARPEGGIOS and your SOLOS will acquire a profound new depth that many players can never dream of. Since ARPEGGIOS contain the same notes as CHORDS, you can use your CHORD FINGERINGS to play ARPEGGIOS (Just pluck the strings in a sequence or FINGER PICK them rather than STRUMMING them all at once). But look at the ARPEGGIO FINGER CHARTS included in with your CHORD CHARTS and you will see a separate set of fingerings for them that differ slightly from your CHORD FINGERINGS. This is because when playing ARPEGGIOS, YOU CAN PLAY TWO NOTES ON THE SAME STRING because you are playing the notes in a SEQUENCE. Obviously you can't play two notes on the same string when STRUMMING CHORDS. Consider this one example from an E FORM MAJOR BAR CHORD. In the CHORD FINGERING, you can't play the MAJOR 3rd on your A STRING because you're already playing the 5th on that string. But in the comparable ARPEGGIO FINGERING you can play both the 3rd and the 5th in SEQUENCE. ARPEGGIOS are actually fun to learn and practice. Make sure you use your DRUM MACHINE to stay in time!

ALTERED TUNINGS

Many of you know that the GUITAR is not always tuned the normal way. There are any number of ALTERED TUNINGS and many popular players use them. Several involve tuning so that strumming the OPEN STRINGS produces a CHORD. We don't have time to go into these here but plenty of information is available on them. ALTERED TUNINGS can produce interesting and exotic CHORD VOICINGS but often make it more difficult to improvise MELODIES on the instrument.

4 PART HARMONY

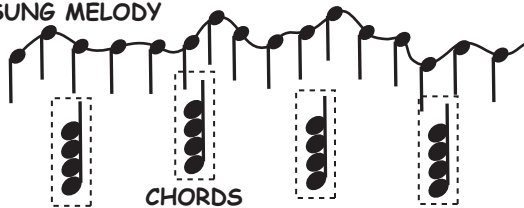
We live in an exciting period in MUSIC HISTORY. I call it the ERA OF THE POLYPHONIC INSTRUMENT (instruments that can play several notes SIMULTANEOUSLY ...CHORDAL INSTRUMENTS like the GUITAR or KEYBOARD). Before this era, POLYPHONIC INSTRUMENTS were more rare and costly and more people were apt to take up MONOPHONIC INSTRUMENTS (most BAND and ORCHESTRAL instruments can only play one note at a time). In order to produce CHORDS, you needed to have several people all playing at the same time (or SINGING. The HUMAN VOICE is, of course, a MONOPHONIC INSTRUMENT. So this would also include CHORAL MUSIC). Music written for ENSEMBLES of MONOPHONIC INSTRUMENTS can be quite different from music composed on a POLYPHONIC INSTRUMENT. For one thing, ENSEMBLE MUSIC doesn't have to have the NOTE COMBINATION RESTRICTIONS imposed by the PHYSICAL LIMITATIONS of POLYPHONIC INSTRUMENTS (not that they don't have their own limitations). Composers writing for

ENSEMBLES must think about CHORDS and CHORD PROGRESSIONS in a more IN DEPTH way because they have to pay attention to how each individual instrument (or VOICE) MOVES in PITCH as it proceeds through the piece. Each instrument is in fact producing it's own MELODY while at the same time it is combining with the other instruments to produce CHORDS. The process of guiding each note in these PARALLEL MELODIES from CHORD to CHORD is called VOICE LEADING. POP/ROCK GUITAR PLAYERS can often be blissfully unaware of the complexities of this type of COMPOSING and ARRANGING. As we strum our familiar CHORD FINGER FORMS, the GUITAR itself has in some sense made our VOICE LEADING decisions for us. But what if your band has a HORN SECTION? Someone has to understand the RULES OF MOTION in order to get the players to coordinate their individual notes into a smooth and interesting ARRANGEMENT as they proceed through the CHORD PROGRESSION. As POLYPHONIC INSTRUMENTS proliferate, this skill is more and more becoming a lost art.

In conventional HARMONY, no matter how many players there are, the ENSEMBLE is generally divided into 4 GROUPS. This is known as 4 PART HARMONY. The low part is called the BASS. Up from that is the TENOR ...then the ALTO and finally, the high part is called the SOPRANO. It is beyond the scope of this program to familiarize you with all the complexities of PROPER VOICE LEADING IN 4 PART HARMONY but hopefully we can get enough of it to you so that the standard MUSIC THEORY TEXTS will make sense to you and you can take it from there. We are going to view VOICE LEADING as our last stop on our tour of the subject of CHORD VOICING because that's all it is. No big deal.

For example, here is how a modern SINGER/SONGWRITER might compose on the guitar. First they write a POEM and create a MELODY to sing it to.

SUNG MELODY



Then they look for a set of CHORDS that sounds good with that MELODY, often unaware of what NOTES they are playing or what INTERVALS separate them.

A person writing for a STRING QUARTET must create a separate, smoothly flowing MELODY for each PART. The SPACING of the notes is crucial.

VIOLIN (SOPRANO)

VIOLIN (ALTO)

VIOLA (TENOR)

CELLO (BASS)

CHORDS are formed by the vertical conjunction of these 4 individual parts. The movement of each NOTE and INTERVAL must be carefully controlled.

SMOOTH VOICE LEADING CAN BE ACCOMPLISHED BY OBSERVING CERTAIN "RULES".

DOUBLING

When playing TRIADS containing only 3 notes, one note must obviously be DOUBLED or played twice in two separate VOICES generally in two different OCTAVES. As a rule, you would DOUBLE THE ROOT NOTE particularly in CLOSED VOICED TRIADS.

SPACING

The distance between your UPPER VOICES should not be too great...never more than an OCTAVE. Your BASS and TENOR parts can be any distance apart and should not be too close together or the CHORD will sound "mucky" or "bottom heavy".

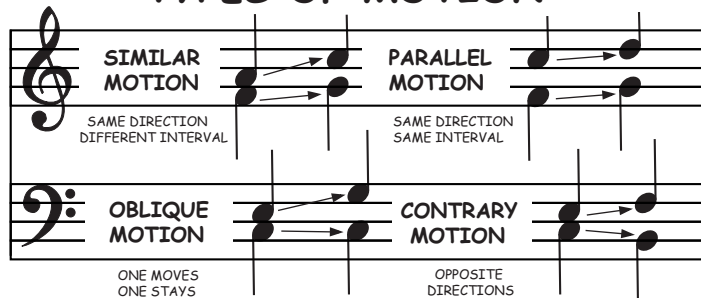
CROSSING

The PITCHES of one VOICE should never CROSS by going above or below the notes in the adjacent VOICES at any particular moment in the piece. For example, your ALTO part should never descend so far that it goes below the note in the TENOR part.

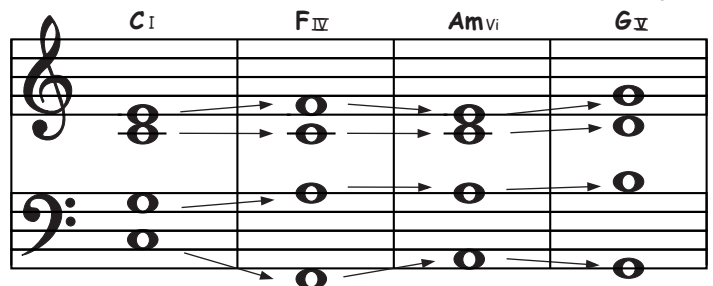
MOVEMENT

THE RULES of MOTION tell us how the individual NOTES in each VOICE move from CHORD to CHORD relative to the NOTES in the adjacent VOICES. Once again, certain guidelines are prescribed. The SOPRANO part will be heard as the MAIN MELODY and is afforded more freedom of movement. The notes in the INNER VOICES (TENOR and ALTO) should move as little as possible from CHORD to CHORD (2nds and 3rds) and if two consecutive CHORDS share a note, that note is generally CARRIED OVER from one CHORD to the next in the same VOICE. CONTRARY or OBLIQUE MOVEMENT between parts serves to retain the INDIVIDUALITY of the VOICES more than SIMILAR MOTION. PARALLEL MOTION particularly by OCTAVES or 5ths is strongly discouraged. The BASS will often move CONTRARY to the MELODY or UPPER VOICES and rarely do ALL parts move in SIMILAR MOTION. VOICE LEADING of this sort yields the smoothest transition between CHORDS.

TYPES OF MOTION

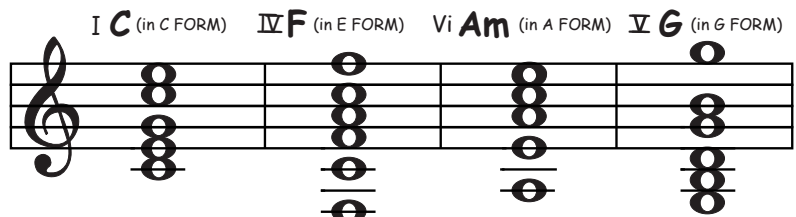


NOTICE THE MOTION IN THIS EXAMPLE IN C MAJOR



WHY DO POP/ROCK GUITARISTS THINK THEY DON'T NEED TO KNOW PROPER VOICE LEADING?

The VOICINGS imposed on us by our 5 FINGER FORMS yield relatively smooth CHORD TRANSITIONS anyway as you can see in this example. We are also very often playing more than 4 notes. In POP/ROCK you sometimes want more "jarring" changes. The NOTES in a guitar CHORD all tend to merge together into a single "BLOB" since they all have the same basic TAMBRE. The individual VOICES in a STRING QUARTET are much more apparent because of the subtle TONE differences between any given violin, viola or cello so VOICE LEADING must be taken more seriously.



CHORD PROGRESSIONS

LESSON 30

In PART 2 of this program (beginning with LESSON 11) we have been working our way through what I call the 6 LEVELS OF PITCH THEORY. In this section we have been looking at our 50 PITCH PATTERNS IN ORDER OF COMPLEXITY in 6 different increments. They are ...

① SILENCE ② THE NOTE ③ INTERVALS ④ SCALES ⑤ CHORDS

The 6th level of complexity is CHORD PROGRESSIONS. Interestingly enough, we have already learned much about PROGRESSIONS in LEVEL 4 (SCALES AND MODES) where we saw that conventional CHORD PROGRESSIONS often evolve out of systems of MODES through the process of HARMONIZATION. So in many ways, this lesson is a review of the various types of HARMONIZATIONS that we have already examined.

THE MAJOR DIATONIC SCALE IS THE CENTRAL PITCH PATTERN IN WESTERN MUSIC.

The CHORD PROGRESSIONS in a huge percentage of songs are based on the HARMONIZATION of this SCALE. You will hear so many people talking about a I-IV-V PROGRESSION but always remember that this is only part of a larger structure

MAJOR KEY HARMONIZATION

IONIAN

1 **2** **3 4** **5** **6** **7 8** I MAJOR

DORIAN

1 **2 b3** **4** **5** **6 b7** **8** ii MINOR

PHRYGIAN

1 b2 **b3** **4** **5 b6** **b7** **8** iii MINOR

LYDIAN

1 **2** **3** **#4 5** **6** **7 8** IV MAJOR

MIXOLYDIAN

1 **2** **3 4** **5** **6 b7** **8** V MAJOR

AEOLIAN

Vi MINOR **1** **2 b3** **4** **5 b6** **b7** **8**

LOCRIAN

Vii DIM. **1 b2** **b3** **4 b5** **b6** **b7** **8**

EVERY MAJOR KEY HAS IT'S RELATIVE MINOR KEY ROOTED ON IT'S 6th DEGREE.

Sad sounding songs are most often based on HARMONIZATIONS of the NATURAL and HARMONIC MINOR SCALES.

NATURAL MINOR HARMONIZATION

AEOLIAN

1 **2 b3** **4** **5 b6** **b7** **8** i MINOR

LOCRIAN

1 b2 **b3** **4 b5** **b6** **b7** **8** ii DIM.

IONIAN

1 **2** **3 4** **5** **6** **7 8** III MAJOR

DORIAN

1 **2 b3** **4** **5** **6 b7** **8** iv MINOR

PHRYGIAN

1 b2 **b3** **4** **5 b6** **b7** **8** v MINOR

LYDIAN

VI MAJOR **1** **2** **3** **#4 5** **6** **7 8**

MIXOLYDIAN

VII MAJOR **1** **2** **3 4** **5** **6 b7** **8**

HARMONIC MINOR HARMONIZATION

1 **2 b3** **4** **5 b6** — **7 8** i MINOR

1 b2 **b3** **4 b5** — **6 b7** **8** ii DIM.

1 **2** **3 4** — **#5 6** **7 8** III AUG.

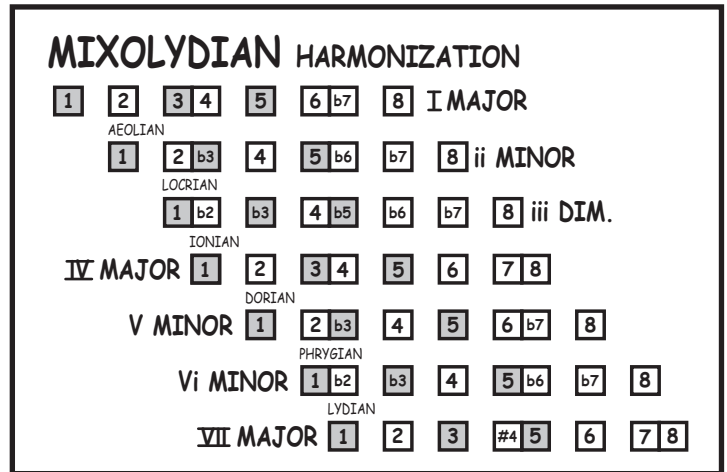
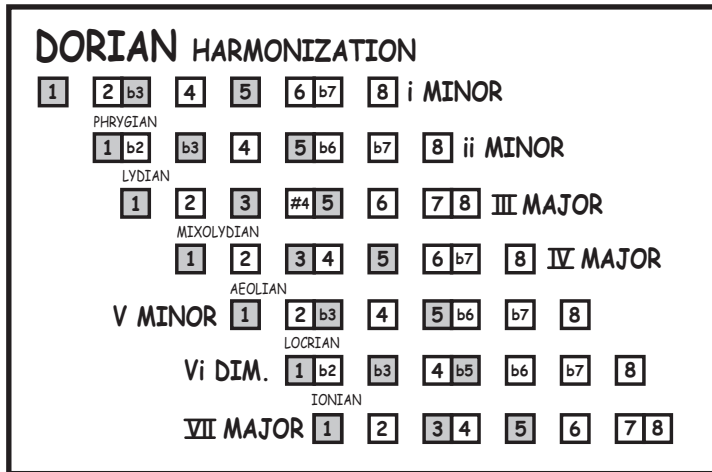
1 **2 b3** — **#4 5** **6 b7** **8** iv MINOR

1 b2 — **3 4** **5 b6** **b7** **8** V MAJOR

VI MAJOR **1** — **#2 3** **#4 5** **6** **7 8**

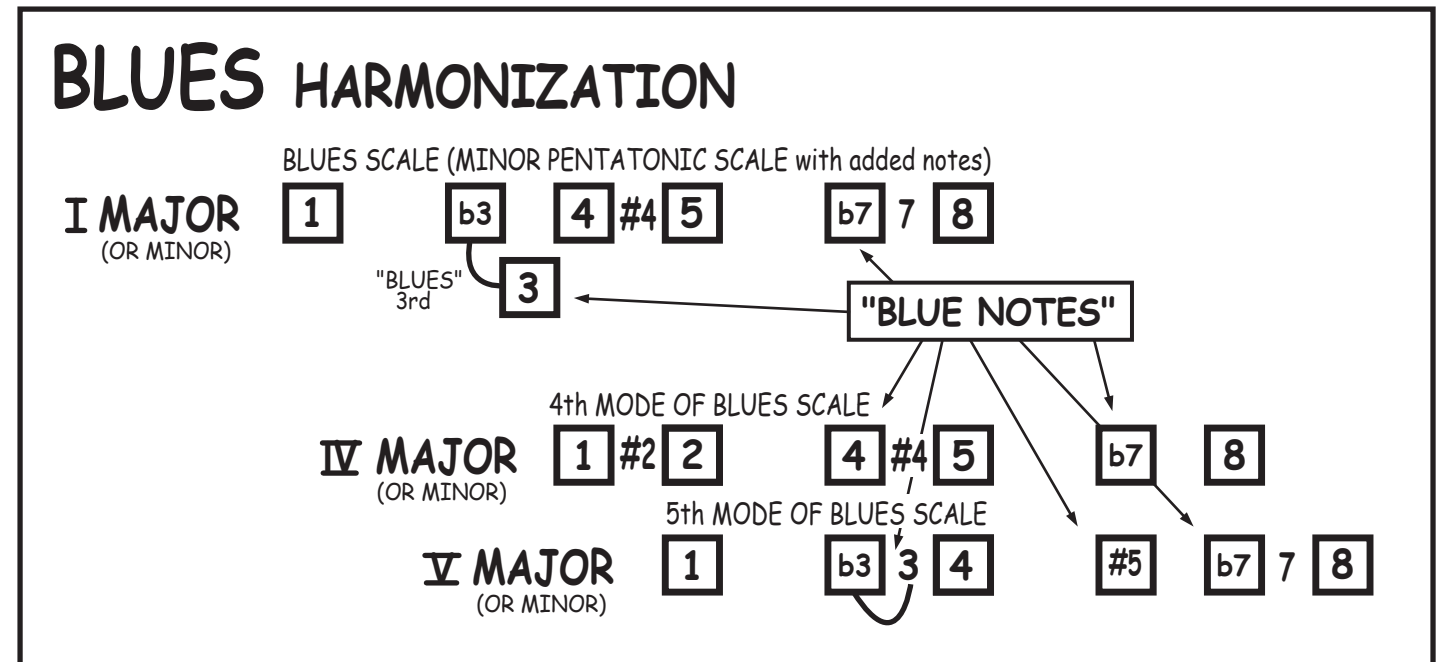
Vii DIM. **1 b2** **b3 b4** **b5** **b6 bb7** — **8**

MODAL MUSIC INVOLVES SONGS BASED ON THE HARMONIZATION OF OTHER DIATONIC MODES.
DORIAN and MIXOLYDIAN HARMONIZATIONS account for the vast majority of MODAL COMPOSITIONS.



BLUES MUSIC IS A FAIRLY RECENT BUT ENORMOUSLY POPULAR STYLE OF HARMONIZATION.

This also includes a lot of **HARD ROCK**, **HEAVY METAL** and **ALTERNATIVE MUSIC**. **JAZZ** is said to have evolved out of **THE BLUES**. **BLUES HARMONY** is related to **DIATONIC HARMONY** but also has some significant differences including the presence of **BLUE NOTES**.



These styles of **HARMONIZATION** account for the vast majority of the **KEY/CHORD PROGRESSION/ SCALE/ MELODY** relationships in **POPULAR MUSIC**. There certainly are styles of music that do not conform to these conventions but you might be surprised as to what they are. Certain styles of **DEATH METAL** involve **POWER CHORD PROGRESSIONS** that contain an extended **CHROMATICISM** not found in **DIATONIC**, **PENTATONIC** or **BLUES HARMONY**. But overall, **HEAVY METAL MUSIC** is much more conventional than many people think. As you know, **POWER CHORDS** do not contain a 3rd of any kind and so can not be labeled as either **MAJOR** or **MINOR**. However, their placement within the **PROGRESSION** will often imply their **FUNCTION** as either a **MAJOR** or **MINOR CHORD**. For example, let's say a **PROGRESSION** contains a **C5 CHORD** and an **A5 CHORD**. This implies a **I-Vi** **PROGRESSION** in the **KEY OF C**. Even though these are **POWER CHORDS**, your ear may hear the **C5** as a **MAJOR CHORD** and the **A5** as a **MINOR CHORD** because of the **RELATIVE MAJOR/MINOR** relationship that exists between these two **CHORDS**. Other clues may be provided by the **MELODY** as well. The **SINGER** may be singing 3rds that essentially convert these **POWER CHORDS** into **MAJOR** or **MINOR CHORDS**. The sad, negative, angry tenor of **HEAVY METAL** and **HARD ROCK MUSIC** comes from the fact that a lot of it is in **MINOR KEYS** and might involve **BLUES HARMONY**. **PUNK ROCK** is most often incredibly conventional music employing standard **DIATONIC CHORD PROGRESSIONS** and **MELODIES** but replacing the normal **MAJOR** and **MINOR CHORDS** with **POWER CHORDS**.

I'm often asked whether the people that write these styles of music actually understand what they are doing. The answer is often **NO** and the reason that this music is ultimately so conventional is that the song writers do not understand the **LANGUAGE OF MUSIC** well enough to do something truly revolutionary and creative. It is interesting to note that the audience for these styles of music is largely comprised of **NON PLAYERS**. Most real musicians eventually lose interest in these styles. This is not to say that this is inferior or bad art. There are more and less interesting players and composers in all styles of music.

MODULATION AND SUBSTITUTION

LESSON 31

This is HUGE! Understanding HARMONIZATION is just the beginning! That just tells you what to do when playing in one defined KEY. But who says a song has to stay in one KEY? A lot of the most fascinating music changes KEY ...sometimes often. This is known as MODULATION.

MODULATION

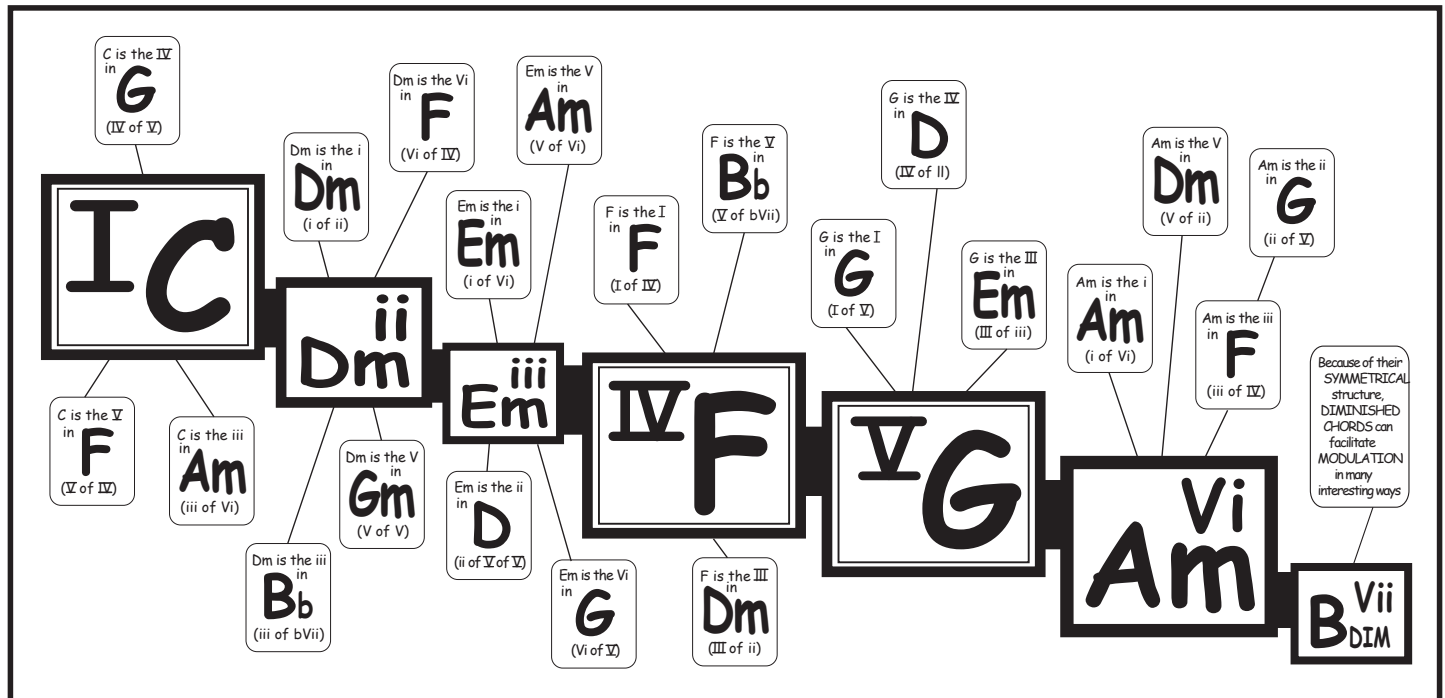
You will come to find that the possibilities here are quite enormous. I'm not sure it is possible to know everything about MODULATION. That's one of the things that keeps music fresh and interesting. Even though composers are always finding new and unique ways to

string CHORDS together and change KEYS, there are also observed theories and conventions that you will want to learn about. Even the known concepts can generate copious variations. In the end, composers rely on their ears to tell them what works and what doesn't within the context of what they are trying to accomplish aesthetically. Different individuals and cultures embrace different styles of movement.

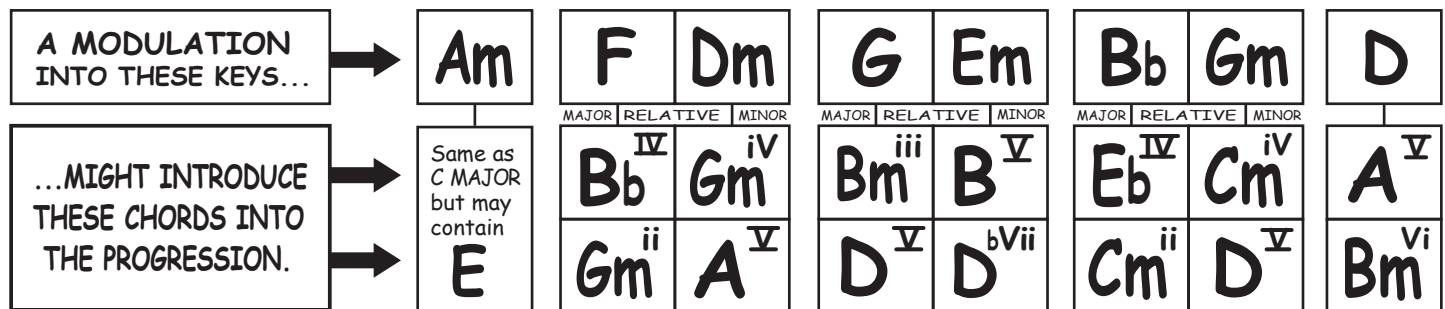
How do you know that a MODULATION has taken place? Once you go through EAR TRAINING, you may simply be able to tell by LISTENING. Songs that change KEY sound different than those that don't. Grab your SLIDERULE and look at the CHORDS in the song. You may be able to identify many of the CHORDS as ones you might expect in a certain KEY. But then you notice some CHORDS whose presence you can't immediately explain. If you look carefully enough you may be able to see that these CHORDS relate to a second KEY.

OFTEN TIMES THESE TWO KEYS ARE LINKED BY COMMON CHORDS

AND THESE CHORDS "PIVOT" US OUT OF ONE KEY AND INTO THE OTHER INTRODUCING THE UNEXPECTED CHORDS. AS AN EXAMPLE, LETS LOOK AT THE KEY OF C. HERE ARE THE NORMAL CHORDS YOU WOULD EXPECT TO FIND AND THE KEYS THAT SHARE THESE COMMON CHORDS.



SO THE KEYS THAT SHARE COMMON CHORDS WITH THE KEY OF C ARE Am-F-Dm-G-Em-Bb-Gm-D.



The KEYS that share COMMON CHORDS with the KEY OF C (or any KEY for that matter) have something else in common. They are CLOSELY RELATED KEYS ...that is that they share a significant number of notes. The KEY OF A MINOR is the RELATIVE MINOR KEY of C and these two KEYS share all seven notes. The KEY OF F and the KEY OF G share six notes with the KEY OF C as do their RELATIVE MINOR KEYS, D MINOR and E MINOR. The KEY OF Bb and the KEY OF D share five notes with the KEY OF C. You can see which KEYS are more or less related to one another by looking at the CIRCLE OF FIFTHS on the front of LESSON 32. KEYS that are closest to one another on the CIRCLE OF FIFTHS share the most notes and KEYS that are farthest apart share the least. This includes the RELATIVE MINOR KEYS which are noted on the inner circle.

What is a CIRCLE OF FIFTHS? As you go clockwise around the circle you are always proceeding to the 5th of the previous note or KEY. So G is the 5th of C ...D is the 5th of G ...A is the 5th of D ...E is the 5th of A ...etc. Another interesting thing to note is that if you go around the circle in the opposite direction ..counterclockwise ...it becomes a CIRCLE OF FOURTHS because FOURTHS and FIFTHS are the INVERTED INTERVALS of one another. Notice that all the KEYS in the above example are quite close together on the circle including their RELATIVE MINORS. You can also see that their KEY SIGNATURES are quite similar.

OTHER TYPES OF MODULATION

Sometimes a CHORD PROGRESSION will uproot as a whole and move into a new KEY. This is called a TRANSPOSITIONAL MODULATION. The PROGRESSION and MELODY remain the same but they have been TRANPOSED into the new KEY ...most often UP a WHOLE OR HALF STEP. These types of MODULATION are often used to add drama to

the ending of some pop ballads or to indicate the passage of time in certain story telling type songs.

The addition of an unexpected CHORD into a PROGRESSION does not always mean that a FULL MODULATION has taken place. There are what are called TEMPORARY or PARTIAL MODULATIONS as well.

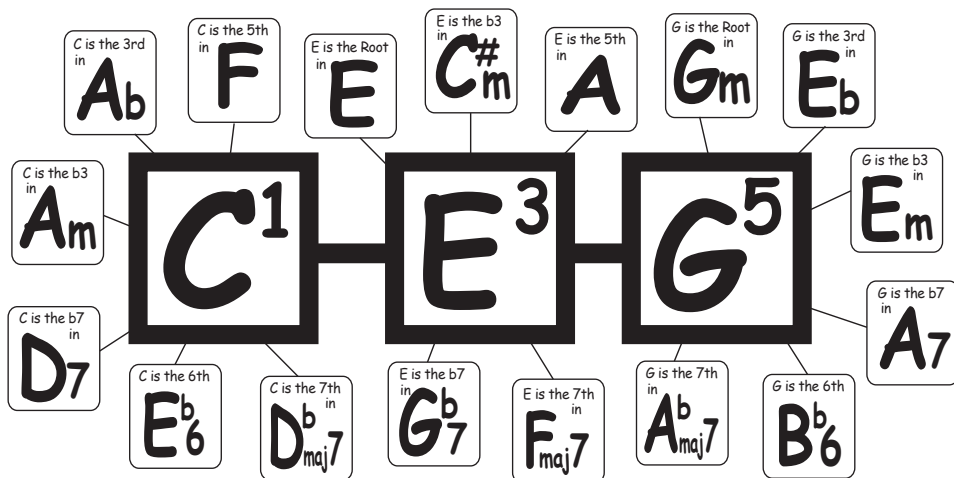
Even the movement of individual NOTES can facilitate MODULATION. A MAJOR CHORD switches to a MINOR CHORD by FLATTING the 3rd and suddenly we are in a new and unexpected KEY. I've heard it said that you can change from any CHORD to any other CHORD as long as the two share a note. That's a lot of possibilities! In the chart below we see just some of the CHORDS that share notes with a C MAJOR CHORD. These CHORDS can be altered in any number of ways and occupy various TONAL CENTERS within the KEY that they pivot us into. The shared note could also be a 9th or 11th.

DIMINISHED CHORDS are EQUAL INTERVAL CHORDS with many special properties. They have a foot in at least three different KEYS and can facilitate many interesting MODULATIONS. I've heard it said that you can MODULATE from any KEY into any other KEY using one or another DIMINISHED CHORD as the PIVOT CHORD.

Sometimes we see series of MODULATIONS. We may progress through several KEYS before we settle on one. Try playing a PROGRESSION based on a CIRCLE OF 4ths. Start on C ...go to F ...then to Bb ...then Eb ...and so on. This is a perfectly normal sounding PROGRESSION but note that you keep MODULATING to the KEY OF THE 4 CHORD. Here again there are just too many possibilities to completely explain.

Deciding which SCALES to use to construct MELODIES over the CHORDS in a MODULATING PROGRESSION can be challenging. In the end, your ears have to tell you what works and what doesn't. These are places where we might encounter certain of the MODES OF MELODIC MINOR ...notably LYDIAN b7 and MIXOLYDIAN b13. Many players who can solo quite well over PROGRESSIONS that stay in one KEY (and therefore one set of MODES) become totally lost when jamming over CHORD PROGRESSIONS that MODULATE.

SOME CHORDS THAT SHARE NOTES WITH A C MAJOR CHORD



SUBSTITUTION

In more complex music, not only do we see players creating alternate MELODIES (solos are IMPROVISED MELODIES) for a given song, we hear them altering the CHORD PROGRESSION itself at different points in the tune. This is called CHORD SUBSTITUTION. This is another broad subject with too many variables to completely cover here but understand that there are TWO BASIC KINDS OF SUBSTITUTION.

LITERAL CHORD REPLACEMENT INVOLVING TOTAL OR PARTIAL CHORD SYNONYMS

In our discussion of MODES we learned that the same set of notes can function as several different SCALES. The same thing is true of CHORDS. There are many CHORDS that share identical or very similar spellings and these CHORDS can sometimes be swapped to add variety to a CHORD PROGRESSION. Consider an Am7 and a C6 CHORD.

Am7 A-C-E-G
 1 b3 5 b7

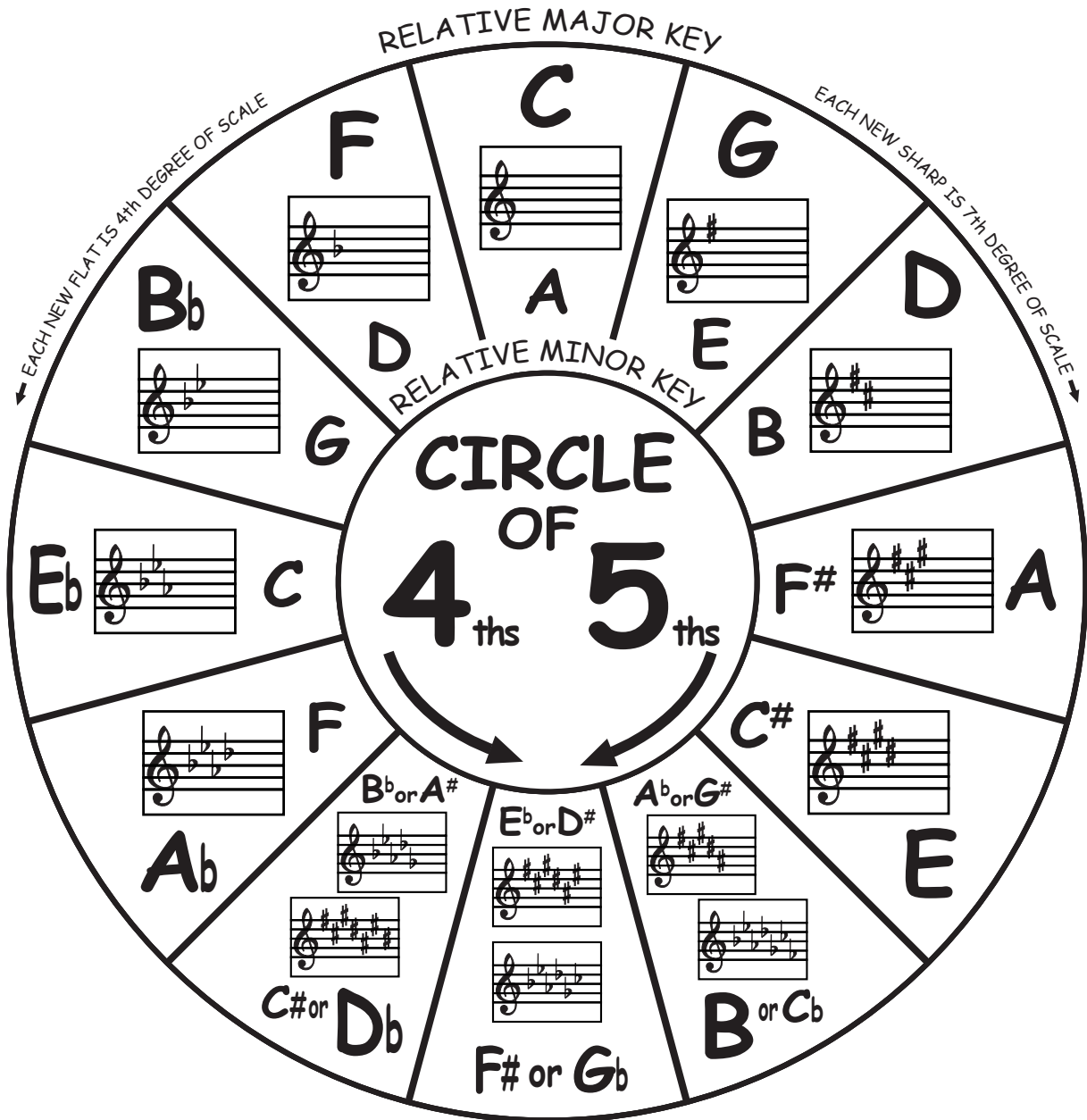
C6 C-E-G-A
 1 3 5 6

These two CHORDS contain the same notes and are therefore SYNONYMS and might replace one another in a PROGRESSION.

Most of these SYNONYMS involve complex EXTENDED CHORDS which we do not see so much in POP/ROCK MUSIC so we won't go into them here. If you've gotten this far and want to go further into JAZZ HARMONY, there are many fine texts that you can acquire. I recommend THE GUITAR HANDBOOK by RALPH DENYER as a start.

INSERTING ADDITIONAL CHORDS BETWEEN THE EXISTING CHORDS IN A PROGRESSION

This can be as simple as inserting SECONDARY CHORDS in the KEY (ii-iii-vi-vii) between the PRIMARY CHORDS (I-IV-V). DIMINISHED CHORDS can be used to bridge other CHORDS. ALTERED DOMINANT CHORDS (7 CHORDS where the 5 or 9 has been SHARPED or FLATTED) can be introduced as PASSING TONES in specific situations. To control this level of playing requires much memorization, experience and EAR TRAINING.



CONGRATULATIONS!

You have completed the ABSOLUTELY UNDERSTAND GUITAR SUPER COURSE! Spend some time and take the quiz on the back of this page. You now know more about MUSIC and more about the GUITAR than most anyone you will ever meet! This gives you the best shot at doing interesting, fulfilling and creative things with your instrument and maintaining a lifelong relationship with it. You will begin to believe that you know most everything about music when you begin doing what I've been telling you to do all along. That is ...don't just PLAY songs ...take a minute and try to ANALYZE them. You will see that the vast majority of POPULAR SONGS are structured in the ways that we have suggested and that understanding these structures is the QUICKEST and surest way to get where you want in your career.

Keep in touch with us by PHONE or through our WEB SITE. We will try to answer any questions you might still have. If you have any suggestions as to how we might improve the program, please let us know. Keep an eye out for additional products that we will be introducing. Our contact info is printed on the back cover of this book. Tell a friend about how much you enjoyed our program. Thanks again for choosing us and best of luck to you!

KEEP ON ROCKIN!
Scary

ABSOLUTELY UNDERSTAND GUITAR FINAL EXAM

Before taking this exam, make sure you can still answer the quiz questions at the beginning of Lesson 11. Your ability to use this information in any practical way requires that you retain it. So if you haven't gotten down to doing your specific memorization, you might want to start doing it now.

SILENCE AND THE NOTE

WHY IS SILENCE SO IMPORTANT IN MUSIC?

HOW MANY DIFFERENT NAMES CAN YOU RECITE FOR THE "NUCLEUS NOTE"?

INTERVALS

WHAT ARE THE 13 SIMPLE INTERVALS AND HOW LARGE ARE THEY ALL?

CAN YOU FINGER ALL 13 INTERVALS ANYWHERE ON THE NECK? WHICH INTERVALS HAVE 2 FINGERINGS?

DO YOU UNDERSTAND THE DIFFERENCE BETWEEN MAJOR AND MINOR INTERVALS?

HOW ABOUT PERFECT, AUGMENTED AND DIMINISHED INTERVALS?

WHAT IS A MELODY? WHAT IS HARMONY? SO WHAT ARE MELODIC AND HARMONIC INTERVALS?

DO YOU KNOW YOUR INVERTED INTERVALS? WHAT'S AN INVERTED MAJOR 3rd? WHAT'S AN INVERTED PERFECT 4th?

SCALES, MODES AND HARMONIZATION

CAN YOU NAME AND SPELL ALL 14 OF THE MUST KNOW SCALES? HOW DOES AEOLIAN MODE GO? MIXOLYDIAN?

CAN YOU NAME THE 3 SYMMETRICAL SCALES ON THE LIST?

HOW DO YOU KNOW WHICH SCALES ARE MAJOR AND WHICH ARE MINOR? WHAT ABOUT DIMINISHED SCALES?

WHAT IS HARMONIZATION AND WHAT DOES IT HAVE TO DO WITH COMPOSITION AND IMPROVISATION?

WHAT IS A MODE? WHAT IS THE RELATIONSHIP BETWEEN MODES AND SCALES? ARE THEY THE SAME?

CAN YOU RECITE THE 7 DIATONIC MODES IN ORDER AND SPELL THEM ALL? WHICH NUMBER TRIADS ARE MINOR?

WHAT IS THE RELATIONSHIP BETWEEN MODES AND CHORD PROGRESSIONS?

IF ALL 7 MODES HAVE THE SAME NOTES IN THEM, WHY IS IT SO IMPORTANT TO VIEW THEM AS SEPARATE SCALES?

HOW MANY OF YOUR MODE AND SCALE FINGERINGS HAVE YOU MASTERED? WHY ARE THERE SO MANY?

ARE YOU WORKING OUT WITH THE MODE JAM TRACKS EVERYDAY? CAN YOU MAKE YOUR OWN JAM TRACKS?

WHAT OTHER SCALES CAN GENERATE MEANINGFUL SYSTEMS OF MODES? DO PENTATONIC SCALES HAVE MODES?

WHERE MIGHT YOU USE THE MODES OF MELODIC MINOR?

WHAT DOES IT MEAN FOR MUSIC TO BE IN A KEY? HOW DO YOU TELL WHAT KEY A SONG IS IN?

THE CONCEPT OF MINOR KEYS CAN BASICALLY BE SUMMED UP IN 3 WORDS. WHAT ARE THEY?

CAN YOU RECITE THE MAJORNESS OR MINORNESS OF THE TRIADS IN A NATURAL MINOR HARMONIZATION?

HOW DOES HARMONIC MINOR DIFFER FROM NATURAL MINOR AND WHY IS IT SO SIGNIFICANT?

HOW DOES MODAL MUSIC DIFFER FROM MAJOR AND MINOR KEY MUSIC?

HOW IS BLUES HARMONY SIMILAR TO DIATONIC HARMONY? HOW IS IT DIFFERENT? WHAT IS A BLUE NOTE?

CHORDS AND HARMONY

WHAT IS TERTIAN HARMONY? WHAT IS QUARTEL HARMONY?

CAN YOU RECITE THE SPELLING AND INTERVAL SEQUENCE FOR EACH OF THE 4 TYPES OF TRIADS?

WHICH ARE THE SYMMETRICAL TRIADS AND WHAT SPECIAL PROPERTIES DO THEY HAVE?

HOW DO YOU TURN A TRIAD INTO A CHORD? HOW DOES THIS RELATE TO THE NUMBER SYSTEM?

WHAT DO YOU HAVE TO REMEMBER ABOUT DOMINANT 7 CHORDS AND HOW THEY DIFFER FROM MAJOR 7 CHORDS?

WHAT ARE EXTENDED CHORDS AND WHAT DO YOU HAVE TO REMEMBER ABOUT THEIR SPELLINGS?

WHAT IS A SUSPENDED CHORD? WHAT IS AN EXTENDED DOMINANT CHORD?

CAN YOU RECITE THE SPELLING OF ALL THE CHORDS ON THE LIST WE GAVE YOU? HOW IS A DIMINISHED 7 SPELLED?

HAVE YOU BEGUN LOOKING AT ALL THE MULTIPLE FINGERINGS FOR YOUR CHORDS?

WHAT ARE TRIAD AND CHORD INVERSIONS? HOW MANY INVERSIONS DOES A DOMINANT 7 CHORD HAVE?

WHAT IS THE DIFFERENCE BETWEEN CLOSED VOICING AND OPEN VOICING?

WHAT IS A SLASH CHORD AND HOW DO THEY RELATE TO INVERSIONS?

WHAT IS A POLYCHORD?

CAN YOU EXPLAIN THE CONCEPT OF 4 PART HARMONY? WHAT IS VOICE LEADING? WHAT ARE THE RULES OF MOTION?

WHY DO POP/ROCK GUITAR PLAYERS THINK THAT THEY DON'T NEED TO UNDERSTAND PROPER VOICE LEADING?

CHORD PROGRESSIONS

WHAT IS THE RELATIONSHIP BETWEEN HARMONIZATION AND CHORD PROGRESSIONS?

HOW MANY HARMONIZATIONS CAN YOU RECITE? HOW IS BLUES HARMONY DIFFERENT FROM DIATONIC HARMONY?

WHAT IS MODULATION? CAN YOU DESCRIBE THE DIFFERENT TYPES OF MODULATION?

HOW CAN YOU TELL THAT A MODULATION HAS TAKEN PLACE?

WHAT ARE CLOSELY RELATED KEYS AND HOW DO YOU TELL WHICH KEYS ARE MORE OR LESS RELATED?

WHAT KEYS SHARE CHORDS WITH THE KEY OF F? WHAT IS THE RELATIVE MINOR KEY OF F MAJOR?

WHAT IS SUBSTITUTION? CAN YOU DESCRIBE THE 2 DIFFERENT TYPES OF SUBSTITUTIONS?

FINALLY... REMEMBER THAT MUSIC THEORY IS LARGELY USELESS WITHOUT EAR TRAINING !